

# Locally Robust Commensurate BART for Borrowing External Controls

## Abstract

Real-world data (RWD) can supplement the control arm of a randomized controlled trial (RCT), but the comparability of external and trial controls may vary with patient characteristics. Study-level meta-analytic predictive, commensurate, and power priors apply a common borrowing parameter across patients. We propose locally robust commensurate Bayesian additive regression trees (LRC-BART) to model the RCT control surface as a pooled BART surface plus a sparse discrepancy ensemble. A spike-and-slab commensurate prior on each discrepancy leaf allows borrowing to vary across covariate regions. The spike represents commensurability and the slab serves as the vague component of a robust meta-analytic predictive prior. Tree partitions determine the regions over which each component applies. All leaf marginals are exact, so the sampler targets the exact posterior. The prior effective sample size is defined at the estimand, the RCT-standardized control mean, and calibrated against the RWD posterior with an explicit ceiling. We prove that the RWD-induced shift of a leaf estimate is bounded and vanishes at a Gaussian rate under conflict, that the posterior probability of a small local discrepancy is consistent under fixed partitions and representability of the discrepancy, and that the calibration is monotone and well posed. A covariate-resolved map reports posterior compatibility at each patient profile; realized regional effective sizes summarize borrowing. Simulations with Gaussian and survival outcomes and a single-arm myeloma trial with a registry control illustrate the method.

## 1 Introduction

Real-world data can supplement a randomized trial when collecting a sufficiently large control arm is difficult. Their usefulness depends on whether external outcomes represent the outcomes trial patients would have experienced under control. Trial selection, treatment practice and outcome ascertainment may vary with age, disease stage and prior therapy. External and trial controls may therefore be comparable for some patient profiles and differ substantially for others.

A common borrowing parameter cannot fully accommodate this variation. It may allow excessive borrowing where the sources disagree or discard useful information where they agree. We propose a Bayesian tree model that learns regions of covariate space from the data and allows borrowing to differ across them. We accompany the treatment-effect estimate with profile-specific summaries of compatibility and external information.

The established priors for historical borrowing act on a study-level parameter. The meta-analytic predictive (MAP) prior of Neuenschwander et al. [17] and Schmidli et al. [23] treats the control parameters of the historical studies and of the new study as exchangeable draws from a common population and takes the posterior predictive distribution of the new study’s parameter as its prior; the robust MAP prior mixes that predictive component with a vague one so that a current study in conflict with the historical data can switch borrowing off. The commensurate prior of Hobbs et al. [9, 10] conditions the current parameter on the historical one,  $\theta | \theta_0 \sim \mathcal{N}(\theta_0, 1/\tau)$ , and learns the commensurability precision  $\tau$  from the data, with a spike-and-slab hyperprior so that the posterior can move between near-complete borrowing and near-none. The power prior of Ibrahim and Chen [11] raises the historical likelihood to a power  $a_0$ , normalized by Duan et al. [7] and Neuenschwander et al. [16] so that  $a_0$  can carry its own prior. Röver and Friede [21, 22] show that with a single historical study, the case we treat here, the normal MAP, commensurate, and power-prior forms nearly coincide. In these study-level formulations, a common heterogeneity or discount scale determines borrowing without reference to individual covariates.

Locally robust commensurate BART (LRC-BART) models the two control sources jointly as a pooled Bayesian additive regression tree (BART) surface  $f$  fitted to RCT and RWD controls alike, plus a sparse discrepancy ensemble  $g$  that only the RCT controls inform, so that the RWD follow  $f$  and the RCT controls follow  $f + g$ . Each leaf of  $g$  carries a spike-and-slab prior whose spike is a commensurate component of small scale and whose slab is a vague component on the outcome scale, so the tree prior of  $g$  decides, region by region, whether the RCT control surface is the RWD surface up to a calibrated discrepancy or a free local offset.

Our construction combines three established ideas. The decomposition into a pooled surface and a discrepancy ensemble multiplied by an indicator is the Bayesian causal forest of Hahn et al. [8], with the source indicator in the place of the treatment indicator. The leaf prior is the commensurate prior of Hobbs et al. [9] in spike-and-slab form, with the slab in the role of the vague component of Schmidli et al. [23]; conditioning on the pooled surface is the commensurate conditioning on the historical parameter, and since with one external source this is close to a single-study MAP prior [22], we do not claim a methodological distinction on that basis. The calibration uses the expected local-information-ratio effective sample size of Neuenschwander et al. [18].

We combine these components by placing the spike-and-slab commensurate prior on the leaves

of a discrepancy ensemble so that borrowing varies across covariate regions learned jointly with the surface, which a single ensemble with two means per leaf cannot do (Web Appendix G.6). Every leaf marginal is in closed form, so the tree moves target the exact posterior and no surrogate likelihood is needed. The prior effective sample size is defined at the estimand, the RCT-standardized control mean, and calibrated against the RWD posterior; it has a ceiling equal to the information the RWD carries about that estimand, and a target can be stated as a fraction of the ceiling. We prove that the RWD-induced shift of a leaf estimate is bounded and vanishes at a Gaussian rate as the local conflict grows, that the posterior probability of a small local discrepancy is consistent under fixed partitions and representability of the discrepancy, and that the calibration is monotone and well posed. The posterior of  $g$  then gives a covariate-resolved borrowing map, which reports at each patient profile the posterior probability that the local discrepancy is below a pre-specified threshold, together with a pointwise effective sample size. The model extends to right-censored outcomes through an accelerated failure time (AFT) likelihood with data augmentation, and to the single-arm setting in which no RCT control exists.

## 1.1 Related work

A second group of methods lets the amount of borrowing vary with covariates, through propensity-score strata [4, 25], a commensurate precision that depends on the covariates through kernel local regression [13], exchangeability over sources or over latent patient classes [1, 14, 15], a global congruence statistic [12], or clustering of the patients on the covariates before the outcome model is fitted [2, 3]; Web Appendix G.0 reviews them, and the closest parametric relative of our borrowing map is the covariate-adaptive hybrid borrowing of Jin et al. [13].

Among tree methods, Zhou and Ji [27] is the most directly related approach. They fit one BART ensemble [5] to RCT and external controls with the data source as a splitting covariate, so that trees pool the sources wherever they do not split on the source; borrowing is implicit and has no parameter, and their discussion proposes as future work a prior discouraging source splits together with a source offset whose scale controls borrowing. We model the proposed offset nonparametrically and give it a calibratable prior. Pan et al. [19] apply BART to patient-level borrowing across subgroups. HE-BART [26] places group-specific random effects inside each terminal node with a shared precision, the structural precedent for a hierarchical parameter inside a leaf, with the aim of

mixed-effects prediction rather than controlling historical borrowing for a current study. flexBART [6] lets a tree assign several levels of a categorical predictor to one branch, which with the source as predictor gives the implicit borrowing of Zhou and Ji [27] in a richer split space. None of these places a commensurate prior on a leaf, and among the covariate-adaptive methods only Jin et al. [13] and Alt et al. [1] report borrowing at the resolution of a patient profile, neither on a nonparametric outcome surface.

Section 2 sets out the model, the sampler, the effective sample size and its calibration, the borrowing map with its guarantees, and the single-arm variant. Section 3 reports simulation studies with Gaussian and survival outcomes, including scenarios in which the external data are compatible on part of the covariate space only, Section 4 applies the method to a single-arm myeloma trial with a registry control, and Section 5 concludes. Proofs are in the Web Appendix.

## 2 Methods

### 2.1 Setting and estimands

We observe an RCT with  $N$  subjects, of whom  $n_1$  are controls and  $N - n_1$  are treated, and an external source of  $n_2$  RWD controls. For subject  $i$  we record the outcome  $y_i$ , the covariate vector  $x_i \in \mathbb{R}^p$ , and the source  $g_i \in \{1, 2\}$ , with  $g_i = 1$  for an RCT control and  $g_i = 2$  for an RWD control;  $D_i = \mathbf{1}\{g_i = 1\}$  indicates an RCT control. Continuous outcomes are modelled as Gaussian given the covariates. Survival outcomes are modelled on the log scale under a log-normal AFT model with right censoring, and  $y_i$  then denotes the log survival time, observed or imputed as described in Section 2.3.

The estimands are standardized to the RCT covariate distribution. Let  $f_T(x)$  be the mean outcome under treatment and  $f_1(x)$  the mean outcome of an RCT control at covariate  $x$ . For Gaussian outcomes the average treatment effect is  $\text{ATE} = N^{-1} \sum_{i \leq N} \{f_T(x_i) - f_1(x_i)\}$ , the average over all  $N$  RCT profiles, treated and control. For survival outcomes we report the ratio of population median survival times  $q_{0.5}(\bar{S}_T)/q_{0.5}(\bar{S}_1)$ , where  $\bar{S}_a(t) = N^{-1} \sum_i S_a(t | x_i)$  is the arm- $a$  survival function standardized to the RCT profiles, and the ratio of restricted mean survival times  $\int_0^{t^*} \bar{S}_T(t) dt / \int_0^{t^*} \bar{S}_1(t) dt$  at a horizon  $t^*$  inside the follow-up. The treatment surface  $f_T$  is estimated from the RCT treated subjects alone with standard BART or AFT-BART. External observations

inform the control surface  $f_1$ . We define effective sample size through the RCT-standardized control mean  $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$ .

Both ensembles below are BART models [5]. A BART surface is a sum of  $H$  regression trees,  $\sum_{h=1}^H m(x; T_h, \theta_h)$ , where  $T_h$  is a binary tree whose internal nodes split on one covariate at one cutpoint and whose terminal nodes (leaves)  $\ell = 1, \dots, L_h$  carry means  $\theta_{h\ell}$ , and  $m(x; T_h, \theta_h)$  returns the mean of the leaf to which  $x$  is routed. The tree prior splits a node at depth  $d$  with probability  $\alpha(1+d)^{-\beta}$ , chooses the splitting variable and cutpoint uniformly, and the leaf means are independent  $\mathcal{N}(0, \lambda^2)$  with  $\lambda = (\max y - \min y)/(2k\sqrt{H})$  and  $k = 2$ , the range rule, so that the sum of  $H$  trees spans the outcome range with high prior probability.

## 2.2 The model

The two control sources are modelled jointly as

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad (1)$$

Here  $f(x)$  is the external-control response mean and  $f_1(x) = f(x) + g(x)$  is the RCT-control response mean, with a separate residual variance for each source. Both surfaces are estimated jointly.

The pooled surface  $f$  is a standard BART ensemble of  $H_f = 50$  trees with the default split prior  $(\alpha_f, \beta_f) = (0.95, 2)$  and leaf prior  $\mathcal{N}(0, \lambda_f^2)$  with  $\lambda_f$  from the range rule. The discrepancy surface  $g$  is a small, shallow ensemble of  $H_g = 5$  trees with split prior  $(\alpha_g, \beta_g) = (0.5, 3)$ , so that a  $g$  tree is a stump a priori with probability one half and rarely deeper than two levels, and its leaves represent differences from the external-control response mean. The commensurate component centers  $f_1(x)$  at  $f(x)$ . In particular, conditional on  $f$ , the discrepancy-tree partitions  $T_g$  and the spike variance, if all  $H_g$  leaves reached by  $x$  are in the spike, then

$$f_1(x) | f, T_g, \{z_{h, \ell_h(x)} = 1\}_{h=1}^{H_g}, \tau_0^2 \sim \mathcal{N}(f(x), H_g \tau_0^2).$$

Thus agreement means that the RCT-control response mean is close to the external-control response mean. We implement this prior with zero-centered leaves for  $g(x) = f_1(x) - f(x)$ :

$$\theta_{h\ell} | z_{h\ell} \sim z_{h\ell} \mathcal{N}(0, \tau_0^2) + (1 - z_{h\ell}) \mathcal{N}(0, \tau_1^2), \quad z_{h\ell} \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w), \quad (2)$$

with  $w \sim \text{Beta}(1, 1)$ ,  $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$  with  $\nu_0 = 3$ , and  $\tau_1^2 = (\max y - \min y)^2 / (4k^2 H_g)$  fixed by the range rule for an  $H_g$ -tree ensemble. We require  $\tau_0^2 < \tau_1^2$ , Assumption A1 of the Web Appendix, and the sampler draws  $\tau_0^2$  from its conditional truncated to  $(0, \tau_1^2)$ . The residual variances carry independent inverse-gamma priors. The spike,  $z = 1$ , shrinks an individual discrepancy contribution toward zero and thereby shrinks the RCT response mean toward the external response mean. Section 2.4 calibrates its leaf scale  $\tau_0$  through the effective sample size of the full ensemble. The slab,  $z = 0$ , is the vague component of the robust MAP prior placed in a leaf of  $g$ : a free local discrepancy on the outcome scale, sized by the range rule. If the scale prior is left untruncated, integrating  $z$  and  $\tau_0^2$  gives the leaf marginal  $w t_{\nu_0}(\theta; 0, s_0^2) + (1 - w)\mathcal{N}(\theta; 0, \tau_1^2)$ . The sampler uses the truncated scale prior, for which the spike marginal is not exactly Student- $t$ . The local discrepancy  $g(x) = f_1(x) - f(x)$  sums  $H_g$  leaf contributions, one per tree. More generally, conditional on  $f$ ,  $T_g$ , the reached-leaf indicators and both scales,  $f_1(x)$  is normal with mean  $f(x)$  and variance  $\sum_{h=1}^{H_g} \{z_{h, \ell_h(x)} \tau_0^2 + (1 - z_{h, \ell_h(x)}) \tau_1^2\}$ . The slab increases the permitted departure from  $f(x)$  while retaining the same prior center. These distributions describe the latent control mean; observation-level residual variance enters through (1).

Three special cases place the model among its comparators. With  $w = 1$  every leaf is in the spike and the model is a commensurate BART (C-BART) with a single commensurability scale  $\tau_0^2$ ; with  $w = 1$  and  $\tau_0^2 \rightarrow 0$  the discrepancy vanishes and the model is BART fitted to the pooled controls (BART-CP); with  $w = 0$  every leaf is in the slab and  $g$  is a free nonparametric source offset regularized by the range rule, the tree analogue of BART with the source indicator as a covariate (BART-PP). LRC-BART with  $w$  learned from the data moves between these per leaf.

The decomposition is what lets a regional shift be detected: in the alternative design, one ensemble with two means per leaf coupled through a common variance, the cheapest fit spreads a shift across all  $H$  trees under the spike, and a prior-cost calculation (Web Appendix G.6) shows that at  $H = 50$  no shift of clinical size flips an indicator, whereas in (1) the shift lands in one or two leaves of a small  $g$  ensemble after a single split, where the spike cannot hide it by spreading it. Identification is by support: where the RCT has no support  $g$  is a prior draw and  $f$  is the RWD surface, and where both sources have support  $f$  and  $g$  separate because  $f$  is pinned by the RWD and  $g$  is shrunk by the spike (Web Appendix E).

### 2.3 Posterior computation

We use Markov chain Monte Carlo with the backfitting scheme of Chipman et al. [5], one tree at a time with the others held fixed. The factors in each tree-move ratio have closed forms because conditionally on  $(\tau_0^2, \tau_1^2, w)$  the leaf prior (2) is a finite Gaussian mixture. For a leaf of an  $f$  tree the partial residuals  $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$  come from both sources with their own variances; with  $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$  and  $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$ , where  $n_g$  and  $S_g$  are the count and sum of the source- $g$  residuals in the leaf, the observation-level factors  $\prod_i \phi(r_i; 0, \sigma_{g_i}^2)$  cancel from every tree-move ratio and the leaf factor is

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad (3)$$

the heteroscedastic form of the standard BART marginal. For a leaf of a  $g$  tree the partial residuals  $r_i = y_i - f(x_i) - g_{-h}(x_i)$  use only RCT controls in that leaf, with count  $n_1$  and mean  $\bar{y}_1$ , and writing  $v_1 = \sigma_1^2/n_1$ ,

$$M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)}, \quad (4)$$

a two-component Gaussian mixture in the leaf mean, with  $M_g = 1$  in a leaf holding no RCT controls. Grow, prune, and change moves for both ensembles use the standard tree-prior and proposal ratios times the ratio of products of  $M_f$  or  $M_g$  over the affected leaves, a grow move contributing  $M_L M_R / M_P$ . Since these are the exact collapsed ratios, the tree kernels leave the exact posterior invariant with the leaf parameters and indicators integrated out, and the full sweep is a partially collapsed Gibbs sampler with the joint posterior as its invariant law (Lemma A1 and Proposition A1 of Web Appendix A).

Conditional on the trees, we update the leaf parameters conjugately. An  $f$  leaf draws  $\theta \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$ . A  $g$  leaf is updated as a block: first

$$P(z = 1 | \bar{y}_1) = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1)}{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}, \quad (5)$$

then  $\theta | z \sim \mathcal{N}(\tau_z^2 \bar{y}_1 / (\tau_z^2 + v_1), \tau_z^2 v_1 / (\tau_z^2 + v_1))$  with  $\tau_z^2$  equal to  $\tau_0^2$  in the spike and  $\tau_1^2$  in the slab. With  $K_0$  spike leaves among the  $L_g$  leaves of  $g$  and  $S_0$  the sum of their squared means, the hyperparameters update as  $\tau_0^2 | \cdot \sim \text{scaled-Inv-}\chi^2(\nu_0 + K_0, (\nu_0 s_0^2 + S_0) / (\nu_0 + K_0))$  truncated to  $(0, \tau_1^2)$  and  $w | \cdot \sim \text{Beta}(1 + K_0, 1 + L_g - K_0)$ , and the residual variances update from their inverse-gamma

conditionals using the residuals  $y_i - f(x_i) - D_i g(x_i)$  of each source. We fix  $\tau_1^2$  by the range rule because slab leaves are few and the slab must allow discrepancies on the outcome scale. An  $f$  leaf holding only one source is an ordinary BART leaf, and the minimum leaf size of five is enforced on each leaf's own data, pooled for  $f$  and RCT-only for  $g$ . A sweep costs a standard BART on  $n_1 + n_2$  observations plus a small ensemble on  $n_1$ .

For right-censored survival outcomes we use the AFT model  $\log y_i = f(x_i) + D_i g(x_i) + \varepsilon_i$  with data augmentation. At each iteration a censored subject's log survival time is imputed from  $\mathcal{N}(f(x_i) + D_i g(x_i), \sigma_{g_i}^2)$  truncated to  $(\log C_i, \infty)$ , where  $C_i$  is the censoring time, and treated as observed in the tree and parameter updates that follow; the treatment-arm AFT-BART is imputed the same way. The remaining tree and parameter updates are unchanged.

## 2.4 Prior effective sample size and calibration

The borrowing induced by a spike scale  $s_0^2$  depends on the residual variance, leaf sizes and  $H_g$ . This makes the scale difficult to specify directly. We calibrate it through a prior effective sample size (ESS) defined at the estimand. Conditionally on the RWD, the prior for the RCT control surface is  $f_1 = f + g$  with  $f | \text{RWD}$  the posterior of a standard BART fitted to the RWD alone and  $g$  drawn from its prior. For a scalar parameter with prior variance  $V$  and unit Fisher information  $1/\sigma_1^2$ , the expected local-information-ratio ESS of Neuenschwander et al. [18] is  $\sigma_1^2/V$ , the number of RCT control observations carrying the same information as the prior, and we apply this to  $\mu$ . Write  $V_\mu^f = \text{Var}(N^{-1} \sum_{i \leq N} f(x_i) | \text{RWD})$ , which we estimate from posterior draws of the RWD-only BART evaluated at the RCT profiles, and  $c_g = \mathbb{E}[N^{-2} \sum_{i,j} \mathbf{1}\{x_i, x_j \text{ share a leaf}\}]$  under the  $g$ -tree prior, the leaf-sharing constant, which equals one for stumps and is computed once by Monte Carlo. Given the spike and slab states,  $g$  adds  $c_g \sum_h \tau_{z_h}^2$  to the prior variance of  $\mu$ . The ESS of the commensurate component, with every  $g$  leaf in the spike, is

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[ \frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right], \quad (6)$$

a one-dimensional integral, and the prior-averaged ESS is

$$\text{ESS}(w, s_0^2) = \sum_{k=0}^{H_g} \binom{H_g}{k} w^k (1-w)^{H_g-k} \sigma_1^2 \mathbb{E}[\{V_\mu^f + c_g(k\tau_0^2 + (H_g - k)\tau_1^2)\}^{-1}].$$

We calibrate  $\text{ESS}_0$  and report  $w$  separately, as robust MAP reports the ESS of its informative component and the robust weight separately: conditional on  $w$ , the prior probability that every leaf covering a given profile is commensurate is  $w^{H_g}$ , so the averaged ESS is dominated by  $w$  and cannot reach a target of the order of  $n_2$  whatever  $s_0^2$  is.

The prior ESS has a finite upper limit: as  $s_0^2 \rightarrow 0$ ,  $\text{ESS}_0 \rightarrow \sigma_1^2/V_\mu^f$ , the information the RWD posterior carries about  $\mu$  before any discrepancy is added. For a single tree with RCT leaf proportions equal to the RWD ones,  $V_\mu^f = \sigma_2^2/n_2$  and the ceiling is  $n_2\sigma_1^2/\sigma_2^2$  exactly, the RWD sample size in RCT units; under covariate shift it is lower, since RWD in regions without RCT patients carries no information about  $\mu$  (Theorem 3(d)). The regularization of  $f$  itself contributes a term of order  $\sigma_1^2/(H_f\lambda_f^2)$  that is negligible at  $H_f = 50$  and is not subtracted. A target  $N_{\text{target}}$  may be given as an absolute number of RCT controls, feasible only below the ceiling, or as a fraction of the ceiling, which is always feasible and which we recommend as the primary sensitivity ladder when the overlap is poor.

We first fit BART to the RWD and estimate  $V_\mu^f$  from draws of the standardized surface. We divide the draws into  $B$  blocks and rescale their variances  $V^{(b)}$  to have mean  $V_\mu^f$ . For a grid  $\mathcal{S}$  of spike scales, we use the Pr-rule

$$\hat{s}_{0\star}^2 = \inf\{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}, \quad (7)$$

with  $q = 0.95$ . This selects the most informative spike whose prior ESS is at most  $N_{\text{target}}$  in a fraction  $q$  of the blocks. Targets at or above the whole-chain ceiling are flagged as capped. We report the prior  $\text{ESS}_0$  and the realized ESS,  $\mathbb{E}_{\text{post}}[\sigma_1^2/\{V_\mu^f + c_g \sum_h \tau_{z_h}^2\}]$ . The pointwise quantity  $\text{ESS}(x) = \sigma_1^2/\{V_f(x) + H_g\tau_0^2\}$  describes local prior information, where  $V_f(x)$  is the RWD posterior variance at  $x$ . Web Appendix D gives the calibration mechanics.

## 2.5 The borrowing map and its guarantees

We summarize compatibility over covariate space by the posterior probability that the local discrepancy is smaller than a threshold  $c > 0$  fixed in advance on the outcome scale,

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}), \quad (8)$$

computed as the fraction of posterior draws, together with the posterior mean and interval of  $g(x)$  and the pointwise ESS of Section 2.4. We also retain the all-spike probability  $B(x)$ , the posterior probability that every leaf covering  $x$  belongs to the spike, as a diagnostic; its prior scale is  $w^{H_g}$  and, as Theorem 2 explains, its limiting value need not identify the true compatibility state, whereas  $\mathcal{B}_c$  is consistent under the stated conditions. In the simulation study  $c = 0.5$ , one third of the RCT residual standard deviation for Gaussian outcomes and 0.5 on the log-time scale for survival outcomes.

We give three theoretical results. The first two concern leaf-level inference and condition on the pooled surface  $f$ , on the partition of the  $g$  trees, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ , and consider a  $g$  leaf with  $n$  RCT controls, residual mean  $\bar{y}$ , and  $v = \sigma_1^2/n$ . Write  $\rho_j = \tau_j^2/(\tau_j^2 + v)$  for  $j \in \{0, 1\}$ , so that  $\hat{\theta}_1 = \rho_1 \bar{y}$  is the slab estimate, close to the RCT-only residual mean, and  $\hat{\theta}_0 = \rho_0 \bar{y}$  the spike-shrunk estimate, close to zero, that is, to the pooled surface. Let  $p = P(z = 0 | \bar{y})$  be the posterior slab probability and

$$\kappa = \frac{1}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a}. \quad (9)$$

**Theorem 1** (Bounded local influence). *Under Assumption A1 and the conditioning above,  $\theta | \bar{y}$  is the mixture  $p\mathcal{N}(\hat{\theta}_1, \rho_1 v) + (1-p)\mathcal{N}(\hat{\theta}_0, \rho_0 v)$  with  $\text{logit} p = a + \kappa \bar{y}^2$ , and the shift of the posterior mean relative to the slab estimate,  $I(\bar{y}) = \mathbb{E}[\theta | \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0)\bar{y}$ , satisfies*

$$\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}, \quad 1-p \leq C e^{-\kappa \bar{y}^2},$$

so  $I(\bar{y}) \rightarrow 0$  at a Gaussian rate as  $|\bar{y}| \rightarrow \infty$ . With  $w = 1$  (C-BART) the shift is  $-(\rho_1 - \rho_0)\bar{y}$ , linear and unbounded, and with  $w = 1, \tau_0^2 \rightarrow 0$  (BART-CP) it is  $-\rho_1 \bar{y}$ .

Theorem 1 is Theorem A1 of Web Appendix B. The commensurate component changes a leaf estimate relative to the RCT-only estimate by a bounded amount, of the order of one RCT standard error of the leaf, and this borrowing-induced shift vanishes as local conflict grows. The bound is on the borrowing-specific part; the slab estimate itself is shrunk toward the pooled surface by the factor  $\rho_1$ , the range-rule regularization any BART leaf applies, which is slight at ordinary leaf sizes. C-BART and BART-CP have no such bound, and their bias grows linearly with the

conflict. Corollary A1 carries the bound to the estimand: at fixed partitions of the  $g$  ensemble the RWD-induced shift of the standardized control mean relative to the free-offset model is bounded uniformly in the RCT data, whereas under C-BART and BART-CP it is a nonzero linear functional of the data and hence unbounded.

**Theorem 2** (Detection and the borrowing map). *Under Assumption A1 and the conditioning above, let the RCT residuals in the leaf be  $d + \epsilon_i$  with  $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_1^2)$ , where  $d$  is the true local discrepancy.*

- (a) *The spike probability  $P(z = 1 | \bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$  is decreasing in  $|\bar{y}|$ , and when  $\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$  it equals one half at*

$$d_{1/2} = \left[ \frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\} \right]^{1/2},$$

*a few multiples of  $(\tau_0^2 + \sigma_1^2/n)^{1/2}$ , so the probability that the leaf abstains at true shift  $d$  is  $\bar{\Phi}\{(d_{1/2} - d)/\sqrt{v}\} + \bar{\Phi}\{(d_{1/2} + d)/\sqrt{v}\}$ , with  $\bar{\Phi}$  the standard normal survival function.*

- (b) *For a single  $g$  tree,  $\mathcal{B}_c(x) = p \Psi_1(c) + (1-p) \Psi_0(c)$  with  $\Psi_j(c) = P\{|\mathcal{N}(\rho_j \bar{y}, \rho_j v)| < c\}$ , and as  $n \rightarrow \infty$  with  $|d| \neq c$ ,  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$  almost surely.*

- (c) *As  $n \rightarrow \infty$  with  $\tau_0^2$  fixed, the spike probability converges to a limit in  $(0, 1)$  that is at most  $\frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$  for  $d \neq 0$  and exceeds  $w$  for  $d = 0$ ; it tends to zero only as  $|d|/\tau_0 \rightarrow \infty$ , or for every fixed  $d \neq 0$  if  $\tau_0^2 \rightarrow 0$  with  $n$ .*

Theorem 2 combines Theorems A2 and A3 of Web Appendix C. The ensemble extension requires fixed partitions and a discrepancy representable on their additive span. The leaf results also condition on  $f$ ; they do not establish consistency under the full posterior over both ensembles. Further interpretation is given in Web Appendix G.3.

**Theorem 3** (Monotone, well-posed calibration). *Fix  $\sigma_1^2 > 0$ ,  $V > 0$ , and  $c_g H_g > 0$ , and write  $\text{ESS}_0(V; s)$  for (6) at  $V_\mu^f = V$  and  $s = s_0^2$ .*

- (a)  *$s \mapsto \text{ESS}_0(V; s)$  is continuous, convex, and strictly decreasing on  $(0, \infty)$ , with limits  $\sigma_1^2/V$  as  $s \downarrow 0$  and 0 as  $s \uparrow \infty$ , under the untruncated calibration prior in (6). Under truncation to  $(0, \tau_1^2)$ , continuity and strict decrease remain, but the limit as  $s \uparrow \infty$  is  $\sigma_1^2/(V + c_g H_g \tau_1^2) > 0$ ; convexity is asserted only for the untruncated law.*

- (b) For each  $V$  the sublevel set  $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\}$  is a half-line  $[s^\dagger(V), \infty)$ , so the population rule  $G(s) = P\{s^\dagger(V) \leq s\}$  is a distribution function and  $\{s : G(s) \geq q\} = [s_\star, \infty)$  has a unique attained infimum;  $s_\star > 0$  if and only if  $P(N_{\text{target}} < \sigma_1^2/V) > 1 - q$ , and otherwise  $s_\star = 0$ , the point-mass limit.
- (c) Under the stationary, aperiodic Harris-ergodic chain assumptions and fixed block-length regime specified in Web Appendix D,  $\widehat{G}(s) \rightarrow G(s)$  almost surely for each  $s$ , uniformly in  $s$  when  $G$  is continuous, and on a finite grid with  $G \neq q$  at every grid point the empirical selection equals  $s_\star$  for all  $B$  large enough, almost surely; under the additional geometric-ergodicity and regularity conditions in that appendix, a central limit theorem includes the block-rescaling term. With a fixed number of blocks and increasing block length, the rule instead converges to the grid approximation of the whole-chain ESS root.
- (d) For a single  $f$  tree with fixed partition, RCT profile proportions  $a_\ell$  and RWD proportions  $\pi_\ell$  over its leaves,  $\lim_{s \downarrow 0} \text{ESS}_0(s) \geq n_2 \sigma_1^2 / \sigma_2^2 / \sum_\ell a_\ell^2 / \pi_\ell$ , with equality in the flat-prior limit; the denominator is at least one, with equality if and only if  $a = \pi$ .

Theorem 3 is Theorem A4 of Web Appendix D. Parts (a) and (b) establish that the Pr-rule is well defined: the ESS decreases monotonically with the spike scale, so there is exactly one most informative admissible scale, and it is strictly positive exactly when the target is below the ceiling in enough Stage-1 blocks. A leaf-level ESS diverges as the scale shrinks and is feasible for any target; here the ceiling is finite and feasibility depends on the data and the target, which motivates specifying a fractional target. Part (c) gives conditions under which the Monte Carlo rule recovers the population selection. Part (d) relates the ceiling to the sample sizes: with matched covariate distributions the ceiling is the RWD sample size in RCT units, and covariate shift between the sources lowers it by the factor  $\sum_\ell a_\ell^2 / \pi_\ell$ , which measures how unevenly the RCT profiles are spread over the leaves relative to the RWD.

## 2.6 The single-arm variant

When the trial has no control arm,  $n_1 = 0$ , every  $g$  leaf holds no RCT controls,  $M_g = 1$ , and the  $f$  moves are exactly those of standard BART on the RWD. The control surface is then  $f_1 = f + g$  with  $f$  the RWD surface and  $g$  a prior draw, a zero-mean discrepancy process of pointwise variance

$H_g\{w\tau_0^2 + (1-w)\tau_1^2\}$  at a profile covered by  $H_g$  distinct leaves, and  $\tau_0^2$  and  $w$  stay at their priors. The indicators receive no likelihood information, and slab draws only add prior noise of variance  $(1-w)H_g\tau_1^2$  to the counterfactual control, so we report the  $w = 1$  model, the commensurate prior with the discrepancy carried by  $g$  at the calibrated scale, and use  $w < 1$  as a sensitivity analysis. The calibration is unchanged: Stage 1 is the RWD-only fit,  $\sigma_1^2$  is set to  $\sigma_2^2$  in the absence of RCT outcomes, the ceiling is at most about  $n_2$ , and the target is stated as an absolute number of controls or a fraction of the ceiling. With  $w = 1$  the single-arm model is a hierarchically centered commensurate AFT-BART,  $f_1(x) = f(x) + \sum_h e_h$  with  $e_h \sim \mathcal{N}(0, \tau_0^2)$ , with the discrepancy defined on the partition of  $g$  instead of that of  $f$ . Because the borrowing map is flat by construction here, the covariate-resolved display for a single-arm analysis is the local ESS map  $\text{ESS}(x)$ , which describes variation in RWD support across trial profiles.

### 3 Simulation studies

#### 3.1 Design

Each replicate consists of an RCT of  $N = 300$  subjects and an external source of  $n_2 = 300$  RWD controls, with  $p = 10$  covariates  $X_j \sim \mathcal{N}(\mu_j, 1)$ ,  $\boldsymbol{\mu} = (1, -1, 0.5, -1, 2, -2, 2, -3, 1, 0)$ , and  $X_1$  replaced with probability 0.15 by a draw from  $\mathcal{N}(\pm 3, 0.5^2)$  so that the sources overlap imperfectly. The RCT randomizes 2:1 for Gaussian outcomes, giving about  $n_1 = 100$  controls, and 3:1 for survival outcomes, giving about 75, as in a hybrid design with a smaller control arm. The RCT control and RWD outcome surfaces are the same nonlinear function

$$f_0(\mathbf{W}) = \alpha + \beta_1 W_1^2 + \beta_2 e^{-(W_2+1)^2/2} + \beta_3 |W_3 - 0.5| + \beta_4 |W_4 + 1| + \beta_5 \tanh(W_5 - 2) + \beta_6 \tanh(W_6 + 2) + 2\{b(W_5 - 2) + b(W_6 + 2)\} + \sum_{j=7}^{10} \beta_j W_j, \quad (10)$$

with  $b(u) = u^2 / \{1 + (u/1.5)^2\}$  and

$$\boldsymbol{\beta} = (-0.50, -0.75, -0.50, -0.50, -1.35, -0.80, -0.01, -0.01, -0.01, -0.01),$$

and  $\mathbf{W} = \mathbf{X}$  except where stated; the treated RCT subjects add  $\delta = 1$  for Gaussian outcomes and  $\delta = \log 1.4$  on the log-time scale for survival outcomes. Residual standard deviations are  $\sigma_1 = 1.5$

and  $\sigma_2 = 1.8$  for Gaussian outcomes and 1.4 and 1.8 for survival outcomes, where the survival time is log-normal, dropout is Exponential(0.02), entry is Uniform(0, 1) and administrative censoring occurs at calendar time 3.

We consider five patterns of agreement between sources. Sc1 assigns the source at random. Sc2 assigns it through  $P(D = 1 | \mathbf{X}) = \text{logit}^{-1}\{(\ell - q_{0.5}(\ell))/0.5\}$  with  $\ell = -1.20(X_5 + X_6)$ , so that source membership depends on the two covariates with the largest outcome effects. Sc3 drives the same assignment by unmeasured  $(U_1, U_2)$  that also replace  $(X_5, X_6)$  in the outcome, with the observed  $(X_5, X_6)$  correlated with  $\mathbf{U}$  at  $\rho \in \{-0.5, 0, 0.5\}$ , so the RWD surface differs from the RCT surface everywhere in a way the covariates cannot remove; Web Appendix G lists the intercepts and the assignment constants.

Sc4 and Sc5 evaluate borrowing under regional and global outcome shifts. Sc4 keeps the assignment of Sc1 and raises the RWD surface by  $\delta_2$  outside a compatible region  $\mathcal{R} = \{X_5 > 2\}$ , so that the true discrepancy  $g = f_1 - f$  is  $-\delta_2$  there and 0 inside. The region holds half of the RCT population and lies on the covariate with the dominant nonlinear effect, so the shift has to be separated from a slope that  $f$  already models. We take  $\delta_2 \in \{0.5, 1, 2\}$ , one third to 1.3 RCT residual standard deviations, chosen around the leaf resolution  $d_{1/2}$  of Theorem 2(a), which at  $\sigma_1 = 1.5$ , fifty RCT controls per leaf and the spike scale calibrated in Sc1 is about 0.5. Sc5 raises the RWD surface by  $\delta_2 \in \{1, 2\}$  everywhere, so the compatible region is empty and borrowing should be suppressed throughout the covariate space. Survival outcomes use Sc4 with  $\delta_2 \in \{0.5, 1\}$  on the log scale and Sc5 with  $\delta_2 = 1$ .

The estimands are those of Section 2.1, standardized to the replicate’s RCT profiles: the ATE for Gaussian outcomes, and for survival outcomes the ratio of standardized population medians, whose true value is 1.4, and the ratio of standardized restricted mean survival times at  $t^* = 3$ . Over  $R = 500$  replicates we report the bias of the posterior mean, the mean posterior SD, the Monte Carlo RMSE of the posterior mean, the coverage of the central 95% credible interval, and power, the fraction of replicates whose 95% interval excludes the null value, 0 for the ATE and 1 for the ratios; the arm-specific means are summarized the same way. Monte Carlo standard errors are about 0.009 for a Gaussian ATE bias and 0.01 for a coverage or power, and the replicate data are paired across methods and scenarios, so paired standard errors of a contrast between two methods are much smaller. For Sc4 and Sc5 we add the RMSE of the posterior mean control surface over all

$N$  RCT profiles inside and outside  $\mathcal{R}$ , the mean of the borrowing map  $\mathcal{B}_c(x)$  over the RCT control profiles in each region at  $c = 0.5$  on the outcome scale, and the mean of the posterior mean of  $g(x)$  in each region, whose true values are 0 in  $\mathcal{R}$  and  $-\delta_2$  outside it. For every LRC-BART fit we also record the prior  $\text{ESS}_0$ , the realized ESS, the ceiling, and whether the target was capped. Since  $\text{ESS}_0$  averages  $\sigma_1^2 / \{V^{(b)} + c_g H_g \tau_0^2\}$  over the Stage-1 blocks while the ceiling uses the whole-chain variance, Jensen’s inequality puts the printed  $\text{ESS}_0$  of a capped fit a few percent above the printed ceiling.

We compare LRC-BART with Bayesian linear and log-normal AFT regressions fitted to the RCT controls alone (LM-NP, AFT-NP), to the pooled controls (LM-CP, AFT-CP) and to the pooled controls with the source indicator as a covariate (LM-PP, AFT-PP); standard BART and AFT-BART [5, 24] in the same three configurations (BART-NP, BART-CP, BART-PP), the last being the implicit borrowing of Zhou and Ji [27]; the propensity-score composite likelihood PSCL of Chen et al. [4]; the MAP prior for the control mean without covariate adjustment; hierarchical linear and AFT models (HierLM, HierAFT); and the two nested versions of the proposed model, C-BART ( $w = 1$ ) and the free-offset LRC-BART ( $w = 0$ ) of Section 2.2. Web Appendix G.2 gives their priors, samplers and implementation details. We also transfer the analysis component of CAHB [13] to Sc1, Sc4 and Sc5 in  $R = 200$  Gaussian replicates paired with the first 200 full-study replicates. Our choices for the historical smoother, covariate selection, bandwidth, precision scales and joint draws are described in Web Appendix G.2, together with the ten-covariate and literal-constant variants.

LRC-BART is fitted as in Section 2.2, with  $H_f = 50$ ,  $H_g = 5$ ,  $(\alpha_g, \beta_g) = (0.5, 3)$ , the range-rule slab,  $\nu_0 = 3$ ,  $w \sim \text{Beta}(1, 1)$ , and a treated-arm BART with 50 trees. In Stage 1, we fit the calibration model to the RWD with 1,000 retained draws cut into  $B = 20$  blocks, take  $\sigma_1^2$  from a plain BART fitted to the RCT controls, compute  $c_g$  once by Monte Carlo under the  $g$ -tree prior, and search a geometric grid of 141 spike scales from  $10^{-6}$  to 10. We report an absolute ladder  $N_{\text{target}} \in \{50, 75, 100\}$ , capped at the ceiling where infeasible, and a fractional ladder at 0.25, 0.5 and 0.9 of the ceiling, with LRC-BART (100) as the reference configuration and LRC-BART (0.9) as the fractional reference; LRC-BART ( $H_g = 10$ ) doubles the discrepancy ensemble. Averaged over scenarios and replicates in the full study, run on 16 forked workers, a full LRC-BART fit with calibration and treated arm took 2.3 seconds for Gaussian and 2.5 seconds for survival outcomes, against 1.6 and 2.1 seconds for BART-PP and 1.1 and 1.6 for BART-NP, and the whole study ran in 5.6 hours. The configuration of  $g$  was chosen on a development run of  $R = 200$  paired with the

first 200 replicates here (Web Appendix G.3).

### 3.2 Gaussian outcomes

Table 1 reports the two-arm Gaussian results for selected scenarios and methods, and Web Table S1 the full set.

Under agreement between sources (Sc1), the RMSE of LRC-BART (100) is 0.172 against 0.205 for BART-PP and 0.220 for BART-NP, with coverage 0.94; the paired difference in mean squared error from BART-PP is  $-0.012$  with standard error 0.001. The RMSEs reflect the extent of borrowing: BART-CP pools without adjustment and has RMSE 0.142, followed by C-BART at 0.150, LRC-BART at 0.172 and BART-PP at 0.205. The prior  $ESS_0$  of LRC-BART (100) averages 82 RCT controls, below the target because the ceiling, 127 on average, is exceeded by the target in 20% of the replicates and the calibration then returns the point-mass limit, and the realized ESS, computed with the indicators and  $\tau_0^2$  drawn from the posterior, is 53. Power at  $\delta = 1$  is 1.00 for every tree model and distinguishes only the linear methods, and the same holds at a 90% interval (Web Appendix G).

Under measured confounding (Sc2), the covariate-adjusted methods have small biases: LRC-BART (100) with bias 0.023 and RMSE 0.192 against 0.007 and 0.204 for BART-PP, whereas BART-CP and C-BART, which cannot let the sources differ, carry biases of 0.058 and 0.042 with coverage 0.91. The ESS ceiling is substantially lower in Sc2, consistent with the covariate-overlap calculation in Theorem 3(d): the RCT profiles concentrate in regions with limited RWD support, the ceiling falls from 127 in Sc1 to 23, every absolute target is capped in every replicate and the three absolute targets give the same fit (Web Table S3), whereas the fractional ladder still spans prior ESS 5, 10 and 16 with biases 0.012, 0.016 and 0.019. We therefore recommend reporting fractional targets when covariate overlap is poor.

Under unmeasured confounding, Sc3, the RWD surface differs from the RCT surface everywhere by an amount the covariates cannot explain, and the two tree models with fixed commensurate components have substantial bias, as do their linear counterparts: BART-CP has bias 0.98 at  $\rho = 0$  with coverage 0, and C-BART, with one commensurability scale and no slab, has bias 0.23 with coverage 0.87, consistent with the unbounded influence under fixed components in Theorem 1.

LRC-BART (100) largely suppresses borrowing. Its realized ESS is 1.6, its map averages 0.07,

and its bias is 0.041 with coverage 0.94 and RMSE 0.280, against 0.029 and 0.273 for BART-PP and  $-0.002$  and 0.289 for BART-NP, with similar results at the three values of  $\rho$ . The residual bias of 0.04 lies 0.012 above BART-PP's, with a paired Monte Carlo standard error of 0.001, so the difference exceeds Monte Carlo uncertainty. The spike provides a possible explanation: a leaf whose RCT residual mean is not far beyond  $d_{1/2}$  keeps a small but nonzero posterior spike probability, and Theorem 1 bounds the resulting shift by about one RCT standard error of the leaf. Doubling the discrepancy ensemble raises the bias to 0.055, as  $H_g\tau_0^2$  increases the variation permitted under the spike. This supports our use of a small  $H_g$ .

Table 2 and Figure 1 report compatibility and control-surface estimation under Sc4 and Sc5, together with the ATE. At  $\delta_2 = 2$  the map separates the regions:  $\bar{B}_c$  is 0.64 over the RCT control profiles in  $\mathcal{R}$  and 0.05 outside it, and the posterior mean of  $g$  averages  $-0.26$  in  $\mathcal{R}$  and  $-1.71$  outside it against true values of 0 and  $-2$ .

Figure 1 shows how discrepancy estimates vary across the boundary. Away from the boundary at  $X_5 = 2$  the posterior mean of  $g$  is  $-1.8$  outside  $\mathcal{R}$  and  $-0.15$  inside it, short of the truth by about a tenth of the shift on each side, and the map is 0.02 outside and 0.7 inside; the transition occupies the unit around the boundary, with  $g$  averaging  $-1.66$  in the half-unit bin below it and  $-0.44$  in the half-unit bin above it. The boundary is learned from the RCT controls alone, about four of them per 0.1 of  $X_5$ , and the pooled surface  $f$ , which is estimated from both sources, smooths the RWD step across it, leaving the shallow discrepancy ensemble  $g$  to estimate the remaining difference. The discrepancy underestimation persisted with  $f$  fixed at the truth and under every  $g$  prior we tried (Web Appendix G.3), so the configurations examined did not resolve it.

The ATE is unbiased at  $\delta_2 = 2$  (0.000, coverage 0.92), because the errors on the two sides of the boundary have opposite signs and the estimand averages over both. Control-surface estimation is less accurate: its RMSE over the RCT profiles in  $\mathcal{R}$  is 0.824 against 0.785 for BART-PP, and outside  $\mathcal{R}$  it is 0.857 against 0.822, so at  $\delta_2 = 2$  LRC-BART does not beat the free-offset model on the surface, while C-BART, at 1.087 outside  $\mathcal{R}$ , has a larger error than either method. A secondary version of Sc4 places the shift on  $X_7$ , a covariate with negligible effect, and the map and ATE results are similar to those on  $X_5$  at both  $\delta_2 = 2$  and  $\delta_2 = 1$  (Web Appendix G.3).

At  $\delta_2 = 1$ , a shift of about twice  $d_{1/2}$ , the map separates only partially (0.62 against 0.52), the posterior mean of  $g$  is  $-0.6$  outside  $\mathcal{R}$  and  $-0.25$  inside it (Figure 1), and LRC-BART borrows too

much outside  $\mathcal{R}$ : the ATE bias is  $-0.041$  (paired difference from BART-PP  $-0.039$ , standard error  $0.003$ ), the RMSE  $0.223$  against  $0.207$ , and the coverage  $0.89$ , whereas C-BART and BART-CP have biases of  $-0.21$  and  $-0.34$ .

At  $\delta_2 = 0.5$ , below the stated resolution, the map is flat at  $0.84$ ,  $g$  averages  $-0.15$  in  $\mathcal{R}$  and  $-0.18$  outside it, and the bias is  $-0.043$  with RMSE  $0.200$ , below BART-PP's  $0.203$  because the variance reduction offsets the bias in this scenario. Lower fractional targets reduce bias under the moderate shift,  $-0.032$  at a quarter of the ceiling against  $-0.041$ , at a cost of  $0.01$  in RMSE in Sc1 (Web Table S3).

Under the global shift of Sc5, LRC-BART suppresses borrowing: at  $\delta_2 = 2$  the map averages  $0.01$ , the realized ESS is  $0.2$ , the bias is  $-0.001$  and the RMSE  $0.203$ , against BART-NP's  $0.220$  and BART-PP's  $0.205$ , whereas C-BART and BART-CP have biases of  $-0.32$  and  $-1.39$ ; at  $\delta_2 = 1$  the map is  $0.10$  and the bias  $-0.005$ .

The realized regional ESS distinguishes where borrowing occurs (Table 2). Across the first 200 replicates, it is  $32.7$  and  $31.9$  on the two sides of the Sc4 partition under Sc1. Under Sc4 it is  $15.1$  in  $\mathcal{R}$  and  $12.7$  outside at  $\delta_2 = 1$ , and  $9.5$  and  $0.44$  at  $\delta_2 = 2$ . Under Sc5 with  $\delta_2 = 2$ , it is about  $0.2$  on both sides. These quantities use the control mean standardized within each region and are not additive. The prior pointwise ESS uses the RWD fit and cannot establish that trial outcomes support regional borrowing.

The surface loss persists away from the boundary. On RCT control profiles with  $X_5 > 2.5$ , the Sc4  $\delta_2 = 2$  RMSE is  $0.747$  for LRC-BART and  $0.723$  for BART-PP, a paired difference of  $0.024$  with standard error  $0.005$  over 200 replicates. Using all RCT profiles in  $\mathcal{R}$  over those same replicates gives a difference of  $0.039$  with standard error  $0.006$ . Web Appendix G.3 reports both profile sets. Thus the evidence for LRC-BART is its efficiency under Sc1 and its regional compatibility and ESS summaries, with a surface-estimation loss under Sc4.

The transferred CAHB analysis has Sc1 RMSE  $0.258$ , compared with  $0.175$  for LRC-BART and  $0.223$  for BART-NP over the same 200 replicates. At Sc4  $\delta_2 = 2$ , its bias is  $-0.333$  and its historical precision shares average  $0.49$  and  $0.41$  in the two regions. At Sc5  $\delta_2 = 1$ , its bias is  $-0.437$ . These results show the difficulty of transferring kernel-local borrowing to this nonlinear ten-covariate design. They depend on our transfer choices and do not establish a general ranking of CAHB and tree-based borrowing.

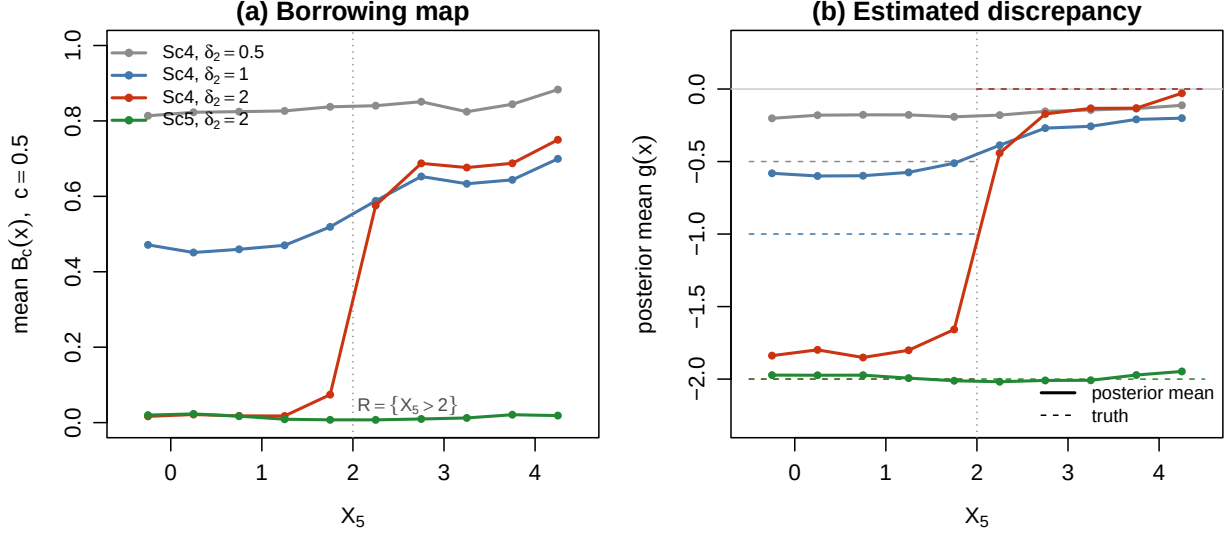


Figure 1: Regional and global shifts, Gaussian outcomes, LRC-BART (100): (a) the borrowing map  $B_c(x) = P(|g(x)| < 0.5 | \text{data})$  and (b) the posterior mean of  $g(x)$ , both averaged over the RCT control profiles in bins of width 0.5 on  $X_5$  across the first 40 replicates. The RWD surface is raised by  $\delta_2$  where  $X_5 \leq 2$  (Sc4) or everywhere (Sc5), so the true discrepancy, dashed in (b), is  $-\delta_2$  there.

Table 1: Gaussian outcomes, two-arm design,  $R = 500$  replicates: average treatment effect (ATE) bias, mean posterior SD, Monte Carlo RMSE, coverage of the 95% credible interval, and power (95% interval excludes zero at  $\delta = 1$ ), with bias and RMSE of the RCT-standardized control mean. Sc4 raises the RWD control surface by  $\delta_2$  outside  $R = \{X_5 > 2\}$ ; Sc5 raises it everywhere. LRC-BART (100) targets a prior ESS of 100 RCT controls and LRC-BART (0.9) targets 0.9 of the ESS ceiling; CAHB uses  $R = 200$  paired replicates and our transfer described in Web Appendix G.2; other rows use  $R = 500$ . C-BART is LRC-BART with  $w = 1$ . PSCL reports no arm-specific means. The full table with every scenario and method is Web Table S1.

| Scenario | Method         | ATE    |       |       |      |      | Control mean |       |
|----------|----------------|--------|-------|-------|------|------|--------------|-------|
|          |                | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  |
| Sc1      | LM-PP          | -0.002 | 0.347 | 0.337 | 0.94 | 0.85 | 0.001        | 0.243 |
|          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|          | BART-CP        | 0.008  | 0.145 | 0.142 | 0.96 | 1.00 | -0.003       | 0.096 |
|          | BART-PP        | 0.006  | 0.207 | 0.205 | 0.94 | 1.00 | -0.001       | 0.170 |
|          | PSCL           | 0.005  | 0.269 | 0.261 | 0.95 | 0.97 | -0.005       | 0.171 |
|          | MAP (100)      | 0.006  | 0.292 | 0.265 | 0.97 | 0.95 | -0.005       | 0.174 |
|          | HierLM         | -0.003 | 0.338 | 0.341 | 0.93 | 0.85 | 0.003        | 0.249 |
|          | C-BART         | 0.007  | 0.155 | 0.150 | 0.95 | 1.00 | -0.002       | 0.106 |
|          | LRC-BART (100) | 0.009  | 0.171 | 0.172 | 0.94 | 1.00 | -0.004       | 0.134 |
|          | LRC-BART (0.9) | 0.011  | 0.171 | 0.171 | 0.95 | 1.00 | -0.005       | 0.132 |
|          | CAHB           | -0.003 | 0.410 | 0.258 | 1.00 | 0.78 | -0.023       | 0.184 |

Table 1, continued

| Scenario                 | Method         | ATE    |       |       |      |      | Control mean |       |
|--------------------------|----------------|--------|-------|-------|------|------|--------------|-------|
|                          |                | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  |
| Sc2                      | LM-PP          | 0.010  | 0.340 | 0.331 | 0.95 | 0.84 | -0.010       | 0.248 |
|                          | BART-NP        | -0.008 | 0.201 | 0.219 | 0.93 | 1.00 | 0.009        | 0.180 |
|                          | BART-CP        | 0.058  | 0.177 | 0.195 | 0.91 | 1.00 | -0.056       | 0.151 |
|                          | BART-PP        | 0.007  | 0.208 | 0.204 | 0.96 | 1.00 | -0.005       | 0.163 |
|                          | PSCL           | 0.024  | 0.321 | 0.292 | 0.97 | 0.91 | -0.029       | 0.205 |
|                          | MAP (100)      | 0.332  | 0.360 | 0.491 | 0.83 | 0.97 | -0.337       | 0.424 |
|                          | HierLM         | -0.011 | 0.318 | 0.321 | 0.95 | 0.88 | 0.011        | 0.238 |
|                          | C-BART         | 0.042  | 0.168 | 0.186 | 0.91 | 1.00 | -0.041       | 0.141 |
|                          | LRC-BART (100) | 0.023  | 0.181 | 0.192 | 0.94 | 1.00 | -0.021       | 0.148 |
|                          | LRC-BART (0.9) | 0.019  | 0.183 | 0.193 | 0.93 | 1.00 | -0.017       | 0.151 |
| Sc3, $\rho = 0$          | LM-PP          | 0.043  | 0.356 | 0.346 | 0.95 | 0.85 | -0.027       | 0.241 |
|                          | BART-NP        | -0.002 | 0.292 | 0.289 | 0.94 | 0.92 | 0.024        | 0.219 |
|                          | BART-CP        | 0.984  | 0.202 | 1.003 | 0.00 | 1.00 | -0.962       | 0.971 |
|                          | BART-PP        | 0.029  | 0.275 | 0.273 | 0.94 | 0.96 | -0.007       | 0.197 |
|                          | PSCL           | 0.704  | 0.269 | 0.753 | 0.26 | 1.00 | -0.693       | 0.713 |
|                          | MAP (100)      | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | -0.352       | 0.431 |
|                          | HierLM         | 0.039  | 0.362 | 0.354 | 0.96 | 0.82 | -0.023       | 0.253 |
|                          | C-BART         | 0.226  | 0.300 | 0.380 | 0.87 | 0.98 | -0.204       | 0.311 |
|                          | LRC-BART (100) | 0.041  | 0.284 | 0.280 | 0.94 | 0.96 | -0.018       | 0.206 |
|                          | LRC-BART (0.9) | 0.037  | 0.282 | 0.277 | 0.94 | 0.96 | -0.015       | 0.203 |
| Sc4, $X_5, \delta_2 = 1$ | LM-PP          | -0.012 | 0.349 | 0.339 | 0.95 | 0.83 | 0.011        | 0.245 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -0.338 | 0.146 | 0.368 | 0.35 | 1.00 | 0.343        | 0.357 |
|                          | BART-PP        | -0.002 | 0.209 | 0.207 | 0.95 | 1.00 | 0.007        | 0.173 |
|                          | PSCL           | -0.246 | 0.271 | 0.358 | 0.87 | 0.80 | 0.247        | 0.300 |
|                          | MAP (100)      | -0.224 | 0.306 | 0.361 | 0.91 | 0.75 | 0.224        | 0.292 |
|                          | HierLM         | -0.025 | 0.338 | 0.343 | 0.94 | 0.83 | 0.024        | 0.250 |
|                          | C-BART         | -0.207 | 0.173 | 0.273 | 0.73 | 1.00 | 0.212        | 0.258 |
|                          | LRC-BART (100) | -0.041 | 0.195 | 0.223 | 0.89 | 1.00 | 0.046        | 0.194 |
|                          | LRC-BART (0.9) | -0.041 | 0.196 | 0.224 | 0.90 | 1.00 | 0.047        | 0.194 |
| Sc4, $X_5, \delta_2 = 2$ | CAHB           | -0.208 | 0.413 | 0.333 | 0.98 | 0.42 | 0.181        | 0.262 |
|                          | LM-PP          | -0.023 | 0.354 | 0.343 | 0.95 | 0.81 | 0.022        | 0.249 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -0.685 | 0.149 | 0.701 | 0.00 | 0.56 | 0.690        | 0.698 |
|                          | BART-PP        | -0.006 | 0.209 | 0.205 | 0.96 | 1.00 | 0.011        | 0.170 |
|                          | PSCL           | -0.498 | 0.273 | 0.562 | 0.54 | 0.44 | 0.498        | 0.528 |
|                          | MAP (100)      | -0.377 | 0.333 | 0.495 | 0.78 | 0.46 | 0.378        | 0.437 |
|                          | HierLM         | -0.044 | 0.339 | 0.345 | 0.94 | 0.81 | 0.043        | 0.253 |
|                          | C-BART         | -0.227 | 0.197 | 0.391 | 0.66 | 0.90 | 0.232        | 0.383 |
|                          | LRC-BART (100) | 0.000  | 0.189 | 0.202 | 0.92 | 1.00 | 0.005        | 0.168 |
| Sc5, $\delta_2 = 2$      | LRC-BART (0.9) | -0.000 | 0.189 | 0.202 | 0.93 | 1.00 | 0.006        | 0.170 |
|                          | CAHB           | -0.333 | 0.422 | 0.453 | 0.95 | 0.25 | 0.309        | 0.398 |
|                          | LM-PP          | -0.042 | 0.347 | 0.339 | 0.94 | 0.80 | 0.042        | 0.246 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -1.390 | 0.152 | 1.398 | 0.00 | 0.72 | 1.395        | 1.399 |
|                          | BART-PP        | -0.012 | 0.208 | 0.205 | 0.95 | 1.00 | 0.017        | 0.171 |
|                          | PSCL           | -1.002 | 0.269 | 1.035 | 0.04 | 0.04 | 1.002        | 1.017 |
|                          | MAP (100)      | -0.395 | 0.348 | 0.525 | 0.78 | 0.41 | 0.395        | 0.465 |
|                          | HierLM         | -0.038 | 0.338 | 0.344 | 0.94 | 0.82 | 0.037        | 0.251 |
|                          | C-BART         | -0.316 | 0.194 | 0.650 | 0.76 | 0.93 | 0.322        | 0.648 |
|                          | LRC-BART (100) | -0.001 | 0.190 | 0.203 | 0.92 | 1.00 | 0.005        | 0.171 |
|                          | LRC-BART (0.9) | -0.001 | 0.190 | 0.205 | 0.92 | 1.00 | 0.006        | 0.171 |
|                          | CAHB           | -0.067 | 0.479 | 0.400 | 0.95 | 0.52 | 0.041        | 0.326 |

Table 2: Regional and global shifts, Gaussian outcomes,  $R = 500$ : ATE bias and coverage; RMSE of the posterior mean control surface over all  $N$  RCT profiles (treated and control) inside the compatible region  $\mathcal{R}$  and outside it; the borrowing map  $\mathcal{B}_c(x) = P(|g(x)| < 0.5 \mid \text{data})$  averaged over the RCT control profiles in each region; the posterior mean of  $g(x)$  averaged over each region (true value 0 in  $\mathcal{R}$  and  $-\delta_2$  in  $\mathcal{R}^c$ ); and the realized ESS. In Sc5 the compatible region is empty. Sc4' places the shift on the null covariate  $X_7$ . LRC-BART ( $w = 0$ ) has no spike, so its map reflects the slab alone. CAHB rows use  $R = 200$ ; all other rows use  $R = 500$ . CAHB precision shares and effective sizes have different definitions and are reported separately in Web Appendix G.2. For Sc1, the two columns use the Sc4 partition although both regions are compatible.

| Scenario  | Method               | ATE    |      | Surface RMSE  |                 | $\tilde{\mathcal{B}}_c$ |                 | Mean $\hat{g}$ |                 | ESS  |
|-----------|----------------------|--------|------|---------------|-----------------|-------------------------|-----------------|----------------|-----------------|------|
|           |                      | Bias   | Cov  | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$           | $\mathcal{R}^c$ | $\mathcal{R}$  | $\mathcal{R}^c$ |      |
| Sc1       | BART-PP              | 0.006  | 0.94 | 0.750         | 0.780           | —                       | —               | —              | —               | —    |
|           | C-BART               | 0.007  | 0.95 | 0.728         | 0.756           | 1.00                    | 1.00            | 0.00           | 0.00            | 81.0 |
|           | LRC-BART ( $w = 0$ ) | 0.009  | 0.92 | 0.755         | 0.783           | 0.79                    | 0.79            | -0.00          | 0.01            | 0.1  |
|           | LRC-BART (100)       | 0.009  | 0.94 | 0.736         | 0.763           | 0.92                    | 0.92            | -0.00          | 0.00            | 52.5 |
|           | CAHB                 | -0.003 | 1.00 | 2.071         | 2.273           | —                       | —               | —              | —               | —    |
| Sc4 (0.5) | BART-PP              | 0.000  | 0.95 | 0.758         | 0.795           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.109 | 0.90 | 0.736         | 0.799           | 0.98                    | 0.98            | -0.06          | -0.06           | 79.4 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.93 | 0.768         | 0.795           | 0.71                    | 0.69            | -0.21          | -0.27           | 0.1  |
|           | LRC-BART (100)       | -0.043 | 0.92 | 0.751         | 0.796           | 0.84                    | 0.83            | -0.15          | -0.18           | 42.9 |
| Sc4 (1)   | BART-PP              | -0.002 | 0.95 | 0.784         | 0.820           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.207 | 0.73 | 0.749         | 0.915           | 0.91                    | 0.90            | -0.14          | -0.16           | 73.6 |
|           | LRC-BART ( $w = 0$ ) | 0.006  | 0.93 | 0.801         | 0.827           | 0.51                    | 0.40            | -0.35          | -0.63           | 0.1  |
|           | LRC-BART (100)       | -0.041 | 0.89 | 0.783         | 0.843           | 0.62                    | 0.52            | -0.32          | -0.52           | 20.8 |
|           | CAHB                 | -0.208 | 0.98 | 2.133         | 2.207           | —                       | —               | —              | —               | —    |
| Sc4 (2)   | BART-PP              | -0.006 | 0.96 | 0.785         | 0.822           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.227 | 0.66 | 0.801         | 1.087           | 0.66                    | 0.39            | -0.30          | -0.97           | 45.7 |
|           | LRC-BART ( $w = 0$ ) | 0.004  | 0.92 | 0.836         | 0.858           | 0.57                    | 0.05            | -0.25          | -1.73           | 0.1  |
|           | LRC-BART (100)       | 0.000  | 0.92 | 0.824         | 0.857           | 0.64                    | 0.05            | -0.26          | -1.71           | 0.8  |
|           | CAHB                 | -0.333 | 0.95 | 2.172         | 2.157           | —                       | —               | —              | —               | —    |
| Sc4' (1)  | BART-PP              | -0.001 | 0.96 | 0.807         | 0.810           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.206 | 0.73 | 0.767         | 0.899           | 0.91                    | 0.90            | -0.15          | -0.16           | 73.3 |
|           | LRC-BART ( $w = 0$ ) | 0.009  | 0.92 | 0.820         | 0.817           | 0.52                    | 0.41            | -0.37          | -0.63           | 0.1  |
|           | LRC-BART (100)       | -0.038 | 0.90 | 0.805         | 0.833           | 0.62                    | 0.52            | -0.33          | -0.52           | 20.5 |
| Sc4' (2)  | BART-PP              | -0.002 | 0.95 | 0.805         | 0.806           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.218 | 0.69 | 0.828         | 1.075           | 0.64                    | 0.38            | -0.33          | -0.97           | 44.7 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.92 | 0.859         | 0.848           | 0.57                    | 0.05            | -0.25          | -1.74           | 0.1  |
|           | LRC-BART (100)       | 0.001  | 0.93 | 0.850         | 0.848           | 0.64                    | 0.05            | -0.26          | -1.71           | 0.8  |
| Sc5 (1)   | BART-PP              | -0.004 | 0.95 | —             | 0.769           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.244 | 0.68 | —             | 0.821           | —                       | 0.43            | —              | -0.60           | 41.5 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.92 | —             | 0.769           | —                       | 0.10            | —              | -1.00           | 0.1  |
|           | LRC-BART (100)       | -0.005 | 0.91 | —             | 0.768           | —                       | 0.10            | —              | -0.98           | 1.2  |
|           | CAHB                 | -0.437 | 0.91 | —             | 2.203           | —                       | —               | —              | —               | —    |
| Sc5 (2)   | BART-PP              | -0.012 | 0.95 | —             | 0.771           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.316 | 0.76 | —             | 0.932           | —                       | 0.19            | —              | -1.54           | 45.4 |
|           | LRC-BART ( $w = 0$ ) | 0.006  | 0.93 | —             | 0.771           | —                       | 0.01            | —              | -2.00           | 0.1  |
|           | LRC-BART (100)       | -0.001 | 0.92 | —             | 0.771           | —                       | 0.01            | —              | -1.99           | 0.2  |
|           | CAHB                 | -0.067 | 0.95 | —             | 1.983           | —                       | —               | —              | —               | —    |

| Scenario | Regional ESS, $X_5 > 2$ | Regional ESS, $X_5 \leq 2$ |
|----------|-------------------------|----------------------------|
| Sc1      | 32.66                   | 31.86                      |
| Sc4 (1)  | 15.14                   | 12.74                      |
| Sc4 (2)  | 9.51                    | 0.44                       |
| Sc5 (2)  | 0.24                    | 0.24                       |

Regional realized ESS for LRC-BART (100), averaged over the first 200 paired replicates. Each estimand is the control mean standardized to all RCT profiles in that region. The partition is held fixed for Sc1 and Sc5; both regions are compatible in Sc1 and neither is compatible in Sc5. Regional effective sizes are not additive.

### 3.3 Survival outcomes

Table 3 reports the survival study for the median ratio and the RMST ratio side by side, and Web Table S2 the full set. With 75 RCT controls and about a quarter of the times censored, the median ratio has a wide and skewed posterior under every method, and its posterior mean carries an upward bias that grows with the posterior spread: BART-NP has bias 0.26 and RMSE 0.51 in Sc1, where its RMST-ratio bias is 0.13, and AFT-NP is wider still. Borrowing reduces this uncertainty at the studied sample size.

In Sc1 LRC-BART (100) has median-ratio bias 0.045 and RMSE 0.312 against 0.032 and 0.345 for BART-PP, with power 0.34 against 0.25, and its RMST-ratio RMSE is 0.097 against 0.106; in Sc2, where the ceiling falls to 40 and the absolute targets are capped, the two are within 0.03 of each other on both estimands.

Under unmeasured confounding (Sc3), the tree models with a fixed commensurate component have large biases, as in the Gaussian study: BART-CP with median-ratio bias 2.3 and C-BART with 0.58 at  $\rho = 0$ , and LRC-BART abstains with a realized ESS of 2.3. Its median-ratio bias is then 0.15 against 0.04 for BART-PP and 0.19 for BART-NP, with RMSE 0.558 against 0.451 and coverage 0.94; the paired difference in bias from BART-PP is 0.11 with standard error 0.005, the largest loss to BART-PP in the study. The control log-median is biased by only 0.02, against  $-0.006$  for BART-PP, so most of the difference on the ratio scale comes from the wider posterior LRC-BART has once it has switched borrowing off with 75 controls, with posterior SD 0.52 against 0.39, together with the nonlinear transformation to the ratio scale. On the RMST ratio the biases are 0.031 and  $-0.005$  and the RMSEs 0.146 and 0.137, a paired difference in mean squared error of 0.003 with standard error 0.001. We report both estimands because the median ratio amplifies a small control bias and a wide posterior in a way the RMST ratio does not, and the RMST ratio is

the estimand of the application.

Under the regional shift, Sc4 at  $\delta_2 = 1$  on the log scale with a residual SD of 1.4, the map separates weakly, 0.64 in  $\mathcal{R}$  against 0.58 outside, and the treatment-effect biases are  $-0.011$  on the median ratio and  $-0.037$  on the RMST ratio where C-BART and BART-CP have biases of  $-0.20$  and  $-0.37$ ; under the global shift of Sc5 the map is 0.16, the median-ratio bias 0.03, and C-BART's ladder behaves as in the Gaussian study.

Table 3: Survival outcomes, two-arm design,  $R = 500$ : bias, Monte Carlo RMSE, coverage and power (95% interval excludes one) for the ratio of standardized population medians (true value 1.4) and the ratio of standardized restricted mean survival times at  $t^* = 3$ . Sc4 raises the RWD log-time surface by  $\delta_2 = 1$  outside  $\mathcal{R} = \{X_5 > 2\}$  and Sc5 by 1 everywhere. Point estimates are posterior means. The full table is Web Table S2.

| Scenario                 | Method         | Median ratio |       |      |      | RMST ratio, $t^* = 3$ |       |      |      |
|--------------------------|----------------|--------------|-------|------|------|-----------------------|-------|------|------|
|                          |                | Bias         | RMSE  | Cov  | Pow  | Bias                  | RMSE  | Cov  | Pow  |
| Sc1                      | AFT-PP         | 0.261        | 0.737 | 0.95 | 0.12 | -0.009                | 0.142 | 0.94 | 0.10 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | 0.064        | 0.232 | 0.96 | 0.63 | -0.011                | 0.072 | 0.95 | 0.35 |
|                          | BART-PP        | 0.032        | 0.345 | 0.95 | 0.25 | -0.023                | 0.106 | 0.94 | 0.17 |
|                          | HierAFT        | 0.246        | 0.733 | 0.94 | 0.12 | -0.008                | 0.152 | 0.92 | 0.11 |
|                          | C-BART         | 0.034        | 0.248 | 0.97 | 0.42 | 0.003                 | 0.079 | 0.95 | 0.32 |
|                          | LRC-BART (100) | 0.045        | 0.312 | 0.95 | 0.34 | 0.003                 | 0.097 | 0.94 | 0.28 |
|                          | LRC-BART (0.9) | 0.042        | 0.304 | 0.96 | 0.38 | 0.002                 | 0.095 | 0.95 | 0.30 |
| Sc2                      | AFT-PP         | 0.389        | 0.818 | 0.95 | 0.19 | 0.050                 | 0.154 | 0.95 | 0.20 |
|                          | BART-NP        | 0.251        | 0.495 | 0.91 | 0.43 | 0.110                 | 0.177 | 0.88 | 0.45 |
|                          | BART-CP        | 0.194        | 0.383 | 0.93 | 0.56 | 0.037                 | 0.100 | 0.94 | 0.43 |
|                          | BART-PP        | 0.036        | 0.335 | 0.95 | 0.24 | -0.002                | 0.097 | 0.95 | 0.21 |
|                          | HierAFT        | 0.233        | 0.695 | 0.97 | 0.14 | 0.001                 | 0.134 | 0.95 | 0.12 |
|                          | C-BART         | 0.138        | 0.331 | 0.93 | 0.55 | 0.043                 | 0.104 | 0.92 | 0.49 |
|                          | LRC-BART (100) | 0.061        | 0.322 | 0.94 | 0.34 | 0.018                 | 0.100 | 0.93 | 0.31 |
|                          | LRC-BART (0.9) | 0.058        | 0.325 | 0.94 | 0.33 | 0.017                 | 0.100 | 0.94 | 0.30 |
| Sc3, $\rho = 0$          | AFT-PP         | 0.440        | 0.919 | 0.94 | 0.19 | 0.054                 | 0.158 | 0.95 | 0.19 |
|                          | BART-NP        | 0.186        | 0.571 | 0.90 | 0.30 | 0.058                 | 0.173 | 0.92 | 0.21 |
|                          | BART-CP        | 2.308        | 2.412 | 0.00 | 1.00 | 0.588                 | 0.607 | 0.01 | 1.00 |
|                          | BART-PP        | 0.041        | 0.451 | 0.91 | 0.20 | -0.005                | 0.137 | 0.92 | 0.15 |
|                          | HierAFT        | 0.337        | 0.890 | 0.95 | 0.13 | 0.006                 | 0.147 | 0.95 | 0.11 |
|                          | C-BART         | 0.576        | 0.909 | 0.88 | 0.40 | 0.139                 | 0.222 | 0.85 | 0.38 |
|                          | LRC-BART (100) | 0.155        | 0.558 | 0.94 | 0.17 | 0.031                 | 0.146 | 0.94 | 0.18 |
|                          | LRC-BART (0.9) | 0.148        | 0.554 | 0.94 | 0.17 | 0.029                 | 0.144 | 0.93 | 0.18 |
| Sc4, $X_5, \delta_2 = 1$ | AFT-PP         | 0.191        | 0.691 | 0.95 | 0.10 | -0.039                | 0.139 | 0.92 | 0.07 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | -0.367       | 0.401 | 0.50 | 0.04 | -0.156                | 0.168 | 0.32 | 0.07 |
|                          | BART-PP        | -0.012       | 0.339 | 0.97 | 0.17 | -0.059                | 0.113 | 0.91 | 0.07 |
|                          | HierAFT        | 0.193        | 0.700 | 0.94 | 0.10 | -0.023                | 0.149 | 0.91 | 0.09 |
|                          | C-BART         | -0.202       | 0.305 | 0.84 | 0.08 | -0.093                | 0.120 | 0.79 | 0.04 |
|                          | LRC-BART (100) | -0.011       | 0.340 | 0.93 | 0.24 | -0.037                | 0.108 | 0.90 | 0.16 |
|                          | LRC-BART (0.9) | -0.018       | 0.342 | 0.93 | 0.24 | -0.040                | 0.110 | 0.90 | 0.16 |
| Sc5, $\delta_2 = 1$      | AFT-PP         | 0.193        | 0.695 | 0.94 | 0.09 | -0.029                | 0.141 | 0.93 | 0.08 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | -0.703       | 0.712 | 0.00 | 0.64 | -0.256                | 0.262 | 0.01 | 0.65 |
|                          | BART-PP        | -0.029       | 0.333 | 0.95 | 0.17 | -0.041                | 0.110 | 0.93 | 0.11 |
|                          | HierAFT        | 0.204        | 0.701 | 0.94 | 0.10 | -0.018                | 0.150 | 0.92 | 0.09 |
|                          | C-BART         | -0.233       | 0.405 | 0.77 | 0.13 | -0.085                | 0.143 | 0.77 | 0.11 |
|                          | LRC-BART (100) | 0.032        | 0.353 | 0.93 | 0.30 | 0.001                 | 0.112 | 0.93 | 0.26 |
|                          | LRC-BART (0.9) | 0.029        | 0.350 | 0.94 | 0.31 | 0.001                 | 0.111 | 0.94 | 0.26 |

### 3.4 The single-arm design

To assess the single-arm setting of Section 4, we use the same mechanisms with no RCT control arm: the trial contributes  $n_T$  treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and LRC-BART runs in the single-arm mode of Section 2.6 with  $w = 1$  against the complete-pooling models, the hierarchical models and BART-CP; Sc3 is not identifiable without an internal control and is omitted. Table 4 reports the results and Web Appendix G.7 discusses them in detail. Under Sc1 the tree models have Gaussian RMSE 0.16 with coverage 0.94 to 1.00.

Under Sc2, all methods have substantial bias because without RCT controls the data cannot identify a control discrepancy at trial profiles without RWD support: on the Gaussian ATE, BART-CP has bias 0.41 with coverage 0.75 and LRC-BART 0.42. In every design LRC-BART (100) gives the point estimate of the pooled model within 0.005, with a wider posterior where the target is not capped, and the  $w = 0.9$  sensitivity only adds prior noise to the counterfactual control, which is why we report  $w = 1$  in single-arm mode.

Table 4: Single-arm design,  $R = 500$ : the trial contributes  $n_T$  treated subjects only and the  $n_2 = 300$  RWD controls are the sole control source, under Sc1 (no confounding) and Sc2 (measured confounding). LRC-BART (pooled) is  $w = 1$  with the spike scale at the grid minimum, LRC-BART (0.9) is  $w = 1$  at 0.9 of the ESS ceiling, and LRC-BART ( $w = 0.9$ ) is the slab sensitivity at the absolute target 100. HierLM and HierAFT use the stated hyperprior  $\tau_j^2 \sim \text{Inv-Gamma}(1.5, 0.75)$  with no RCT controls, under which the control surface is drawn from the hierarchy; values above  $10^3$  are printed as such. Control columns are on the outcome scale (Gaussian) or the log-median scale (survival). Web Table S6 adds the absolute-target rows and the median ratio at  $n_T = 200$ .

| Design                              | Method                 | Treatment effect |       |       |      |      | Control |       |       |     |
|-------------------------------------|------------------------|------------------|-------|-------|------|------|---------|-------|-------|-----|
|                                     |                        | Bias             | SD    | RMSE  | Cov  | Pow  | Bias    | RMSE  | Prior | ESS |
| Gaussian, $n_T = 200$<br>ATE<br>Sc1 | LM-CP                  | 0.007            | 0.259 | 0.254 | 0.95 | 0.97 | -0.003  | 0.233 | —     |     |
|                                     | HierLM                 | -0.014           | 8.239 | 0.497 | 1.00 | 0.00 | 0.018   | 0.487 | —     |     |
|                                     | BART-CP                | 0.010            | 0.163 | 0.162 | 0.94 | 1.00 | -0.003  | 0.127 | —     |     |
|                                     | LRC-BART (pooled)      | 0.007            | 0.165 | 0.162 | 0.94 | 1.00 | 0.001   | 0.127 | 172.3 |     |
|                                     | LRC-BART (0.9)         | 0.007            | 0.202 | 0.163 | 0.98 | 1.00 | 0.000   | 0.128 | 115.8 |     |
|                                     | LRC-BART ( $w = 0.9$ ) | 0.010            | 1.383 | 0.169 | 1.00 | 0.00 | -0.003  | 0.136 | 81.9  |     |
| Gaussian, $n_T = 200$<br>ATE<br>Sc2 | LM-CP                  | 1.682            | 0.417 | 1.734 | 0.01 | 1.00 | -1.676  | 1.726 | —     |     |
|                                     | HierLM                 | 1.642            | 8.317 | 1.754 | 1.00 | 0.00 | -1.636  | 1.747 | —     |     |
|                                     | BART-CP                | 0.408            | 0.310 | 0.506 | 0.75 | 1.00 | -0.399  | 0.492 | —     |     |
|                                     | LRC-BART (pooled)      | 0.420            | 0.316 | 0.515 | 0.75 | 1.00 | -0.412  | 0.501 | 34.8  |     |
|                                     | LRC-BART (0.9)         | 0.417            | 0.416 | 0.511 | 0.89 | 0.98 | -0.409  | 0.498 | 23.1  |     |
|                                     | LRC-BART ( $w = 0.9$ ) | 0.416            | 1.329 | 0.511 | 1.00 | 0.00 | -0.408  | 0.498 | 34.8  |     |

Table 4, continued

| Design                                      | Method                 | Treatment effect |          |          |      |      | Control  |          |           |
|---------------------------------------------|------------------------|------------------|----------|----------|------|------|----------|----------|-----------|
|                                             |                        | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior ESS |
| Survival, $n_T = 30$<br>median ratio<br>Sc1 | AFT-CP                 | 1.540            | 4.678    | 4.444    | 0.92 | 0.15 | -0.006   | 0.204    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                             | BART-CP                | -0.013           | 0.580    | 0.482    | 0.97 | 0.05 | -0.026   | 0.103    | —         |
|                                             | LRC-BART (pooled)      | 0.182            | 0.918    | 0.686    | 0.98 | 0.02 | -0.017   | 0.102    | 80.2      |
|                                             | LRC-BART (0.9)         | 0.222            | 1.174    | 0.711    | 0.99 | 0.01 | -0.005   | 0.133    | 51.8      |
|                                             | LRC-BART ( $w = 0.9$ ) | 1.423            | 11.199   | 1.959    | 1.00 | 0.00 | 0.318    | 0.433    | 79.3      |
| Survival, $n_T = 30$<br>median ratio<br>Sc2 | AFT-CP                 | 20.944           | 46.014   | 51.645   | 0.19 | 0.89 | -0.441   | 0.526    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                             | BART-CP                | 0.814            | 1.109    | 1.231    | 0.92 | 0.25 | -0.199   | 0.277    | —         |
|                                             | LRC-BART (pooled)      | 1.324            | 1.859    | 1.864    | 0.95 | 0.21 | -0.207   | 0.284    | 29.2      |
|                                             | LRC-BART (0.9)         | 1.511            | 3.461    | 2.055    | 0.97 | 0.12 | -0.180   | 0.279    | 19.3      |
|                                             | LRC-BART ( $w = 0.9$ ) | 4.243            | 28.669   | 5.305    | 1.00 | 0.00 | 0.179    | 0.537    | 28.9      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc1   | AFT-CP                 | 0.036            | 0.178    | 0.202    | 0.90 | 0.16 | 0.079    | 0.187    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.311    | 0.355    | —         |
|                                             | BART-CP                | -0.111           | 0.176    | 0.186    | 0.92 | 0.02 | -0.021   | 0.089    | —         |
|                                             | LRC-BART (pooled)      | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.085    | 80.2      |
|                                             | LRC-BART (0.9)         | -0.040           | 0.218    | 0.156    | 0.98 | 0.01 | 0.002    | 0.085    | 51.8      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.080            | 0.998    | 0.200    | 1.00 | 0.00 | 0.020    | 0.089    | 79.3      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc2   | AFT-CP                 | 1.216            | 0.522    | 1.336    | 0.13 | 0.94 | -0.518   | 0.559    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.017    | 0.201    | —         |
|                                             | BART-CP                | 0.149            | 0.271    | 0.286    | 0.95 | 0.14 | -0.219   | 0.260    | —         |
|                                             | LRC-BART (pooled)      | 0.234            | 0.297    | 0.339    | 0.94 | 0.19 | -0.210   | 0.251    | 29.2      |
|                                             | LRC-BART (0.9)         | 0.256            | 0.452    | 0.358    | 0.97 | 0.13 | -0.207   | 0.248    | 19.3      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.527            | 2.689    | 0.643    | 1.00 | 0.00 | -0.185   | 0.228    | 28.9      |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc1  | AFT-CP                 | -0.023           | 0.101    | 0.098    | 0.94 | 0.15 | 0.074    | 0.114    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.305    | 0.317    | —         |
|                                             | BART-CP                | -0.001           | 0.084    | 0.082    | 0.97 | 0.32 | -0.021   | 0.061    | —         |
|                                             | LRC-BART (pooled)      | -0.024           | 0.078    | 0.076    | 0.95 | 0.25 | 0.001    | 0.054    | 169.5     |
|                                             | LRC-BART (0.9)         | -0.022           | 0.094    | 0.076    | 0.99 | 0.17 | 0.002    | 0.054    | 114.0     |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.109            | 1.179    | 0.275    | 1.00 | 0.00 | 0.021    | 0.057    | 81.8      |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc2  | AFT-CP                 | 1.164            | 0.428    | 1.244    | 0.01 | 1.00 | -0.535   | 0.550    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.001    | 0.101    | —         |
|                                             | BART-CP                | 0.305            | 0.206    | 0.361    | 0.61 | 0.77 | -0.233   | 0.261    | —         |
|                                             | LRC-BART (pooled)      | 0.284            | 0.206    | 0.336    | 0.70 | 0.74 | -0.224   | 0.250    | 35.9      |
|                                             | LRC-BART (0.9)         | 0.301            | 0.350    | 0.353    | 0.85 | 0.50 | -0.222   | 0.248    | 23.9      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.562            | 2.419    | 0.633    | 1.00 | 0.00 | -0.200   | 0.228    | 36.2      |

## 4 Application: a single-arm myeloma trial with a registry control

### 4.1 Data and harmonization

EloKRd is a single-arm phase II trial of elotuzumab with carfilzomib, lenalidomide and dexamethasone in 30 patients with newly diagnosed multiple myeloma, with progression-free survival (PFS) and overall survival (OS) measured from the first day of treatment; 13 patients progressed or died and 10 died over a maximum follow-up of 6.75 years. The external control is the University of Chicago Multiple Myeloma (UCMM) registry, an institutional database of 797 patients from which we retain, in order, those with a recorded induction regimen, positive survival times, no transplant

on or before induction, and a regimen other than elotuzumab with lenalidomide and dexamethasone, leaving 253 patients on standard induction regimens, mostly VRd, Vd, KRd and CyBorD, with survival measured from the start of induction. Six baseline covariates are complete in both sources: age, sex, race, ethnicity, high-risk cytogenetics, and receipt of autologous stem cell transplantation (ASCT) after induction (Web Table S5). The trial cohort is enriched for high-risk cytogenetics (50% against 23%) and male sex (70% against 51%), and far fewer trial patients received ASCT (30% against 80%). The unadjusted Kaplan–Meier curves give a 5-year RMST ratio of 1.015 for PFS and 0.90 for OS, with log-rank  $p$ -values of 0.51 and 0.048.

We harmonize race and ethnicity before fitting the models and retain the original coding as a sensitivity analysis (Web Appendix G.5). The ESS ceilings under the original coding are 12.6 for PFS and 14.6 for OS, compared with 63.7 and 36.7 after harmonization; a target of 30 is capped under the original coding.

## 4.2 Estimand and calibration

Neither trial median is reached, and the registry median PFS of 5.1 years falls near the end of the trial follow-up, so we take the standardized 5-year RMST ratio as the primary estimand and the median ratio as a secondary summary. Writing  $\bar{S}_a(t)$  for the arm- $a$  log-normal survival function averaged over the 30 trial profiles,  $\text{RMST}_a(5) = \int_0^5 \bar{S}_a(t) dt$  has a closed form under the AFT model and is evaluated at every posterior draw. The treated arm is AFT-BART with 50 trees fitted to the 30 trial patients, and the control arm is LRC-BART in single-arm mode with  $f$  an AFT-BART of  $H_f = 10$  trees on the registry,  $H_g = 5$ , the split prior  $(0.5, 3)$ ,  $\nu_0 = 3$  and  $w = 1$ , with  $w = 0.9$  and  $H_f = 50$  as sensitivity analyses. Four chains of 2,000 burn-in and 2,000 retained iterations give a split- $\hat{R}$  of 1.000 and bulk effective sizes above 6,800 of 8,000 draws for the RMST ratio.

The calibration uses the registry alone, with  $\sigma_1 = \sigma_2$  since the trial has no controls. The resulting ceiling measures the information the registry posterior carries about the standardized control log-time mean at the 30 trial profiles: 63.7 RCT-equivalent controls for PFS and 36.7 for OS, far below the 253 registry patients, because the trial profiles, half of them high-risk and 70% without ASCT, lie where registry support is limited. A target of 30 is therefore about half the ceiling for PFS and most of the ceiling for OS, and the Pr-rule returns  $s_0^2 = 0.0056$  and 0.0050 with  $\text{ESS}_0$  of 28.1 and 24.4. We report the absolute ladder  $N_{\text{target}} \in \{30, 22.5, 15\}$  and the fractional

ladder at 0.9, 0.5 and 0.25 of the ceiling; the realized ESS equals  $\text{ESS}_0$  within one unit in every row, consistent with  $g$  remaining a prior draw.

### 4.3 Results

Table 5 reports the standardized 5-year RMST ratio for both endpoints. At  $N_{\text{target}} = 30$  LRC-BART gives 0.97 (0.77, 1.23) for PFS and 0.90 (0.77, 1.04) for OS, with  $P(\text{ratio} > 1)$  of 0.37 and 0.08; lowering the target to 15 widens the intervals to (0.75, 1.40) and (0.76, 1.11) and moves the means by at most 0.02, and the fractional ladder does the same. Lower targets increase uncertainty while leaving the point estimates nearly unchanged.

The point estimates agree with the harmonized reference analyses: AFT-BART on the registry with no discrepancy gives 0.97 and 0.90, and the complete-pooling AFT gives 0.87 (0.69, 1.06) and 0.80 (0.65, 0.94), the only interval in the table that excludes one, while the hierarchical AFT stays near 1.2 for both endpoints (Web Appendix G.5). Under the harmonized coding no nonparametric analysis supports a marginal benefit of EloKRd on either endpoint, and the OS estimate favours the registry, which is the direction of the unadjusted log-rank test.

For comparison, the lower block of Table 5 reports the previous analysis. Its MAP-AFT-BART rows had 97.5% limits of 16 to 33 for PFS and 4.8 to 7.2 for OS; the fitted distributions under the calibrated five-tree discrepancy have shorter tails, and Web Appendix G.5 reports the prior and coding comparisons underlying these changes. The  $w = 0.9$  sensitivity widens the PFS interval to (0.74, 1.48) with the mean unchanged, the slab adding prior noise as Section 2.6 predicts, and  $H_f = 50$  changes the means by less than 0.015.

For this single-arm analysis, Figure 2 reports the local prior ESS  $\text{ESS}(x)$  at the 30 trial profiles. It ranges from 5 to 17 for PFS (median 9.5) and 6 to 15 for OS, highest for White patients with ASCT and standard-risk cytogenetics, where registry support is strongest, and lowest for the one Hispanic patient, the Other/Unknown-race patients, and the high-risk profiles without ASCT. The borrowing map  $\mathcal{B}_c$  has no single-arm counterpart, since  $g$  is a prior draw here, mean zero with SD 0.26 on the log-time scale at  $N_{\text{target}} = 30$  at every profile (Web Figure S1); the patient-specific summary therefore describes available registry information. A target of 30 does not imply equally strong external support at every trial profile.

Table 5: EloKRd versus UCMM: standardized 5-year RMST ratio, posterior mean [posterior median] and 95% credible interval, for progression-free survival (PFS) and overall survival (OS) at the 30 EloKRd profiles. The ESS ceilings are 63.7 (PFS) and 36.7 (OS). Reference rows are rerun under the harmonized covariate coding of Section 4.1; the Kaplan–Meier row is unadjusted and reports the estimate with its large-sample interval from the survRM2 package. The published rows are those of the earlier analysis under the original coding, reported there as posterior medians.

| Method                                                                                         | PFS                      | OS                       | ESS <sub>0</sub> (PFS / OS) |
|------------------------------------------------------------------------------------------------|--------------------------|--------------------------|-----------------------------|
| <i>Reference methods, harmonized coding</i>                                                    |                          |                          |                             |
| Kaplan–Meier, unadjusted                                                                       | 1.01 (0.84, 1.22)        | 0.90 (0.79, 1.03)        |                             |
| AFT, complete pooling                                                                          | 0.87 [0.87] (0.69, 1.06) | 0.80 [0.80] (0.65, 0.94) |                             |
| AFT, hierarchical ( $s^2 = 0.5$ )                                                              | 1.23 [1.21] (0.74, 1.90) | 1.21 [1.19] (0.77, 1.70) |                             |
| AFT-BART on UCMM, no discrepancy                                                               | 0.97 [0.97] (0.79, 1.16) | 0.90 [0.91] (0.78, 1.02) |                             |
| <i>LRC-BART, single-arm mode, harmonized coding, <math>H_f = 10</math>, <math>w = 1</math></i> |                          |                          |                             |
| $N_{\text{target}} = 30$                                                                       | 0.97 [0.96] (0.77, 1.23) | 0.90 [0.90] (0.77, 1.04) | 28.1 / 24.4                 |
| $N_{\text{target}} = 22.5$                                                                     | 0.98 [0.96] (0.76, 1.29) | 0.91 [0.90] (0.76, 1.06) | 20.6 / 19.0                 |
| $N_{\text{target}} = 15$                                                                       | 0.99 [0.96] (0.75, 1.40) | 0.91 [0.90] (0.76, 1.11) | 14.2 / 12.7                 |
| 0.9× ceiling                                                                                   | 0.97 [0.96] (0.79, 1.15) | 0.90 [0.90] (0.77, 1.04) | 52.4 / 26.8                 |
| 0.5× ceiling                                                                                   | 0.97 [0.96] (0.78, 1.22) | 0.91 [0.90] (0.76, 1.09) | 29.6 / 15.3                 |
| 0.25× ceiling                                                                                  | 0.99 [0.96] (0.75, 1.38) | 0.92 [0.91] (0.75, 1.16) | 15.4 / 8.2                  |
| <i>Sensitivity analyses at <math>N_{\text{target}} = 30</math></i>                             |                          |                          |                             |
| $w = 0.9$                                                                                      | 1.00 [0.97] (0.74, 1.48) | 0.91 [0.90] (0.76, 1.10) | 28.1 / 24.4                 |
| $H_f = 50$                                                                                     | 0.98 [0.97] (0.78, 1.23) | 0.91 [0.91] (0.77, 1.05) | 27.1 / 25.3                 |
| Original covariate coding                                                                      | 1.04 [1.01] (0.79, 1.43) | 0.91 [0.90] (0.76, 1.07) | 12.9 / 15.0                 |
| <i>Published rows (original coding, posterior medians)</i>                                     |                          |                          |                             |
| AFT, complete pooling                                                                          | 1.03 (0.71, 1.61)        | 0.90 (0.71, 1.17)        |                             |
| AFT, hierarchical ( $s^2 = 0.5$ )                                                              | 1.21 (0.73, 2.68)        | 1.18 (0.77, 1.89)        |                             |
| Standard BART                                                                                  | 1.06 (0.79, 1.51)        | 0.90 (0.76, 1.08)        |                             |
| MAP-AFT-BART, $N_{\text{target}} = 30$                                                         | 1.08 (0.66, 16.40)       | 0.97 (0.74, 4.77)        |                             |
| MAP-AFT-BART, $N_{\text{target}} = 15$                                                         | 1.10 (0.65, 32.80)       | 1.00 (0.73, 7.22)        |                             |

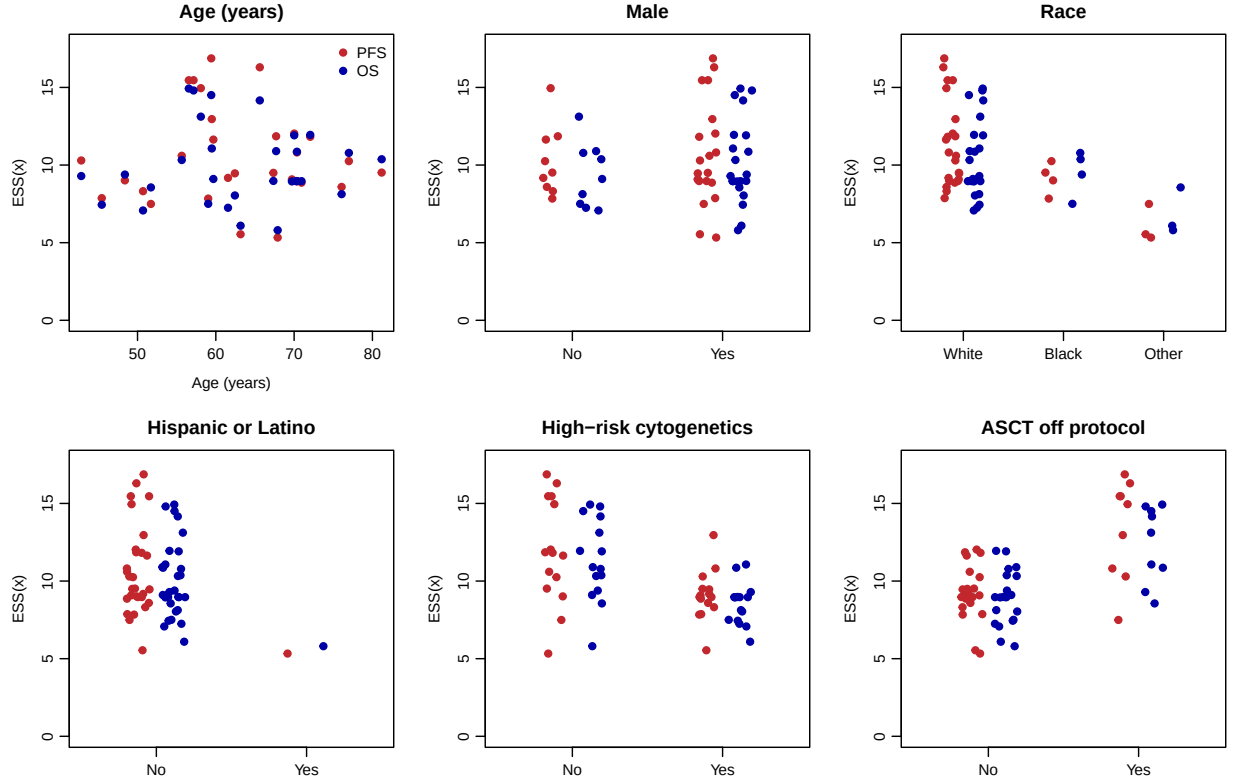


Figure 2: Local prior effective sample size  $ESS(x) = \sigma_1^2 / \{V_f(x) + H_g \tau_0^2\}$  at the 30 EloKRd profiles for PFS (red) and OS (blue) at  $N_{\text{target}} = 30$ , plotted against each harmonized covariate;  $V_f(x)$  is the UCMM posterior variance of the log-time surface at  $x$ .

## 5 Discussion

LRC-BART combines an efficiency gain under agreement with a covariate-resolved assessment of compatibility. In the Gaussian simulations, its RMSE is about 16% lower than BART-PP’s under Sc1. Under global disagreement, its residual bias is about 0.04. For a regional shift of 1.3 residual SDs, the compatibility map and realized regional ESS distinguish the two regions, although the transition spans about one unit of the splitting covariate. The control-surface RMSE remains higher than BART-PP’s, including away from the boundary. At a regional shift of two thirds of a residual SD, excessive borrowing produces bias of 0.04 and coverage of 0.89. The leaf calculation in Theorem 2(a) describes the dependence of detection on the spike scale and local sample size; it does not establish the performance of the full fitted ensemble.

The CAHB comparison illustrates the difficulty of kernel-local borrowing in our nonlinear ten-covariate design. It also depends on our choices for smoothing, precision scaling and joint posterior draws. We therefore interpret it as an assessment of this transfer of CAHB, with the variants reported alongside it. The survival study shows a further limitation: with 75 controls and censoring, the median ratio can amplify control-surface bias and posterior uncertainty. We report the RMST ratio with it. In a single-arm study, trial data cannot identify a control discrepancy. The ceiling and local prior ESS then describe external support at trial profiles, but cannot demonstrate compatibility with an unobserved trial control surface.

The theoretical results condition on features that are learned in the fitted model. Map consistency requires fixed discrepancy partitions and a true discrepancy in their additive span. We have no corresponding result for the posterior over partitions. The leaf-level influence bound also treats  $f$  as fixed; uncertainty in the fitted pooled surface perturbs the residuals used to estimate  $g$ . These limitations prevent a direct extension of the conditional results to the full posterior. The observed surface loss under Sc4 remains an empirical limitation even when profiles near the boundary are excluded.

The spike scale is global, with locality determined by the indicators. Leaf-specific scales could accommodate different degrees of compatibility, at the cost of the  $\text{Inv-}\chi^2$  conjugacy. Multiple external sources would require source-specific discrepancies and a joint calibration that accounts for their information at the trial profiles; additivity of the ceilings has not been established. Binary

and count outcomes require an alternative likelihood, and a nonparametric survival model would require a different RMST calculation. Prior-ESS calibration alone does not control Type I error or power. A confirmatory design would need to evaluate those operating characteristics under specified scenarios and pre-specify how the compatibility map would be used.

## References

- [1] Ethan M. Alt, Xiuya Chang, Xun Jiang, Qing Liu, May Mo, H. Amy Xia, and Joseph G. Ibrahim. LEAP: the latent exchangeability prior for borrowing information from historical data. *Biometrics*, 80(3):ujae083, 2024. doi: 10.1093/biomtc/ujae083.
- [2] Dehua Bi, Tianjian Zhou, Wei Zhong, and Yuan Ji. SAM-HC: a Bayesian nonparametric construction of hybrid control for randomized clinical trials using external data. *Biostatistics*, 27(1):kxag003, 2026. doi: 10.1093/biostatistics/kxag003. Circulated as PAM-HC, arXiv:2310.20087.
- [3] Noirit Kiran Chandra, Abhra Sarkar, John F. de Groot, Ying Yuan, and Peter Müller. Bayesian nonparametric common atoms regression for generating synthetic controls in clinical trials. *Journal of the American Statistical Association*, 118(544):2301–2314, 2023. doi: 10.1080/01621459.2023.2231581.
- [4] Wei-Chen Chen, Chenguang Wang, Heng Li, Nelson Lu, Ram Tiwari, Yunling Xu, and Lilly Q. Yue. Propensity score-integrated composite likelihood approach for augmenting the control arm of a randomized controlled trial by incorporating real-world data. *Journal of Biopharmaceutical Statistics*, 30(3):508–520, 2020. doi: 10.1080/10543406.2020.1730877.
- [5] Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298, 2010. doi: 10.1214/09-AOAS285.
- [6] Sameer K. Deshpande. flexBART: flexible Bayesian regression trees with categorical predictors. *Journal of Computational and Graphical Statistics*, 34(3):1117–1126, 2025. doi: 10.1080/10618600.2024.2431072.
- [7] Yuyan Duan, Keying Ye, and Eric P. Smith. Evaluating water quality using power priors to incorporate historical information. *Environmetrics*, 17(1):95–106, 2006. doi: 10.1002/env.752.

- [8] P. Richard Hahn, Jared S. Murray, and Carlos M. Carvalho. Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3):965–1056, 2020. doi: 10.1214/19-BA1195.
- [9] Brian P. Hobbs, Bradley P. Carlin, Sumithra J. Mandrekar, and Daniel J. Sargent. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics*, 67(3):1047–1056, 2011. doi: 10.1111/j.1541-0420.2011.01564.x.
- [10] Brian P. Hobbs, Daniel J. Sargent, and Bradley P. Carlin. Commensurate priors for incorporating historical information in clinical trials using general and generalized linear models. *Bayesian Analysis*, 7(3):639–674, 2012. doi: 10.1214/12-BA722.
- [11] Joseph G. Ibrahim and Ming-Hui Chen. Power prior distributions for regression models. *Statistical Science*, 15(1):46–60, 2000. doi: 10.1214/ss/1009212673.
- [12] Liyun Jiang, Lei Nie, and Ying Yuan. Elastic priors to dynamically borrow information from historical data in clinical trials. *Biometrics*, 79(1):49–60, 2023. doi: 10.1111/biom.13551.
- [13] Huaqing Jin, Mi-Ok Kim, Aaron Scheffler, and Fei Jiang. Bayesian adaptive design for covariate-adaptive historical control information borrowing. *Statistics in Medicine*, 42(29):5338–5352, 2023. doi: 10.1002/sim.9913.
- [14] Alexander M. Kaizer, Joseph S. Koopmeiners, and Brian P. Hobbs. Bayesian hierarchical modeling based on multisource exchangeability. *Biostatistics*, 19(2):169–184, 2018. doi: 10.1093/biostatistics/kxx031.
- [15] Ales Kotalik, David M. Vock, Eric C. Donny, Dorothy K. Hatsukami, and Joseph S. Koopmeiners. Dynamic borrowing in the presence of treatment effect heterogeneity. *Biostatistics*, 22(4):789–804, 2021. doi: 10.1093/biostatistics/kxz066.
- [16] Beat Neuenschwander, Michael Branson, and David J. Spiegelhalter. A note on the power prior. *Statistics in Medicine*, 28(28):3562–3566, 2009. doi: 10.1002/sim.3722.
- [17] Beat Neuenschwander, Gorana Capkun-Niggli, Michael Branson, and David J. Spiegelhalter. Summarizing historical information on controls in clinical trials. *Clinical Trials*, 7(1):5–18, 2010. doi: 10.1177/1740774509356002.

- [18] Beat Neuenschwander, Sebastian Weber, Heinz Schmidli, and Anthony O’Hagan. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020. doi: 10.1111/biom.13252.
- [19] Jane Pan, Veronica Bunn, Bradley Hupf, and Jianchang Lin. Bayesian additive regression trees (BART) with covariate adjusted borrowing in subgroup analyses. *Journal of Biopharmaceutical Statistics*, 32(4):613–626, 2022. doi: 10.1080/10543406.2022.2089160.
- [20] Veronika Ročková and Stéphanie van der Pas. Posterior concentration for Bayesian regression trees and forests. *The Annals of Statistics*, 48(4):2108–2131, 2020. doi: 10.1214/19-AOS1879.
- [21] Christian Röver and Tim Friede. Dynamically borrowing strength from another study through shrinkage estimation. *Statistical Methods in Medical Research*, 29(1):293–308, 2020. doi: 10.1177/0962280219833079.
- [22] Christian Röver and Tim Friede. Meta-analytic-predictive priors based on a single study, 2025. arXiv:2505.15502, v2, 10 November 2025.
- [23] H. Schmidli, S. Gsteiger, S. Roychoudhury, A. O’Hagan, P. Guo, and B. Neuenschwander. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014. doi: 10.1111/biom.12209.
- [24] Rodney Sparapani, Charles Spanbauer, and Robert McCulloch. Nonparametric machine learning and efficient computation with Bayesian additive regression trees: the BART R package. *Journal of Statistical Software*, 97(1):1–66, 2021. doi: 10.18637/jss.v097.i01.
- [25] Chenguang Wang, Heng Li, Wei-Chen Chen, Nelson Lu, Ram Tiwari, Yunling Xu, and Lilly Q. Yue. Propensity score-integrated power prior approach for incorporating real-world evidence in single-arm clinical studies. *Journal of Biopharmaceutical Statistics*, 29(5):731–748, 2019. doi: 10.1080/10543406.2019.1657133.
- [26] Bruna Wundervald, Andrew Parnell, and Katarina Domijan. Hierarchical embedded Bayesian additive regression trees, 2022. arXiv:2204.07207, v2, 24 April 2023.

- [27] Tianjian Zhou and Yuan Ji. Incorporating external data into the analysis of clinical trials via Bayesian additive regression trees. *Statistics in Medicine*, 40(30):6782–6795, 2021. doi: 10.1002/sim.9196.

## Web Appendices

### Web Appendix A: Model, notation, and exact leaf marginals

This appendix collects the proofs for the locally robust commensurate BART (LRC-BART) model of the main text. We fix notation first. The two control sources are indexed by  $g = 1$  (RCT control) and  $g = 2$  (RWD control), with residual variances  $\sigma_1^2$  and  $\sigma_2^2$ . The control outcomes follow

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad D_i = \mathbf{1}\{i \text{ is an RCT control}\}, \quad (\text{A11})$$

where  $f$  is a standard BART ensemble of  $H_f$  trees with leaf prior  $\mathcal{N}(0, \lambda_f^2)$  and  $g$  is an ensemble of  $H_g$  trees whose leaves carry the two-component prior

$$\theta_{h\ell} \mid z_{h\ell} \sim z_{h\ell} \mathcal{N}(0, \tau_0^2) + (1 - z_{h\ell}) \mathcal{N}(0, \tau_1^2), \quad z_{h\ell} \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w), \quad (\text{A12})$$

with  $z = 1$  the spike (commensurate component) and  $z = 0$  the slab,  $w \sim \text{Beta}(a_w, b_w)$ ,  $\tau_1^2$  fixed by the range rule, and

$$\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2) \text{ truncated to } (0, \tau_1^2), \quad (\text{A13})$$

which is the prior the sampler implements (its conditional for  $\tau_0^2$  is the scaled-Inv- $\chi^2(\nu_0 + K_0, \cdot)$  law truncated to  $(0, \tau_1^2)$ , which is conjugate for (A13) and not for the untruncated prior). Throughout we impose

**Assumption A1** ((A1)).  $0 < \tau_0^2 < \tau_1^2 < \infty$ .

Under (A13), Assumption A1 holds almost surely under the prior and along the chain. The prior-ESS calculation of Web Appendix D uses the untruncated scaled-Inv- $\chi^2(\nu_0, s_0^2)$  law, and we say there where truncation matters; at the calibrated  $s_0^2$  of order  $10^{-3}$  against  $\tau_1^2$  of order one the truncated mass  $P(\chi_{\nu_0}^2 < \nu_0 s_0^2 / \tau_1^2)$  is of order  $10^{-5}$ .

Within a leaf,  $n_g$  and  $\bar{y}_g$  denote the number and the mean of the source- $g$  partial residuals routed to the leaf,  $v_g = \sigma_g^2 / n_g$ , and  $\phi(\cdot; m, V)$  is the  $\mathcal{N}(m, V)$  density;  $\Phi$  is the standard normal distribution function and  $\bar{\Phi} = 1 - \Phi$ . For a  $g$  leaf only RCT controls enter, and we write  $\bar{y} = \bar{y}_1$ ,  $v = v_1$ ,  $n = n_1$  when no confusion arises. The identity we use repeatedly is the Gaussian product

rule

$$\phi(\theta; 0, \tau^2) \phi(\bar{y}; \theta, v) = \phi(\bar{y}; 0, \tau^2 + v) \phi\left(\theta; \frac{\tau^2}{\tau^2 + v} \bar{y}, \frac{\tau^2 v}{\tau^2 + v}\right), \quad (\text{A14})$$

which follows by completing the square in  $\theta$ .

### A.1 Exact leaf marginals

**Lemma A1** (Exact leaf marginals). *Fix all trees other than tree  $h$ , their leaf parameters, the residual variances,  $\lambda_f^2$ , and  $(\tau_0^2, \tau_1^2, w)$ .*

(a) (*f leaf.*) Let  $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$  be the partial residuals of the control observations routed to a leaf of  $f$ -tree  $h$ , so that  $r_i | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(\theta, \sigma_{g_i}^2)$  and  $\theta \sim \mathcal{N}(0, \lambda_f^2)$ . With  $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$  and  $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$ ,  $S_g = \sum_{i:g_i=g} r_i$ ,

$$\int \prod_i \phi(r_i; \theta, \sigma_{g_i}^2) \phi(\theta; 0, \lambda_f^2) d\theta = \left[ \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \right] M_f, \quad M_f = \frac{\exp\{\lambda_f^2 B^2 / \{2(1 + \lambda_f^2 A)\}\}}{(1 + \lambda_f^2 A)^{1/2}}. \quad (\text{A15})$$

(b) (*g leaf.*) Let  $r_i = y_i - f(x_i) - g_{-h}(x_i)$  be the partial residuals of the  $n_1$  RCT controls routed to a leaf of  $g$ -tree  $h$ , with mean  $\bar{y}$ . Under the prior (A12),

$$\sum_{z \in \{0,1\}} P(z) \int \prod_i \phi(r_i; \theta, \sigma_1^2) \phi(\theta; 0, \tau_z^2) d\theta = \left[ \prod_i \phi(r_i; 0, \sigma_1^2) \right] M_g, \quad (\text{A16})$$

$$M_g = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}{\phi(\bar{y}; 0, v)}, \quad (\text{A17})$$

where  $\tau_z^2$  denotes  $\tau_0^2$  for  $z = 1$  and  $\tau_1^2$  for  $z = 0$ . A leaf with  $n_1 = 0$  has  $M_g = 1$ .

(c) (*Leaf conditionals.*) In a  $g$  leaf,

$$P(z = 1 | \bar{y}) = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v)}{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}, \quad \theta | z, \bar{y} \sim \mathcal{N}\left(\frac{\tau_z^2}{\tau_z^2 + v} \bar{y}, \frac{\tau_z^2 v}{\tau_z^2 + v}\right), \quad (\text{A18})$$

and in an  $f$  leaf  $\theta | \cdot \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$ .

*Proof.* (a) Expanding  $-(r_i - \theta)^2 / (2\sigma_{g_i}^2) = -r_i^2 / (2\sigma_{g_i}^2) + r_i \theta / \sigma_{g_i}^2 - \theta^2 / (2\sigma_{g_i}^2)$  and summing over  $i$  gives  $\prod_i \phi(r_i; \theta, \sigma_{g_i}^2) = \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \exp(\theta B - \theta^2 A / 2)$ . Multiplying by  $\phi(\theta; 0, \lambda_f^2)$  and integrating, the

exponent is  $-\frac{1}{2}(A + \lambda_f^{-2})\theta^2 + B\theta$ , a Gaussian integral equal to  $(2\pi)^{1/2}(A + \lambda_f^{-2})^{-1/2} \exp\{B^2/(2(A + \lambda_f^{-2}))\}$ ; dividing by the prior normalizing constant  $(2\pi\lambda_f^2)^{1/2}$  gives  $M_f$ . The conditional of  $\theta$  is read off the same quadratic form.

(b) With a single source, (a) gives  $\prod_i \phi(r_i; \theta, \sigma_1^2) = \prod_i \phi(r_i; 0, \sigma_1^2) \exp\{\theta n \bar{y} / \sigma_1^2 - \theta^2 n / (2\sigma_1^2)\}$ , and the exponential equals  $\phi(\bar{y}; \theta, v) / \phi(\bar{y}; 0, v)$ , since  $-\{(\bar{y} - \theta)^2 - \bar{y}^2\} / (2v) = (2\bar{y}\theta - \theta^2)n / (2\sigma_1^2)$ . Hence the likelihood factors as  $\prod_i \phi(r_i; 0, \sigma_1^2) \times \phi(\bar{y}; \theta, v) / \phi(\bar{y}; 0, v)$ , the sufficiency of  $\bar{y}$ . Integrating  $\phi(\bar{y}; \theta, v)$  against  $\phi(\theta; 0, \tau_z^2)$  by (A14) gives  $\phi(\bar{y}; 0, \tau_z^2 + v)$ , and the sum over  $z$  with weights  $w$  and  $1 - w$  gives  $M_g$ . When  $n_1 = 0$  the leaf has no likelihood factor and the integral of the prior is one.

(c) By (A14), the joint density of  $(z, \theta)$  given  $\bar{y}$  is proportional to  $P(z) \phi(\bar{y}; 0, \tau_z^2 + v) \phi(\theta; \tau_z^2 \bar{y} / (\tau_z^2 + v), \tau_z^2 v / (\tau_z^2 + v))$ ; summing out  $\theta$  gives the stated Bernoulli weight and the remaining factor is the stated conditional of  $\theta$ .  $\square$

**Proposition A1** (The tree moves target the exact posterior). *Condition on  $(\sigma_1^2, \sigma_2^2, \lambda_f^2, \tau_0^2, \tau_1^2, w)$  and on all trees other than  $h$  together with their leaf parameters. Let the update of tree  $h$  consist of a Metropolis–Hastings step on the partition  $T_h$  with grow, prune, and change proposals, acceptance ratio equal to the tree-prior ratio times the proposal ratio times the ratio of products of  $M_f$  (for  $f$  trees) or  $M_g$  (for  $g$  trees) over the affected leaves, followed by a draw of the leaf parameters of  $T_h$  (and, for a  $g$  tree, the indicators) from (A18). Then the composite update leaves the conditional posterior of  $(T_h, \theta_h)$  (resp.  $(T_h, \theta_h, z_h)$ ) invariant. Consequently the full sweep, with the conjugate updates of  $(\sigma_g^2, \tau_0^2, w)$ , the update of  $\tau_0^2$  being the truncated conditional under the prior (A13), is a partially collapsed Gibbs sampler whose invariant law is the exact joint posterior under (A12) and (A13).*

*Proof.* Given the conditioning, the marginal conditional of  $T_h$  with the leaf parameters and indicators integrated out is  $\pi(T_h) \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \prod_{\ell \in T_h} M_\ell$  by Lemma A1, where  $M_\ell$  is  $M_f$  or  $M_g$ . The partial residuals  $r_i$  do not depend on  $T_h$ , and the observation-level product is over all observations of the tree, so it is the same for the current and the proposed partition and cancels from the Metropolis–Hastings ratio. The stated acceptance ratio is therefore the exact ratio for the collapsed target; for a grow move that splits leaf  $P$  into  $L$  and  $R$ , the leaf factor ratio is  $M_L M_R / M_P$ , and the prune and change ratios follow in the same way. A Metropolis–Hastings kernel with the exact ratio leaves the collapsed conditional invariant. Drawing the leaf parameters (and indicators) from their

exact conditional given  $T_h$  then yields a draw from the joint conditional of  $(T_h, \theta_h, z_h)$ , which is the marginal-then-conditional construction of Chipman et al. [5]. The conjugate updates of  $\sigma_g^2$ ,  $\tau_0^2$ , and  $w$  are ordinary Gibbs steps, so the sweep is a partially collapsed Gibbs sampler with the joint posterior as invariant law.  $\square$

**Remark A1** (No surrogate is needed). Every factor entering a tree move is closed form because, conditionally on  $(\tau_0^2, \tau_1^2, w)$ , the leaf prior (A12) is a finite Gaussian mixture, and the Gaussian likelihood integrates against each component in closed form. The previous model placed a leaf-specific discrepancy variance under a scaled-Inv- $\chi^2$  prior and integrated it inside the leaf marginal, which produced an additive scale mixture,  $\int \phi(\bar{y}_1; 0, v_1 + \tau^2) \phi(\bar{y}_2; 0, v_2 + \tau^2) p(\tau^2) d\tau^2$ , with no elementary form; the moment-matched Student- $t$  surrogate and the perturbation bound of the previous Theorem 2 were the response to that integral. In the present model the scale  $\tau_0^2$  is a global parameter updated by its own conjugate step, and the local adaptivity is carried by the per-leaf indicator  $z$ , so the leaf marginal is exact and the tree kernels target the exact posterior (Proposition A1). The surrogate, its error bound, and the compactness assumption that supported it are therefore no longer part of the method; Web Appendix F lists what is dropped.

## Web Appendix B: Bounded local influence (T1)

### B.1 A single $g$ leaf

We fix a  $g$  leaf and condition on  $f$ , on the other  $g$  trees, on the partition, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ , so that  $\bar{y} | \theta \sim \mathcal{N}(\theta, v)$  with  $v = \sigma_1^2/n$ . Write

$$\rho_j = \frac{\tau_j^2}{\tau_j^2 + v}, \quad \hat{\theta}_j = \rho_j \bar{y}, \quad V_j = \rho_j v = \frac{\tau_j^2 v}{\tau_j^2 + v}, \quad j \in \{0, 1\}, \quad (\text{A19})$$

so that  $\hat{\theta}_1$  is the slab estimate (close to the RCT-only residual mean  $\bar{y}$  when  $\tau_1^2 \gg v$ ) and  $\hat{\theta}_0$  the spike-shrunk estimate (close to zero, that is, to the pooled surface  $f$ , when  $\tau_0^2 \ll v$ ). The subscript  $j$  of  $\rho_j$ ,  $\hat{\theta}_j$ ,  $V_j$  indexes the scale  $\tau_j^2$ , so  $j = 1$  is the slab, whereas the indicator convention of (A12) has  $z = 1$  for the spike. Under Assumption A1,  $0 < \rho_0 < \rho_1 < 1$ . Let  $p = P(z = 0 | \bar{y})$  be the posterior

slab probability and define

$$\kappa = \frac{1}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a} = \frac{w}{1-w} \sqrt{\frac{\tau_1^2 + v}{\tau_0^2 + v}}. \quad (\text{A20})$$

Under Assumption A1,  $\kappa > 0$ , and a direct computation gives

$$\rho_1 - \rho_0 = \frac{v(\tau_1^2 - \tau_0^2)}{(\tau_1^2 + v)(\tau_0^2 + v)} = 2v\kappa. \quad (\text{A21})$$

**Theorem A1** (Bounded local influence). *Under Assumption A1 and the conditioning above, the following hold for every  $\bar{y} \in \mathbb{R}$ ; parts (b) to (d) assume  $0 < w < 1$ .*

(a) (Posterior.)  $\theta \mid \bar{y} \sim p\mathcal{N}(\hat{\theta}_1, V_1) + (1-p)\mathcal{N}(\hat{\theta}_0, V_0)$ , and hence

$$\mathbb{E}[\theta \mid \bar{y}] = p \frac{\tau_1^2}{\tau_1^2 + v} \bar{y} + (1-p) \frac{\tau_0^2}{\tau_0^2 + v} \bar{y}. \quad (\text{A22})$$

(b) (Slab probability.)  $\text{logit} p = \log \frac{p}{1-p} = a + \kappa \bar{y}^2$ , with  $a$  and  $\kappa$  as in (A20).

(c) (Bounded spike contribution and bounded RWD-induced shift.) *The spike-shrunk part of (A22) satisfies  $|(1-p)\hat{\theta}_0| \leq |\bar{y}| \tau_0^2 / (\tau_0^2 + v)$ . The shift of the posterior mean relative to the slab estimate,*

$$I(\bar{y}) = \mathbb{E}[\theta \mid \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0) \bar{y}, \quad (\text{A23})$$

*satisfies  $|I(\bar{y})| \leq (\rho_1 - \rho_0) |\bar{y}| \min\{1, Ce^{-\kappa \bar{y}^2}\}$  and*

$$\sup_{\bar{y} \in \mathbb{R}} |I(\bar{y})| \leq 2v \sqrt{\kappa \max\{\log C, \frac{1}{2}\}} \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}. \quad (\text{A24})$$

(d) (Gaussian rate.)  $1-p \leq Ce^{-\kappa \bar{y}^2}$ , so  $p \rightarrow 1$  and  $I(\bar{y}) \rightarrow 0$  as  $|\bar{y}| \rightarrow \infty$ , at a Gaussian rate.

(e) (Nested cases.) *With  $w = 1$  (C-BART),  $p \equiv 0$  and  $I(\bar{y}) = -(\rho_1 - \rho_0)\bar{y}$ , linear and unbounded; if in addition  $\tau_0^2 \rightarrow 0$  (BART-CP),  $\mathbb{E}[\theta \mid \bar{y}] \rightarrow 0$  and  $I(\bar{y}) \rightarrow -\rho_1 \bar{y}$ . With  $w = 0$ ,  $p \equiv 1$  and  $I \equiv 0$ . If Assumption A1 fails with  $\tau_0^2 > \tau_1^2$ , then  $\kappa < 0$ , the roles of the two components are exchanged, and (c), (d) hold with the wide component in the role of the slab; if  $\tau_0^2 = \tau_1^2$ ,  $p \equiv 1-w$  and  $I \equiv 0$ .*

*Proof.* (a) The posterior density is proportional to  $\{w\phi(\theta; 0, \tau_0^2) + (1 - w)\phi(\theta; 0, \tau_1^2)\}\phi(\bar{y}; \theta, v)$ . Applying (A14) to each term gives  $w\phi(\bar{y}; 0, \tau_0^2 + v)\phi(\theta; \hat{\theta}_0, V_0) + (1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)\phi(\theta; \hat{\theta}_1, V_1)$ , a two-component Gaussian mixture with weights proportional to  $w\phi(\bar{y}; 0, \tau_0^2 + v)$  and  $(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)$ ; normalizing gives  $1 - p$  and  $p$  as in (A18), and (A22) is the mean of the mixture.

(b) The log odds of the two weights are

$$\log \frac{(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)}{w\phi(\bar{y}; 0, \tau_0^2 + v)} = \log \frac{1 - w}{w} - \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} + \frac{\bar{y}^2}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right) = a + \kappa \bar{y}^2.$$

(c) The first bound is immediate from  $0 \leq 1 - p \leq 1$ . For  $I$ , subtract  $\hat{\theta}_1$  from (A22). Since  $1 - p = (1 + e^{a + \kappa \bar{y}^2})^{-1} \leq \min\{1, e^{-a - \kappa \bar{y}^2}\} = \min\{1, Ce^{-\kappa \bar{y}^2}\}$ , the pointwise bound follows. For the supremum we show that for  $\kappa > 0$  and  $C > 0$ ,

$$\sup_{u \in \mathbb{R}} |u| \min\{1, Ce^{-\kappa u^2}\} \leq \sqrt{\max\{\log C, \frac{1}{2}\}}/\kappa. \quad (\text{A25})$$

Put  $t = \kappa u^2$ . On  $\{t \leq \log C\}$  (empty if  $C \leq 1$ ) the left side is at most  $|u| \leq \sqrt{\log C/\kappa}$ . On  $\{t > \max(\log C, 0)\}$  the left side equals  $\sqrt{t/\kappa} Ce^{-t}$ . If  $\log C \geq \frac{1}{2}$ , the function  $\sqrt{t} e^{-t}$  is decreasing on  $[\frac{1}{2}, \infty)$ , so the value is at most  $\sqrt{\log C/\kappa} Ce^{-\log C} = \sqrt{\log C/\kappa}$ . If  $\log C < \frac{1}{2}$ , then  $\sup_{t > 0} \sqrt{t} e^{-t} = (2e)^{-1/2}$  gives at most  $C(2e\kappa)^{-1/2} < \sqrt{1/(2\kappa)}$  because  $C < e^{1/2}$ . This proves (A25). Combining with (A21),  $\sup |I| \leq 2v\kappa \sqrt{\max\{\log C, \frac{1}{2}\}}/\kappa = 2v\sqrt{\kappa \max\{\log C, \frac{1}{2}\}}$ , and  $\kappa \leq 1/(2(\tau_0^2 + v)) \leq 1/(2v)$  gives the second inequality in (A24).

(d) is the pointwise bound in (c) together with  $\kappa > 0$ . (e) With  $w = 1$  the slab weight is zero for every  $\bar{y}$ ; with  $w = 0$  the spike weight is zero; the remaining statements are read off (A20) and (A23).  $\square$

**Remark A2** (What is and is not bounded). Theorem A1 bounds the influence of the commensurate component relative to the slab estimate  $\hat{\theta}_1 = \rho_1 \bar{y}$ . The slab estimate itself is shrunk toward zero, that is toward the pooled surface  $f$ , by the factor  $\rho_1$ :  $\mathbb{E}[\theta | \bar{y}] - \bar{y} = -(1 - \rho_1)\bar{y} + I(\bar{y})$  with  $1 - \rho_1 = v/(\tau_1^2 + v)$ . This linear term is the range-rule regularization that any BART leaf with prior variance  $\tau_1^2$  applies, and it is of the same form as the shrinkage of a BART-NP leaf toward its ensemble center; at  $n = 50$ ,  $\sigma_1 = 1.5$  and  $\tau_1^2 = 1.25$  its slope is 0.035. The bounded term  $I$  is the part specific to borrowing.

**Remark A3** (Numerical illustration). With  $\sigma_1 = 1.5$ ,  $n = 50$ ,  $\tau_0^2 = 0.001$  (the scale calibrated for  $N_{\text{target}} = 100$  in the Gaussian study),  $\tau_1^2 = 1.25$  (the range rule at  $H_g = 5$  for an outcome range of 10) and  $w = 0.8$  fixed for illustration,  $v = 0.045$ ,  $\kappa = 10.5$ ,  $C = 21.2$ ,  $\log C = 3.06$ , and  $\rho_1 - \rho_0 = 0.94$ . The bound (A24) gives  $\sup |I| \leq 0.51$ ; the exact supremum of (A23), a one-dimensional maximization, is 0.31, attained at  $\bar{y} = 0.43$ . By comparison C-BART's shift is  $0.94 |\bar{y}|$ , which exceeds 0.51 once  $|\bar{y}| > 0.54$ .

## B.2 The estimand level: fixed partitions of the $g$ ensemble

The estimand is the RCT-standardized control mean  $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$  over the  $N$  RCT covariate profiles, and  $f_1 = f + g$ , so  $\mu = \mu_f + \mu_g$  with  $\mu_g = N^{-1} \sum_{i \leq N} g(x_i)$ . We condition on  $f$ , on the partitions  $\mathcal{T} = (T_1, \dots, T_{H_g})$  of the  $g$  trees, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ ; the RWD then enters  $\mu_g$  only through the shrinkage of  $g$  toward zero. Index the  $L_g$  leaves of all  $g$  trees by  $j$ , let  $\theta \in \mathbb{R}^{L_g}$  collect the leaf parameters and  $z \in \{0, 1\}^{L_g}$  the indicators, and let  $r \in \mathbb{R}^{n_1}$  be the residuals  $y_i - f(x_i)$  of the RCT controls. Let  $X \in \{0, 1\}^{n_1 \times L_g}$  be the incidence matrix ( $X_{ij} = 1$  if control  $i$  falls in leaf  $j$ ), so each row of  $X$  has exactly  $H_g$  ones and  $g(x_i) = (X\theta)_i$ . Let  $a \in \mathbb{R}^{L_g}$  with  $a_j = N_j/N$ ,  $N_j$  the number of estimand profiles in leaf  $j$ , so  $\mu_g = a^\top \theta$ ,  $\|a\|_1 = H_g$ , and  $\|a\|_2^2 \leq \max_j a_j \sum_j a_j \leq H_g$ , so  $\|a\|_2 \leq \sqrt{H_g}$ . Then

$$r | \theta \sim \mathcal{N}(X\theta, \sigma_1^2 I), \quad \theta | z \sim \mathcal{N}(0, D_z), \quad D_z = \text{diag}(\tau_{z_j}^2), \quad \pi(z) = \prod_j w^{z_j} (1-w)^{1-z_j}. \quad (\text{A26})$$

Standard Gaussian algebra gives, with  $V_z = XD_zX^\top + \sigma_1^2 I$  and  $\Sigma_z = (X^\top X/\sigma_1^2 + D_z^{-1})^{-1}$ ,

$$P(z | r) \propto \pi(z) \phi_{n_1}(r; 0, V_z), \quad \theta | z, r \sim \mathcal{N}(\hat{\theta}(z), \Sigma_z), \quad \hat{\theta}(z) = D_z X^\top V_z^{-1} r, \quad (\text{A27})$$

where the form of  $\hat{\theta}(z)$  follows from  $(X^\top X/\sigma^2 + D^{-1})^{-1} X^\top/\sigma^2 = DX^\top(XDX^\top + \sigma^2 I)^{-1}$ . We write  $\mathbf{0}$  for the all-slab configuration; the  $w = 0$  model (the tree analogue of BART-PP, a free source offset regularized by the range rule) has  $\mathbb{E}_{w=0}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{0})$ , and we call

$$\Delta(r) = \mathbb{E}[\mu_g | r] - a^\top \hat{\theta}(\mathbf{0}) = \sum_{z \neq \mathbf{0}} P(z | r) a^\top (\hat{\theta}(z) - \hat{\theta}(\mathbf{0})) \quad (\text{A28})$$

the RWD-induced shift of the estimand at the given partitions.

**Corollary A1** (Bounded shift of the standardized control mean). *Under Assumption A1,  $0 < w < 1$ , and the conditioning of this subsection, let  $\delta = \tau_1^2 - \tau_0^2$  and  $b = w\tau_1/\{(1-w)\tau_0\}$ .*

$$(a) \text{ (General partitions.) } \sup_{r \in \mathbb{R}^{n_1}} |\Delta(r)| \leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) (2^{L_g} - 1) \sqrt{2\delta \max\{L_g \log b, \frac{1}{2}\}} < \infty.$$

(b) (Single tree,  $H_g = 1$ .) *With leaves  $\ell$ , RCT counts  $n_{1\ell}$ ,  $v_\ell = \sigma_1^2/n_{1\ell}$ , leaf means  $\bar{y}_\ell$ , and  $\kappa_\ell, C_\ell$  as in (A20) with  $v = v_\ell$ ,*

$$\Delta(r) = \sum_{\ell: n_{1\ell} > 0} a_\ell I_\ell(\bar{y}_\ell), \quad \sup_r |\Delta(r)| \leq \sum_\ell a_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}},$$

*and the right side is at most  $\max_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}}$ .*

(c) ( $H_g$  trees with a common partition.) *If all  $g$  trees share one partition with leaves  $\ell$ , write  $\vartheta_\ell = \sum_h \theta_{h\ell}$ ,  $T_k = k\tau_0^2 + (H_g - k)\tau_1^2$ ,  $\rho_\ell^{(k)} = T_k/(T_k + v_\ell)$ ,*

$$\kappa_{k\ell} = \frac{1}{2} \left( \frac{1}{T_k + v_\ell} - \frac{1}{T_0 + v_\ell} \right), \quad C_{k\ell} = \binom{H_g}{k} \left( \frac{w}{1-w} \right)^k \sqrt{\frac{T_0 + v_\ell}{T_k + v_\ell}}.$$

*Then  $P(k \text{ spikes in leaf } \ell \mid \bar{y}_\ell) \propto \binom{H_g}{k} w^k (1-w)^{H_g-k} \phi(\bar{y}_\ell; 0, T_k + v_\ell)$ , the leaves are independent a posteriori,*

$$\Delta(r) = - \sum_\ell a_\ell \bar{y}_\ell \sum_{k=1}^{H_g} P(k \mid \bar{y}_\ell) (\rho_\ell^{(0)} - \rho_\ell^{(k)}), \tag{A29}$$

$$\sup_r |\Delta(r)| \leq \max_\ell 2v_\ell \sum_{k=1}^{H_g} \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}} \leq \max_\ell H_g v_\ell \sqrt{\frac{2 \max_k \max\{\log C_{k\ell}, \frac{1}{2}\}}{H_g \tau_0^2 + v_\ell}}. \tag{A30}$$

(d) (Contrasts.) *Under complete pooling of the control surface ( $w = 1$ ,  $\tau_0^2 \rightarrow 0$ ) the shift is  $-a^\top \hat{\theta}(\mathbf{0})$ , and under C-BART ( $w = 1$ ) it is  $a^\top (\hat{\theta}(\mathbf{1}) - \hat{\theta}(\mathbf{0}))$ ; both are linear in  $r$ , and each is unbounded on  $\mathbb{R}^{n_1}$  unless the corresponding linear functional vanishes identically. For a single tree the C-BART functional is  $-\sum_\ell a_\ell (\rho_{1\ell} - \rho_{0\ell}) \bar{y}_\ell$ , which is nonzero whenever some leaf has  $a_\ell > 0$  and  $n_{1\ell} > 0$ .*

*Proof.* (a) Fix  $z \neq \mathbf{0}$ , let  $S = \{j : z_j = 1\}$  be its spike leaves,  $E_S$  the diagonal indicator of  $S$ ,  $X_S$  the corresponding columns of  $X$ , and  $u = X_S^\top V_0^{-1} r \in \mathbb{R}^{|S|}$ . Since  $D_0 - D_z = \delta E_S$ ,

$$V_z^{-1} - V_0^{-1} = V_z^{-1}(V_0 - V_z)V_0^{-1} = \delta V_z^{-1} X_S X_S^\top V_0^{-1}, \quad \text{so} \quad V_z^{-1} r = V_0^{-1} r + \delta V_z^{-1} X_S u.$$

Three consequences follow. First, the quadratic form in the likelihood ratio is

$$r^\top (V_z^{-1} - V_0^{-1}) r = \delta (V_z^{-1} r)^\top X_S u = \delta (u + \delta X_S^\top V_z^{-1} X_S u)^\top u = \delta \|u\|^2 + \delta^2 u^\top X_S^\top V_z^{-1} X_S u \geq \delta \|u\|^2.$$

Second, the difference of posterior means is

$$\hat{\theta}(z) - \hat{\theta}(\mathbf{0}) = (D_z - D_0) X^\top V_0^{-1} r + D_z X^\top (V_z^{-1} - V_0^{-1}) r = -\delta \iota_S u + \delta D_z X^\top V_z^{-1} X_S u,$$

where  $\iota_S$  embeds  $\mathbb{R}^{|S|}$  into the  $S$  coordinates of  $\mathbb{R}^{L_g}$ . The matrix  $D_z X^\top V_z^{-1} X = D_z^{1/2} \{M^\top (M M^\top + \sigma_1^2 I)^{-1} M\} D_z^{-1/2}$  with  $M = X D_z^{1/2}$ , and the bracket is symmetric with eigenvalues  $\lambda_i / (\lambda_i + \sigma_1^2) \in [0, 1)$ , so  $\|D_z X^\top V_z^{-1} X_S\| \leq \|D_z X^\top V_z^{-1} X\| \leq \|D_z^{1/2}\| \|D_z^{-1/2}\| \leq \tau_1 / \tau_0$ , and  $\|\hat{\theta}(z) - \hat{\theta}(\mathbf{0})\| \leq \delta(1 + \tau_1 / \tau_0) \|u\|$ . Third,  $|V_z| = \sigma_1^{2n_1} |D_z| |D_z^{-1} + X^\top X / \sigma_1^2|$  and  $D_z^{-1} \succeq D_0^{-1}$  give  $|V_0| / |V_z| \leq |D_0| / |D_z| = (\tau_1^2 / \tau_0^2)^{|S|}$ , so

$$\begin{aligned} P(z | r) &\leq \frac{P(z | r)}{P(\mathbf{0} | r)} = \frac{\pi(z)}{\pi(\mathbf{0})} \sqrt{\frac{|V_0|}{|V_z|}} \exp\left\{-\frac{1}{2} r^\top (V_z^{-1} - V_0^{-1}) r\right\} \\ &\leq c_z e^{-\delta \|u\|^2 / 2}, \quad c_z = b^{|S|}. \end{aligned}$$

Combining, and using (A25) in the variable  $\|u\|$  with  $\kappa = \delta/2$ ,

$$\begin{aligned} |P(z | r) a^\top (\hat{\theta}(z) - \hat{\theta}(\mathbf{0}))| &\leq \|a\| \delta \left(1 + \frac{\tau_1}{\tau_0}\right) \|u\| \min\{1, c_z e^{-\delta \|u\|^2 / 2}\} \\ &\leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) \sqrt{2\delta \max\{|S| \log b, \frac{1}{2}\}}. \end{aligned}$$

Summing over the  $2^{L_g} - 1$  configurations  $z \neq \mathbf{0}$  and using  $|S| \leq L_g$  gives (a).

(b) For a single tree  $X^\top X$  is diagonal, the residual vector splits by leaf, and (A26) factorizes over leaves, so the leaves are independent a posteriori and each is the single-leaf model of Section B.1 with  $v = v_\ell$ ; leaves with  $n_{1\ell} = 0$  have posterior equal to prior under every  $w$  and contribute zero to

$\Delta$ . Theorem A1(c) bounds each  $|I_\ell|$ , and  $\sum_\ell a_\ell = 1$  gives the last inequality.

(c) With a common partition, the residual in leaf  $\ell$  is  $r_i = \vartheta_\ell + \varepsilon_i$  and, given the indicators,  $\vartheta_\ell \sim \mathcal{N}(0, T_{k_\ell})$  with  $k_\ell$  the number of spikes among the  $H_g$  leaves stacked on  $\ell$ ; the leaves are independent a posteriori and  $\mu_g = \sum_\ell a_\ell \vartheta_\ell$ . Summing  $\pi(z)\phi(\bar{y}_\ell; 0, T_k + v_\ell)$  over the  $\binom{H_g}{k}$  configurations with  $k$  spikes gives the stated posterior of  $k$ , and (A14) gives  $\mathbb{E}[\vartheta_\ell | k, \bar{y}_\ell] = \rho_\ell^{(k)} \bar{y}_\ell$ . Under  $w = 0$ ,  $k = 0$  almost surely, which gives the expression for  $\Delta$ . For  $k \geq 1$ ,  $P(k | \bar{y}_\ell) \leq P(k | \bar{y}_\ell)/P(0 | \bar{y}_\ell) = C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}$ , and  $\rho_\ell^{(0)} - \rho_\ell^{(k)} = v_\ell(T_0 - T_k)/\{(T_0 + v_\ell)(T_k + v_\ell)\} = 2v_\ell \kappa_{k\ell}$ ; the  $k$ -th term is therefore at most  $2v_\ell \kappa_{k\ell} |\bar{y}_\ell| \min\{1, C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}\} \leq 2v_\ell \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}}$  by (A25). The final form uses  $\kappa_{k\ell} \leq \kappa_{H_g \ell} \leq 1/\{2(H_g \tau_0^2 + v_\ell)\}$  and  $\sum_\ell a_\ell = 1$ .

(d) With  $w = 1$  the posterior over  $z$  is degenerate at  $\mathbf{1}$ , so  $\mathbb{E}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{1})$ , linear in  $r$ ; as  $\tau_0^2 \rightarrow 0$ ,  $\hat{\theta}(\mathbf{1}) \rightarrow 0$ . A nonzero linear functional on  $\mathbb{R}^{n_1}$  is unbounded. For a single tree,  $\hat{\theta}_\ell(\mathbf{1}) - \hat{\theta}_\ell(\mathbf{0}) = (\rho_{0\ell} - \rho_{1\ell}) \bar{y}_\ell$  by (b).  $\square$

**Remark A4** (On the constants). The constant in (a) is finite for every design and every partition, which is the content of the statement, but it is far from sharp: it treats each of the  $2^{L_g} - 1$  non-slab configurations separately and bounds the posterior probability of each by its odds against the all-slab configuration. Parts (b) and (c) are the cases in which the sum collapses, and they display the dependence on  $H_g$ , on  $\tau_0$  through  $\kappa$ , and on the leaf shrinkage  $\rho_\ell^{(H_g)} = H_g \tau_0^2 / (H_g \tau_0^2 + v_\ell)$  of the all-spike state. For the numbers of Remark A3 with  $\tau_0^2 = 0.0012$  and the study's configuration  $H_g = 5$ , the common-partition bound in (c) is 1.15 per leaf, while the exact supremum of (A29) is 0.148, attained at  $\bar{y}_\ell = 0.33$  (at  $H_g = 10$  the corresponding values are 1.80 and 0.067 at 0.30); the ensemble abstains sooner than a single leaf because there are  $H_g$  ways to place one slab, and one slab leaf absorbs almost all of the discrepancy. The exact supremum is a one-dimensional maximization and is what we recommend reporting.

## Web Appendix C: Detection and the borrowing map (T2)

### C.1 A single $g$ leaf under a true local shift

We keep the conditioning of Section B.1 and add a data-generating assumption: the RCT residuals in the leaf are  $r_i = d + \epsilon_i$  with  $\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_1^2)$ , where  $d$  is the true local discrepancy between the

RCT control surface and the pooled surface  $f$  (taken as fixed; see the Gap paragraph in Section C.3 for the estimated  $f$ ). Then  $\bar{y} \sim \mathcal{N}(d, v)$ ,  $v = \sigma_1^2/n$ . Let  $P_{\text{spike}}(\bar{y}) = 1 - p = P(z = 1 \mid \bar{y})$ .

**Theorem A2** (Leaf-level detection). *Under Assumption A1 and the conditioning above:*

- (a) (Spike probability.)  $P_{\text{spike}}(\bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$  with  $a, \kappa$  as in (A20); it depends on  $(\bar{y}, n)$  only through  $\bar{y}^2$  and  $v = \sigma_1^2/n$ , is decreasing in  $|\bar{y}|$ , and satisfies  $P_{\text{spike}}(0) = \{1 + e^a\}^{-1}$ .
- (b) (Half-detection discrepancy.) If  $\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$ , the equation  $P_{\text{spike}}(\bar{y}) = \frac{1}{2}$  has the unique positive solution

$$d_{1/2} = \sqrt{\frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\}}; \quad (\text{A31})$$

otherwise  $P_{\text{spike}} \leq \frac{1}{2}$  for every  $\bar{y}$  (with equality only at  $\bar{y} = 0$  when the bracket vanishes) and we set  $d_{1/2} = 0$ . When  $\tau_1^2 \gg \tau_0^2 + v$ ,  $d_{1/2}^2 \approx 2(\tau_0^2 + v) \{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2}{\tau_0^2 + v} \}$ , a few multiples of  $\tau_0^2 + \sigma_1^2/n$ . The probability that the leaf's slab probability exceeds one half, under the true shift  $d$ , is

$$P_d(p(\bar{y}) > \frac{1}{2}) = P_d(|\bar{y}| > d_{1/2}) = \bar{\Phi}\left(\frac{d_{1/2} - d}{\sqrt{v}}\right) + \bar{\Phi}\left(\frac{d_{1/2} + d}{\sqrt{v}}\right). \quad (\text{A32})$$

- (c) (Large  $n$ ,  $\tau_0^2$  fixed.) As  $n \rightarrow \infty$ , almost surely

$$P_{\text{spike}}(\bar{y}) \rightarrow P_{\text{spike}}^\infty(d) = \left\{ 1 + \frac{(1-w)\tau_0}{w\tau_1} \exp\left(\frac{d^2}{2} \left(\frac{1}{\tau_0^2} - \frac{1}{\tau_1^2}\right)\right) \right\}^{-1} \in (0, 1). \quad (\text{A33})$$

In particular, when  $d = 0$  (borrowing is warranted),  $P_{\text{spike}} \rightarrow w\tau_1 / \{w\tau_1 + (1-w)\tau_0\} > w$ ; when  $d \neq 0$ ,  $P_{\text{spike}}^\infty(d) \leq \frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$ , which tends to zero as  $|d|/\tau_0 \rightarrow \infty$  but is strictly positive for every fixed  $d$ .

- (d) (Large  $n$  with a vanishing spike.) If  $\tau_0^2 = \tau_{0,n}^2 \rightarrow 0$  as  $n \rightarrow \infty$  (with  $\tau_1^2$ ,  $w$  fixed), then for every  $d \neq 0$ ,  $P_{\text{spike}}(\bar{y}) \rightarrow 0$  almost surely.

*Proof.* (a) is Theorem A1(b) with  $\kappa > 0$ . (b) Setting  $a + \kappa \bar{y}^2 = 0$  gives  $\bar{y}^2 = -a/\kappa$ , which is (A31) after substituting (A20); the condition is  $-a > 0$ . The approximation drops  $v$  and  $\tau_0^2$  against  $\tau_1^2$  in  $(\tau_1^2 + v)/(\tau_1^2 - \tau_0^2)$ . Since  $p > \frac{1}{2}$  iff  $|\bar{y}| > d_{1/2}$  and  $\bar{y} \sim \mathcal{N}(d, v)$ , (A32) follows.

(c) By the strong law,  $\bar{y} \rightarrow d$  almost surely;  $v \rightarrow 0$  gives  $a \rightarrow a_\infty = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2}{\tau_1^2}$  and  $\kappa \rightarrow \kappa_\infty = \frac{1}{2}(\tau_0^{-2} - \tau_1^{-2})$ , both finite. Hence  $a + \kappa \bar{y}^2 \rightarrow a_\infty + \kappa_\infty d^2$  almost surely, and (A33) follows by continuity of  $t \mapsto (1 + e^t)^{-1}$ ; the limit is strictly between 0 and 1 because the exponent is finite. The bound on  $P_{\text{spike}}^\infty(d)$  is  $(1 + e^t)^{-1} \leq e^{-t}$ .

(d) Write  $\epsilon_n = \tau_{0,n}^2 + v_n \rightarrow 0$ . Then  $a_n + \kappa_n \bar{y}_n^2 \geq \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\epsilon_n}{\tau_1^2 + v_n} + \bar{y}_n^2 \left( \frac{1}{2\epsilon_n} - \frac{1}{2\tau_1^2} \right)$ . Almost surely  $\bar{y}_n^2 \rightarrow d^2 > 0$ , and  $\frac{d^2}{2\epsilon_n} + \frac{1}{2} \log \epsilon_n \rightarrow \infty$ , so the log odds of the slab diverge and  $P_{\text{spike}} \rightarrow 0$ .  $\square$

**Gap (consistency of abstention).** The design specification states that  $P_{\text{spike}} \rightarrow 0$  as  $n \rightarrow \infty$  for any nonzero shift. Theorem A2(c) shows this is false for a fixed spike scale: because the spike is a  $\mathcal{N}(0, \tau_0^2)$  component and not a point mass, a discrepancy of the order of  $\tau_0$  is consistent with it, and the limiting spike probability (A33) is positive for every  $d$ . What is true is (i) the limit is exponentially small in  $d^2/(2\tau_0^2)$ , so abstention is consistent in the regime  $|d|/\tau_0 \rightarrow \infty$ , and (ii) abstention is consistent for every fixed  $d \neq 0$  when the spike scale shrinks with  $n$  (part (d)). In the two-arm sampler  $\tau_0^2$  is learned from the spike leaves under a prior calibrated to the order  $10^{-3}$ , so on the outcome scale (i) is the operative statement: at  $\tau_0^2 = 0.001$  a shift of  $d = 0.5$  has  $d^2/(2\tau_0^2) = 125$ . The consistency statement that does hold for every fixed  $d \neq 0$ , without a vanishing spike, is the one for the discrepancy map in Theorem A3 below.

**Remark A5** (Numerical illustration). With the constants of Remark A3, (A31) gives  $d_{1/2} = 0.54$  (the specification's rough figure was 0.7), and (A32) gives leaf abstention probabilities 0.011, 0.43, 0.98 and 1.00 at true shifts  $d = 0, 0.5, 1$  and  $2$ . The limit (A33) at  $d = 0$  is 0.993.

## C.2 The borrowing map $P(|g(x)| < c \mid \text{data})$

The map we display is the posterior probability that the local discrepancy is smaller than a pre-specified outcome-scale threshold  $c > 0$ ,  $\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data})$ , together with the all-spike map  $B(x) = P(\text{all } H_g \text{ leaves covering } x \text{ are spike} \mid \text{data})$  as a diagnostic. For a single tree and a fixed partition,  $g(x) = \theta_\ell$  for  $x$  in leaf  $\ell$ , and the two are related by the leaf mixture of Theorem A1(a).

**Theorem A3** (The discrepancy map for a single tree). *Under Assumption A1, for a single  $g$  tree with a fixed partition,  $f$  and  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$  fixed, and  $x$  in a leaf with statistics  $(\bar{y}, n)$ :*

(a) (Exact form.)  $\mathcal{B}_c(x) = p \Psi_1(c) + (1 - p) \Psi_0(c)$  with

$$\Psi_j(c) = \Phi\left(\frac{c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right) - \Phi\left(\frac{-c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right), \quad j \in \{0, 1\}.$$

(b) (Relation to the spike probability.)  $\mathcal{B}_c(x) \leq P_{\text{spike}}(\bar{y}) + p \Psi_1(c)$ , and on  $\{\rho_0 |\bar{y}| < c\}$ ,

$$\mathcal{B}_c(x) \geq P_{\text{spike}}(\bar{y}) \left\{ 1 - 2\bar{\Phi}\left(\frac{c - \rho_0 |\bar{y}|}{\tau_0}\right) \right\}.$$

(c) (Consistency.) If the RCT residuals in the leaf are  $d + \epsilon_i$  as in Section C.1 and  $|d| \neq c$ , then as  $n \rightarrow \infty$ , almost surely  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$ , whatever the limit of  $P_{\text{spike}}$ .

*Proof.* (a) By Theorem A1(a), the posterior of  $g(x) = \theta$  is the two-component mixture with components  $\mathcal{N}(\rho_j \bar{y}, \rho_j v)$ , and  $\Psi_j(c)$  is the probability that component  $j$  lies in  $(-c, c)$ .

(b) The upper bound uses  $\Psi_0 \leq 1$ . For the lower bound,  $\mathcal{B}_c \geq (1 - p)\Psi_0(c)$  and, for a  $\mathcal{N}(m, V)$  variable with  $|m| < c$ ,  $P(|N| \geq c) \leq P(|N - m| \geq c - |m|) = 2\bar{\Phi}((c - |m|)/\sqrt{V})$ ; here  $|m| = \rho_0 |\bar{y}|$  and  $V = \rho_0 v = \tau_0^2 v / (\tau_0^2 + v) \leq \tau_0^2$ , and  $\bar{\Phi}$  is decreasing.

(c) Almost surely  $\bar{y} \rightarrow d$ , and  $\rho_j \rightarrow 1$ ,  $\rho_j v \rightarrow 0$  for both  $j$ , so  $\rho_j \bar{y} \rightarrow d$  and each component concentrates at  $d$ :  $\Psi_j(c) \rightarrow \mathbf{1}\{|d| < c\}$  when  $|d| \neq c$ . A convex combination of two sequences with the same limit has that limit, regardless of the weights.  $\square$

**Remark A6.** Parts (b) and (c) are the theoretical counterpart of the empirical finding that  $\mathcal{B}_c$  separates shifted from compatible regions while  $B(x)$  does not. The spike indicator answers whether the leaf discrepancy is of the order of  $\tau_0$ , a question whose answer stays uncertain in the limit (Theorem A2(c));  $\mathcal{B}_c$  answers whether it is smaller than  $c$  on the outcome scale, a question that the RCT controls settle as  $n$  grows. The threshold  $c$  must be pre-specified on the outcome scale; the study uses one third of  $\sigma_{\text{RCT}}$ .

### C.3 The ensemble with fixed partitions, and what is not proved

We return to the fixed-partition linear form (A26). For a profile  $x$ , let  $e_x \in \{0, 1\}^{L_g}$  select the  $H_g$  leaves covering  $x$ , so  $g(x) = e_x^\top \theta$ . The atoms of the common refinement of the  $H_g$  partitions are the cells  $A_1, \dots, A_m$ ;  $e_x$  is constant on each cell, and if  $x$  and RCT control  $i$  lie in the same cell then  $e_x$

is row  $i$  of  $X$ . Let  $n_{1k}$  be the number of RCT controls in cell  $A_k$ .

**Proposition A2** (The discrepancy map for  $H_g$  trees). *Under Assumption A1 and the conditioning of Section B.2:*

- (a) (Exact form.)  $\mathcal{B}_c(x) = \sum_z P(z|r) \Psi_z(c)$  with  $\Psi_z(c) = P\{|\mathcal{N}(e_x^\top \hat{\theta}(z), e_x^\top \Sigma_z e_x)| < c\}$ , and  $B(x) = \sum_{z \in \mathcal{Z}_x} P(z|r)$  where  $\mathcal{Z}_x = \{z : z_j = 1 \text{ for all } j \text{ with } (e_x)_j = 1\}$ .
- (b) (Union bound.)  $1 - B(x) \leq \sum_{h=1}^{H_g} P(z_{h, \ell_h(x)} = 0 | r)$  and  $B(x) \leq \min_h P(z_{h, \ell_h(x)} = 1 | r)$ , where  $\ell_h(x)$  is the leaf of tree  $h$  containing  $x$ . For  $z \in \mathcal{Z}_x$ ,  $e_x^\top \Sigma_z e_x \leq e_x^\top D_z e_x = H_g \tau_0^2$ , so on  $\{|e_x^\top \hat{\theta}(z)| < c\}$ ,  $\Psi_z(c) \geq 1 - 2\bar{\Phi}\{(c - |e_x^\top \hat{\theta}(z)|)/(\sqrt{H_g} \tau_0)\}$ .
- (c) (Consistency at fixed partitions.) Suppose the true residual means are representable on the partitions,  $\mathbb{E}[r|x] = X\theta^*$  for some  $\theta^* \in \mathbb{R}^{L_g}$ , and let  $x \in A_k$  with  $m_k^* = e_x^\top \theta^*$  and  $|m_k^*| \neq c$ . If  $n_{1k} \rightarrow \infty$  (the other cell counts arbitrary), then  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|m_k^*| < c\}$  in probability.

*Proof.* (a) is (A27) and the definition of the two maps. (b) The first two inequalities are the union bound and monotonicity for the events  $\{z_{h, \ell_h(x)} = 0\}$ ,  $h = 1, \dots, H_g$ . For Gaussians the posterior covariance is dominated by the prior covariance,  $\Sigma_z = (X^\top X / \sigma_1^2 + D_z^{-1})^{-1} \preceq D_z$ , and for  $z \in \mathcal{Z}_x$ ,  $e_x^\top D_z e_x = H_g \tau_0^2$ ; the tail bound is as in Theorem A3(b).

(c) Fix  $z$ ; write  $e = e_x$ ,  $G = X^\top X / \sigma_1^2$ ,  $m = e^\top \theta$ . *Variance.* The data enter the Gaussian model (A26) through the cell means  $\bar{r}_{k'} | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(e^{(k')\top} \theta, \sigma_1^2 / n_{1k'})$ , where  $e^{(k')}$  is the common row of  $X$  in cell  $k'$ . In the joint Gaussian law of  $(m, \bar{r}_1, \dots, \bar{r}_m)$  given  $z$ , conditional variances are constants and the law of total variance gives  $\text{Var}(m | \bar{r}_k) = \mathbb{E}[\text{Var}(m | \bar{r}) | \bar{r}_k] + \text{Var}(\mathbb{E}[m | \bar{r}] | \bar{r}_k) \geq \text{Var}(m | \bar{r})$ ; and  $\text{Var}(m | \bar{r}_k) = \omega^2 (\sigma_1^2 / n_{1k}) / (\omega^2 + \sigma_1^2 / n_{1k}) \leq \sigma_1^2 / n_{1k}$  with  $\omega^2 = e^\top D_z e$ . Hence  $e^\top \Sigma_z e \leq \sigma_1^2 / n_{1k} \rightarrow 0$ . *Mean.* With  $r = X\theta^* + \epsilon$ , (A27) gives  $\mathbb{E}[\theta | z, r] = \Sigma_z G \theta^* + \Sigma_z X^\top \epsilon / \sigma_1^2$ , and  $\Sigma_z G = I - \Sigma_z D_z^{-1}$ , so

$$e^\top \mathbb{E}[\theta | z, r] - m_k^* = -e^\top \Sigma_z D_z^{-1} \theta^* + e^\top \Sigma_z X^\top \epsilon / \sigma_1^2.$$

By the Cauchy–Schwarz inequality in the  $\Sigma_z$  inner product,

$$|e^\top \Sigma_z D_z^{-1} \theta^*| \leq \sqrt{e^\top \Sigma_z e} \sqrt{(D_z^{-1} \theta^*)^\top \Sigma_z (D_z^{-1} \theta^*)} \leq \sqrt{\sigma_1^2 / n_{1k}} \tau_1 \|\theta^*\| / \tau_0^2,$$

using  $\Sigma_z \preceq D_z \preceq \tau_1^2 I$  and  $\|D_z^{-1}\| \leq \tau_0^{-2}$ . The noise term has mean zero and variance  $e^\top \Sigma_z G \Sigma_z e \leq$

$e^\top \Sigma_z e \leq \sigma_1^2/n_{1k}$ , because  $G \preceq G + D_z^{-1} = \Sigma_z^{-1}$ . Both terms vanish in probability. *Conclusion.* For each of the finitely many  $z$ ,  $\Psi_z(c) = P\{|\mathcal{N}(\hat{m}_z, \hat{V}_z)| < c\}$  with  $\hat{m}_z \rightarrow m_k^*$  in probability and  $\hat{V}_z \rightarrow 0$ , so  $\Psi_z(c) \rightarrow \mathbf{1}\{|m_k^*| < c\}$  in probability when  $|m_k^*| \neq c$ , uniformly over  $z$ ; the mixture  $\sum_z P(z|r)\Psi_z(c)$  then has the same limit whatever the weights.  $\square$

**Gap (the ensemble with random partitions).** Everything in this appendix about the map is conditional on the partitions of the  $g$  trees. Three things are not proved. First, the representability assumption in Proposition A2(c) is a real restriction: with two trees splitting on different variables the incidence matrix has rank three on four cells, so a discrepancy with an interaction on those cells is not representable, and the fitted cell means then converge to a weighted projection of the truth onto the additive span of the partitions rather than to the truth. Second, we have no statement about the posterior over the partitions themselves, hence none about  $\mathcal{B}_c(x)$  or  $B(x)$  under the full posterior; posterior concentration for a single standard BART ensemble is known [20], but the two-ensemble decomposition with a spike-and-slab discrepancy prior is outside the available theory, and the empirical separation of regions by  $\mathcal{B}_c$  (verification report, check 6c) is the evidence we have. Third, the leaf-level results treat  $f$  as fixed; with  $f$  estimated, the residual in a  $g$  leaf carries the error of  $f$  at the RCT profiles, which is  $O_p$  of the RWD posterior standard deviation of  $f$  there, and enters  $d$  as an additive perturbation. On the estimand, this perturbation is what  $V_\mu^f$  of Web Appendix D measures.

## Web Appendix D: Monotone, well-posed ESS calibration (T3)

The prior effective sample size of the commensurate component is defined at the level of the estimand as

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[ \frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right], \quad (\text{A34})$$

where  $V_\mu^f = \text{Var}(N^{-1} \sum_{i \leq N} f(x_i) | \text{RWD}) > 0$  is the Stage-1 posterior variance of the standardized RWD surface at the RCT profiles,  $c_g \in (0, 1]$  is the leaf-sharing constant of the  $g$ -tree prior, and the prior-averaged version is  $\text{ESS}(w, s_0^2) = \sum_{k=0}^{H_g} \binom{H_g}{k} w^k (1-w)^{H_g-k} \sigma_1^2 \mathbb{E}[\{V_\mu^f + c_g(k\tau_0^2 + (H_g-k)\tau_1^2)\}^{-1}]$ . In the calibration,  $V_\mu^f$  is replaced by a block statistic  $V^{(b)}$  from Stage 1, and the selection is the

Pr-rule

$$\hat{s}_{0*}^2 = \inf\{s_0^2 \in \mathcal{S} : \widehat{G}(s_0^2) \geq q\}, \quad \widehat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}, \quad (\text{A35})$$

with  $\text{ESS}_0(V; s_0^2)$  denoting (A34) at  $V_\mu^f = V$ . Its population counterpart is

$$G(s_0^2) = P_{V \sim \mathbb{P}}\{\text{ESS}_0(V; s_0^2) \leq N_{\text{target}}\},$$

where  $\mathbb{P}$  is the stationary law of the block statistic.

**Theorem A4** (Monotone, well-posed ESS calibration). *Fix  $\sigma_1^2 > 0$ ,  $V > 0$ ,  $c = c_g H_g > 0$ ,  $\nu_0 > 0$ , and write  $s = s_0^2$ .*

(a) (Monotonicity and limits.)  *$s \mapsto \text{ESS}_0(V; s)$  is continuous, differentiable, convex, and strictly decreasing on  $(0, \infty)$ , with  $\text{ESS}_0(V; s) \uparrow \sigma_1^2/V$  as  $s \downarrow 0$  and  $\text{ESS}_0(V; s) \downarrow 0$  as  $s \uparrow \infty$ ; it is a bijection of  $(0, \infty)$  onto  $(0, \sigma_1^2/V)$ . If the prior on  $\tau_0^2$  is truncated to  $(0, \tau_1^2)$  as in (A13),  $\text{ESS}_0(V; s)$  remains continuous and strictly decreasing with limit  $\sigma_1^2/V$  as  $s \downarrow 0$ , but its limit as  $s \uparrow \infty$  is  $\sigma_1^2/(V + c\tau_1^2) > 0$ , so it is a bijection onto  $(\sigma_1^2/(V + c\tau_1^2), \sigma_1^2/V)$ ; convexity and differentiability are stated for the untruncated case only.  $\text{ESS}(w, s)$  is strictly decreasing in  $s$  and in  $\tau_1^2$  (for  $w < 1$ ), and, when the prior on  $\tau_0^2$  is truncated to  $(0, \tau_1^2)$ , strictly increasing in  $w$ .*

(b) (Well-posed selection.) *For each  $V$  the sublevel set  $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\}$  is the half-line  $[s^\dagger(V), \infty)$ , where  $s^\dagger(V)$  is the unique root of  $\text{ESS}_0(V; s) = N_{\text{target}}$  if  $N_{\text{target}} < \sigma_1^2/V$  and  $s^\dagger(V) = 0$  otherwise;  $s^\dagger$  is continuous and strictly decreasing in  $V$  on  $\{V < \sigma_1^2/N_{\text{target}}\}$ . Hence  $G(s) = P_{\mathbb{P}}\{s^\dagger(V) \leq s\}$  is the distribution function of  $s^\dagger(V)$ , non-decreasing and right-continuous, and  $\{s : G(s) \geq q\} = [s_*, \infty)$  is a nonempty closed half-line whose infimum  $s_* = \inf\{s \geq 0 : G(s) \geq q\}$  is attained and unique. The selected scale is strictly positive,  $s_* > 0$ , if and only if*

$$P_{\mathbb{P}}(N_{\text{target}} < \sigma_1^2/V) > 1 - q, \quad (\text{A36})$$

*that is, the target lies below the ceiling  $\sup_s \text{ESS}_0(V; s) = \sigma_1^2/V$  for more than a  $1 - q$  fraction of the Stage-1 blocks; when (A36) fails,  $s_* = 0$ , the point-mass limit (BART-CP), which the*

implementation reports as capped. Under the truncated prior (A13) a finite root  $s^\dagger(V)$  exists only if in addition  $N_{\text{target}} > \sigma_1^2/(V + c\tau_1^2)$ , the floor of part (a); below the floor the sublevel set is empty and no scale meets the target.

(c) (Monte Carlo consistency.) Let the Stage-1 draws form a stationary, aperiodic, Harris ergodic Markov chain, and let  $V_{\text{raw}}^{(b)}$  be the sample variance of  $N^{-1} \sum_i f(x_i)$  over the  $b$ -th block of  $m$  consecutive draws,  $b = 1, \dots, B$ , with  $\mathbb{P}_m$  the stationary law of  $V_{\text{raw}}^{(b)}$  and  $G_m$  the corresponding population function. The implementation uses  $V^{(b)} = \gamma_B V_{\text{raw}}^{(b)}$  with  $\gamma_B = \widehat{V}_{\text{whole}}/(B^{-1} \sum_b V_{\text{raw}}^{(b)})$ , a finite- $m$  mean-matching correction that sets the block mean equal to the whole-chain variance and nothing more. Two regimes apply.

- ( $m \rightarrow \infty$  at fixed  $B$ , the regime of the code, which fixes  $B = 20$  blocks of length  $m = n_{\text{draw}}/20$ .) Under geometric ergodicity every within-block variance converges almost surely to the stationary variance  $V_\mu^f$ ,  $\gamma_B \rightarrow 1$ , every  $V^{(b)} \rightarrow V_\mu^f$ , and  $\widehat{G}(s) \rightarrow \mathbf{1}\{s^\dagger(V_\mu^f) \leq s\}$  for  $s \neq s^\dagger(V_\mu^f)$ ; the Pr-rule then reduces to the root of  $\text{ESS}_0(V_\mu^f; s) = N_{\text{target}}$ , and  $\hat{s}_{0\star}^2$  converges to the smallest grid value at or above that root.
- ( $B \rightarrow \infty$  at fixed  $m$ .) For each  $s$ ,  $\widehat{G}(s) \rightarrow G_m(s)$  almost surely; if  $G_m$  is continuous, the convergence is uniform in  $s$ . On a fixed finite grid  $\mathcal{S}$  with  $G_m(s) \neq q$  at every grid point,  $\hat{s}_{0\star}^2 = \inf\{s \in \mathcal{S} : G_m(s) \geq q\}$  for all  $B$  large enough, almost surely. Under geometric ergodicity,  $\sqrt{B}\{\widehat{G}(s) - G_m(s)\}$  is asymptotically normal, with long-run variance given by the Markov-chain central limit theorem of Theorem 3(c) of the paper for the unscaled indicator plus a delta-method term for the random  $\gamma_B$ , and the  $O_p(B^{-1/2})$  rate of that theorem follows.

(d) (Anchor.) Condition on a single  $f$  tree ( $H_f = 1$ ) with a fixed partition into leaves  $\ell$ , RWD counts  $n_{2\ell}$ , and  $\sigma_2^2$  fixed, and let  $a_\ell = N_\ell/N$  be the RCT profile proportions and  $\pi_\ell = n_{2\ell}/n_2$  the RWD ones. Then

$$V_\mu^f = \sum_\ell a_\ell^2 \frac{\lambda_f^2 \sigma_2^2 / n_{2\ell}}{\lambda_f^2 + \sigma_2^2 / n_{2\ell}} \leq \frac{\sigma_2^2}{n_2} \sum_\ell \frac{a_\ell^2}{\pi_\ell}, \quad \lim_{s \downarrow 0} \text{ESS}_0(s) = \frac{\sigma_1^2}{V_\mu^f} \geq \frac{n_2 \sigma_1^2 / \sigma_2^2}{\sum_\ell a_\ell^2 / \pi_\ell},$$

with equality in the first inequality as  $\lambda_f^2 \rightarrow \infty$ . Since  $\sum_\ell a_\ell^2 / \pi_\ell \geq 1$  with equality if and only if  $a = \pi$ , matched leaf proportions give  $\lim_{s \downarrow 0} \text{ESS}_0(s) = n_2 \sigma_1^2 / \sigma_2^2$  in the flat-prior limit, and any

*covariate shift ( $a \neq \pi$ ) lowers the ceiling.*

*Proof.* (a) Write  $\tau_0^2 = \nu_0 s / Q$  with  $Q \sim \chi_{\nu_0}^2$ , so that  $\text{ESS}_0(V; s) = \sigma_1^2 \mathbb{E}[\psi(s, Q)]$  with  $\psi(s, Q) = \{V + c\nu_0 s / Q\}^{-1}$ . For every  $Q > 0$ ,  $s \mapsto \psi(s, Q)$  is continuous, strictly decreasing, convex, and bounded by  $1/V$ ; dominated convergence (dominating constant  $1/V$ ) gives continuity of the expectation, and strict monotonicity and convexity are preserved by integration against a probability measure. Differentiability:  $\partial_s \psi = -c\nu_0 Q^{-1} \{V + c\nu_0 s / Q\}^{-2}$ , and  $|\partial_s \psi| \leq 1/(4Vs)$  because  $(V + x)^2 \geq 4Vx$  for  $x \geq 0$ , so differentiation under the integral is justified on compact subsets of  $(0, \infty)$ . As  $s \downarrow 0$ ,  $\psi \uparrow 1/V$  pointwise, and as  $s \uparrow \infty$ ,  $\psi \downarrow 0$  pointwise; dominated convergence gives the limits, and a continuous strictly decreasing function with these limits is a bijection onto  $(0, \sigma_1^2/V)$ . For the truncated prior, the scaled-Inv- $\chi^2(\nu_0, s)$  family has monotone likelihood ratio in  $s$  (the density ratio  $p_{s'}(t)/p_s(t) \propto \exp\{-\nu_0(s' - s)/(2t)\}$  is increasing in  $t$  for  $s' > s$ ), truncation to a fixed interval preserves monotone likelihood ratio, monotone likelihood ratio implies stochastic ordering, and  $t \mapsto (V + ct)^{-1}$  is strictly decreasing and bounded, so the expectation is strictly decreasing and continuous in  $s$  by the same dominated-convergence argument. For the limits under truncation, the truncated density is proportional to  $t^{-\nu_0/2-1} \exp\{-\nu_0 s/(2t)\}$  on  $(0, \tau_1^2)$ : as  $s \downarrow 0$  the truncated law converges weakly to the point mass at 0 (the untruncated law does, and the truncation probability tends to one), and as  $s \uparrow \infty$  it converges weakly to the point mass at  $\tau_1^2$ , because for any  $\epsilon > 0$  the ratio of the density at  $t < \tau_1^2 - \epsilon$  to the density at  $t \in (\tau_1^2 - \epsilon/2, \tau_1^2)$  is, for  $s \geq 1$ , at most  $K_\epsilon \exp\{-\nu_0 s \epsilon / (8\tau_1^4)\} \rightarrow 0$  with  $K_\epsilon$  independent of  $s$  (split the exponent in two halves; one half absorbs the power factor uniformly in  $t$ , the other is at least  $\nu_0 s \epsilon / (8\tau_1^4)$ ); bounded continuity of  $t \mapsto (V + ct)^{-1}$  gives the limits  $\sigma_1^2/V$  and  $\sigma_1^2/(V + c\tau_1^2)$ . The scale representation  $\tau_0^2 = \nu_0 s / Q$  is not available under truncation, and we do not claim convexity or differentiability there. For  $\text{ESS}(w, s)$ : each summand with  $k \geq 1$  is strictly decreasing in  $s$  by the argument above and the  $k = 0$  term is constant, so the sum is strictly decreasing when  $w > 0$ ; each summand with  $k < H_g$  is strictly decreasing in  $\tau_1^2$ ; and with  $\tau_0^2 < \tau_1^2$  almost surely,  $\varphi(k) = \sigma_1^2 \mathbb{E}[\{V + c(k\tau_0^2 + (H_g - k)\tau_1^2)\}^{-1}]$  is strictly increasing in  $k$ , so  $\text{ESS}(w, s) = \mathbb{E}_{K \sim \text{Bin}(H_g, w)}[\varphi(K)]$  is strictly increasing in  $w$  because the binomial family is stochastically increasing in  $w$ .

(b) By (a), for  $N_{\text{target}} < \sigma_1^2/V$  the equation  $\text{ESS}_0(V; s) = N_{\text{target}}$  has exactly one root  $s^\dagger(V) \in (0, \infty)$  and the sublevel set is  $[s^\dagger(V), \infty)$ ; for  $N_{\text{target}} \geq \sigma_1^2/V$  every  $s > 0$  satisfies  $\text{ESS}_0(V; s) <$

$\sigma_1^2/V \leq N_{\text{target}}$ , and we set  $s^\dagger(V) = 0$ . The map  $(V, s) \mapsto \text{ESS}_0(V; s)$  is continuous and strictly decreasing in each argument, so  $s^\dagger$  is continuous and strictly decreasing on  $\{V < \sigma_1^2/N_{\text{target}}\}$  (implicit function theorem for a strictly monotone continuous function, or directly:  $V < V'$  gives  $\text{ESS}_0(V'; s^\dagger(V)) < N_{\text{target}}$ , hence  $s^\dagger(V') < s^\dagger(V)$ ). Then  $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\} = \{s : s^\dagger(V) \leq s\}$ , so  $G$  is the distribution function of the random variable  $s^\dagger(V)$ ,  $V \sim \mathbb{P}$ , hence non-decreasing and right-continuous with  $G(s) \rightarrow 1$  as  $s \rightarrow \infty$  (since  $s^\dagger(V) < \infty$  for every  $V$ ). Thus  $\{G \geq q\}$  is a nonempty closed half-line and its infimum is attained and unique. Finally  $s_\star > 0$  if and only if  $G(0) < q$ ;  $G(0) = P\{s^\dagger(V) = 0\} = P\{N_{\text{target}} \geq \sigma_1^2/V\}$ , which is (A36). This is the point at which the argument differs from Theorem 3(b) of the paper: there the leaf ELIR diverges as  $s \rightarrow 0$ , so feasibility was automatic, while here the ceiling  $\sigma_1^2/V$  is finite and feasibility is a condition on the data.

(c) *Regime  $m \rightarrow \infty$ .* A geometrically ergodic chain with a stationary start satisfies a strong law for the bounded-moment functional  $N^{-1} \sum_i f^{(t)}(x_i)$  and its square, so the sample variance over a block of length  $m$  converges almost surely to the stationary variance  $V_\mu^f$  as  $m \rightarrow \infty$ ; the whole-chain variance converges to the same limit, hence  $\gamma_B \rightarrow 1$  and  $V^{(b)} \rightarrow V_\mu^f$  for each of the  $B$  blocks. By continuity of  $s^\dagger$  (part (b)),  $s^\dagger(V^{(b)}) \rightarrow s^\dagger(V_\mu^f)$ , so for  $s \neq s^\dagger(V_\mu^f)$  every indicator converges and  $\widehat{G}(s) \rightarrow \mathbf{1}\{s^\dagger(V_\mu^f) \leq s\}$ ; on a grid avoiding  $s^\dagger(V_\mu^f)$  the selected value is eventually the smallest grid point above the root.

*Regime  $B \rightarrow \infty$ .* The chain being aperiodic and Harris ergodic, the  $m$ -step chain is Harris ergodic (aperiodicity is what excludes the periodic case, in which the  $m$ -block sequence can fail to be ergodic), and the raw block statistics  $(V_{\text{raw}}^{(b)})_{b \geq 1}$ , a fixed measurable function of the non-overlapping  $m$ -blocks, form a stationary ergodic sequence with marginal law  $\mathbb{P}_m$ . The indicator  $\mathbf{1}\{s^\dagger(V_{\text{raw}}^{(b)}) \leq s\}$  is bounded and measurable, so the ergodic theorem gives pointwise almost-sure convergence of the unscaled empirical function to  $G_m(s)$ . The rescaling factor  $\gamma_B$  is a ratio of two ergodic averages and converges almost surely to a constant  $\gamma$  (equal to the ratio of the stationary variance to the mean within-block variance); when the distribution function of  $\gamma V_{\text{raw}}$  is continuous, the empirical distribution function of  $\gamma_B V_{\text{raw}}^{(b)}$  converges to it uniformly, by the Glivenko–Cantelli argument (monotonicity and pointwise convergence on a countable dense set) applied at  $\gamma_B$  and the continuity of the limit in the scale. This gives  $\widehat{G}(s) \rightarrow G_m(s)$  pointwise, uniformly when  $G_m$  is continuous. On a finite grid with  $G_m(s) \neq q$  at every grid point,  $\widehat{G}(s)$  lies almost surely on the same

side of  $q$  as  $G_m(s)$  at every grid point for  $B$  large, so the empirical and population selections coincide. For the central limit theorem, write  $\hat{G}(s) = \hat{F}_B(s; \gamma_B)$  with  $\hat{F}_B(s; \gamma) = B^{-1} \sum_b \mathbf{1}\{s^\dagger(\gamma V_{\text{raw}}^{(b)}) \leq s\}$ . At fixed  $\gamma$ ,  $\sqrt{B}\{\hat{F}_B(s; \gamma) - F(s; \gamma)\}$  satisfies the Markov-chain central limit theorem of Theorem 3(c) of the paper, whose proof uses only boundedness and measurability of the indicator;  $\sqrt{B}(\gamma_B - \gamma)$  is asymptotically normal by the same theorem applied to the two averages and the delta method for the ratio; and when  $\gamma \mapsto F(s; \gamma)$  is differentiable at  $\gamma$ , the delta method applied to  $F(s; \gamma_B) - F(s; \gamma)$  adds the term  $\partial_\gamma F(s; \gamma) \sqrt{B}(\gamma_B - \gamma)$ , so  $\sqrt{B}\{\hat{G}(s) - G_m(s)\}$  is asymptotically normal with the sum of the two contributions and their covariance as long-run variance. The  $O_p(B^{-1/2})$  rate of the selection then follows as in Theorem 3(c). The inner Monte Carlo over  $\tau_0^2$  in (A34) (common random numbers across the grid) adds an error that vanishes as the number of  $\tau_0^2$  draws grows, by the strong law applied to the bounded integrand.

(d) Given the partition and  $\sigma_2^2$ , the leaf means of the single  $f$  tree are independent a posteriori with  $\text{Var}(\theta_\ell | \text{RWD}) = \lambda_f^2 v_{2\ell} / (\lambda_f^2 + v_{2\ell})$ ,  $v_{2\ell} = \sigma_2^2 / n_{2\ell}$  (Lemma A1(c) with one source), and  $N^{-1} \sum_i f(x_i) = \sum_\ell a_\ell \theta_\ell$ , which gives the expression for  $V_\mu^f$ ; the inequality uses  $\lambda_f^2 v / (\lambda_f^2 + v) \leq v$  with equality in the limit, and  $v_{2\ell} = \sigma_2^2 / (n_2 \pi_\ell)$ . By the Cauchy–Schwarz inequality,  $1 = (\sum_\ell a_\ell)^2 = (\sum_\ell (a_\ell / \sqrt{\pi_\ell}) \sqrt{\pi_\ell})^2 \leq \sum_\ell a_\ell^2 / \pi_\ell \cdot \sum_\ell \pi_\ell = \sum_\ell a_\ell^2 / \pi_\ell$ , with equality iff  $a_\ell / \sqrt{\pi_\ell} \propto \sqrt{\pi_\ell}$ , that is  $a = \pi$ . The limit of  $\text{ESS}_0$  is (a).  $\square$

**Remark A7** (Units and the ceiling). The ceiling  $\sigma_1^2 / V_\mu^f$  is the ELIR of the RWD-informed prior for  $\mu$  before any discrepancy is added [18], and it is in RCT units only if  $\sigma_1^2$  is the RCT residual variance; the verification report found the calibration using the RWD variance, which inflates the ceiling by  $\sigma_2^2 / \sigma_1^2$ . Under covariate shift the ceiling can be far below the RWD sample size (the report gives 38 against  $n_2 = 300$  in Scenario 2), and then (A36) fails for  $N_{\text{target}} \in \{75, 100\}$ ; Theorem A4(b) says this is a property of the target, not of the rule. The regularization of  $f$  adds  $\sigma_1^2 / (\Lambda_f^2 c_f)$  to the ceiling in the notation of the specification; this is negligible at  $H_f = 50$  and is not subtracted.

**Remark 4** (Calibration implementation). The calibration has two stages, of which Stage 1 fits standard BART to the RWD alone and records, for each posterior draw, the standardized surface  $N^{-1} \sum_{i \leq N} f^{(b)}(x_i)$  at the RCT profiles and the residual variance; the draws are cut into  $B$  consecutive blocks, and each block gives one value  $V^{(b)}$  of the estimand variance, rescaled so that the block mean equals the whole-chain variance  $V_\mu^f$ . The RCT residual variance  $\sigma_1^2$  is the posterior mean

from a standard BART fitted to the RCT controls alone, which puts  $\text{ESS}_0$  in RCT units; before RCT outcomes exist,  $\sigma_2^2$  or a design value is substituted. Stage 2 evaluates (6) on a grid  $\mathcal{S}$  of candidate scales for each block and selects by the Pr-rule (7) with  $q = 0.95$ : the most informative spike whose prior ESS respects the target in at least a fraction  $q$  of the Stage-1 blocks. Stage 1 does not depend on  $s_0^2$ , so the same draws serve any target, grid, or level  $q$ . When  $N_{\text{target}}$  is at or above the ceiling the rule returns the smallest grid value, the point-mass limit that is BART-CP, and the target is flagged as capped. In the two-arm analysis  $\tau_0^2$  is then updated from the spike leaves, so the calibrated prior is informative but not fixed, and we report both the prior  $\text{ESS}_0$  and the realized ESS,  $\mathbb{E}_{\text{post}}[\sigma_1^2 / \{V_\mu^f + c_g \sum_h \tau_{z_h}^2\}]$  with the indicators and  $\tau_0^2$  drawn from the posterior. A pointwise version,  $\text{ESS}(x) = \sigma_1^2 / \{V_f(x) + H_g \tau_0^2\}$  with  $V_f(x)$  the RWD posterior variance of  $f$  at  $x$ , gives the local ESS map of Section 4.

## Web Appendix E: Identifiability of $f$ and $g$

**Proposition A3** (Population identification). *Let  $\mathcal{X}_1$  and  $\mathcal{X}_2$  be the supports of the covariate distributions of the RCT controls and of the RWD, and suppose the regression functions  $x \mapsto \mathbb{E}[y | x, D = 0]$  on  $\mathcal{X}_2$  and  $x \mapsto \mathbb{E}[y | x, D = 1]$  on  $\mathcal{X}_1$  are known (the infinite-data limit). Under model (A11):*

- (a)  *$f$  is identified on  $\mathcal{X}_2$  and  $f + g$  is identified on  $\mathcal{X}_1$ .*
- (b)  *$g$  is identified on  $\mathcal{X}_1 \cap \mathcal{X}_2$ , as  $g = (f + g) - f$ .*
- (c) *On  $\mathcal{X}_1 \setminus \mathcal{X}_2$  only the sum  $f + g$  is identified; the split between  $f$  and  $g$  is determined by the prior, with  $f$  the BART extrapolation of the RWD surface and  $g$  the remainder.*
- (d) *On  $\mathcal{X}_2 \setminus \mathcal{X}_1$ ,  $f$  is identified and  $g$  is not constrained by the data at all: its law there is the prior (A12) propagated through the tree prior, with hyperparameters  $(\tau_0^2, w)$  learned from the leaves in  $\mathcal{X}_1$ . This region does not enter the RCT-standardized estimand  $\mu$ , whose profiles lie in the RCT support.*

*The single-arm case is  $\mathcal{X}_1 = \emptyset$ :  $f$  is the RWD surface,  $g$  is a prior draw everywhere, and  $f_1 = f + g$  is the RWD surface plus a zero-mean discrepancy process of pointwise variance  $H_g\{w\tau_0^2 + (1-w)\tau_1^2\}$  (at a profile covered by  $H_g$  distinct leaves).*

*Proof.* (a) is the model:  $\mathbb{E}[y | x, D = 0] = f(x)$  on  $\mathcal{X}_2$  and  $\mathbb{E}[y | x, D = 1] = f(x) + g(x)$  on  $\mathcal{X}_1$ . (b), (c), (d) are immediate from (a) and from the fact that no other moment of the data involves  $f$  or  $g$ . For the single-arm variance,  $g(x) = \sum_h \theta_{h, \ell_h(x)}$  is a sum of  $H_g$  independent draws from (A12), each with variance  $w\tau_0^2 + (1 - w)\tau_1^2$ .  $\square$

**Remark A8** (Finite samples). In finite samples the statement is leaf-wise. A  $g$  leaf with no RCT controls has  $M_g = 1$  (Lemma A1(b)) and its  $(z, \theta)$  is drawn from the prior at the current  $(\tau_0^2, w)$ ; a  $g$  leaf that straddles the RCT support extends the RCT-informed discrepancy as a constant into the region without RCT controls, which is a piecewise-constant extrapolation and not a prior draw. On  $\mathcal{X}_1 \cap \mathcal{X}_2$  the separation of  $f$  and  $g$  rests on the two priors:  $f$  is pinned by the RWD, whose observations do not involve  $g$ , and the spike shrinks  $g$  toward zero, so a shift common to both sources is attributed to  $f$  and a shift of the RCT controls alone to  $g$ . The mixture (A12) is identifiable as a two-component scale mixture as long as  $\tau_0^2 \neq \tau_1^2$ , and Assumption A1 fixes which component is the spike, so the label of  $z$  is well defined.

## Web Appendix F: What is dropped from the previous version

The previous supplement contained, besides the results replaced above, four items that supported the moment-matched Student- $t$  surrogate for the tree moves: Lemma S2.1 (the exact node marginal as a one-dimensional  $\tau^2$  integral with no closed form), Definition S2.2 (the moment-matched Student- $t$  node marginal), Assumption S2.4 (finite tree space and bounded leaf statistics), and Theorem 2 (the bound on the surrogate’s log-error and the total-variation perturbation of the stationary law). All four are dropped, for one reason: the leaf marginals of the present model are exact (Lemma A1), so the tree kernels target the exact posterior (Proposition A1) and there is no approximation whose error would need to be bounded. The compactness assumption S2.4 was needed only to make the surrogate’s worst-case error finite and to invoke uniform ergodicity on a finite tree space; neither is needed for a kernel with the exact acceptance ratio, whose invariance holds on the full state space. The old Lemma S2.1 explains why the previous model needed a surrogate (the leaf-specific  $\tau^2$  enters additively in  $v_g + \tau^2$  and is integrated inside the leaf), and Remark A1 records why the present one does not (the scale is global and conditioned on; locality is carried by the indicator). Theorem 1 of the paper (location invariance of the exchangeable prior) is replaced by Theorem A1, whose

Remark S1.2 survives as the  $w = 1$  contrast in Theorem A1(e) and Corollary A1(d). Theorem 3 of the paper is replaced by Theorem A4; parts (b) and (c) of its proof carry over with the changes stated there, and Lemma S3.x (monotonicity of the data-informed predictive ELIR) is no longer needed because the calibration target (A34) has the closed monotone form of Theorem A4(a).

## Web Appendix G: Related work, methodological remarks, and simulation and application details

### G.0 Related work: covariate-adaptive borrowing

In addition to the study-level priors and tree methods discussed in Section 1.1, several approaches allow borrowing to depend on covariates. The propensity-score-integrated power prior [25] and composite likelihood [4] stratify the pooled patients on the estimated probability of belonging to the trial and give each stratum its own discount, so borrowing is constant within strata fixed by a one-dimensional summary before the outcomes are seen. The covariate-adaptive hybrid borrowing method of Jin et al. [13] places a commensurate prior on the historical control mean function with a covariate-dependent precision fitted by kernel local regression and defines a covariate-conditional effective sample size; it is the closest parametric relative of our borrowing map, with locality from a kernel bandwidth and a single covariate-adjusted mean model. Multisource exchangeability models [14] average over the subsets of sources deemed exchangeable with the current study, and Kotalik et al. [15] extend the averaging to treatment effects that vary with covariates. The latent exchangeability prior [1] attaches to each historical patient a latent class, so that a subset of patients is borrowed from, with membership driven by the outcome model. Elastic priors [12] set the prior precision through a global congruence statistic and are not local. SAM-HC [2] and Chandra et al. [3] cluster trial and external patients on the covariates with shared- or common-atoms mixtures and borrow only within clusters both sources occupy; their borrowing regions are determined before fitting the outcome model. In LRC-BART, these regions are learned through the outcome-model partition.

### G.1 Data-generating constants

We choose the intercept  $\alpha$  in (10) to maintain the outcome location after including the bounded bump term:  $\alpha = 0.42$  under Sc1, Sc4 and Sc5 and  $\alpha = -0.08$  under Sc2 and Sc3 for Gaussian outcomes, and  $\alpha = -0.20$  and  $-0.70$  on the log-time scale for survival outcomes. Under Sc2 and Sc3 the source is assigned by  $P(D_i = 1 | \mathbf{W}_i) = \text{logit}^{-1}\{(\ell_i - q_{0.5}(\ell))/0.5\}$  with  $\ell_i = -1.20(W_{i5} + W_{i6})$ , where  $\mathbf{W}_i = (X_{i5}, X_{i6})$  under Sc2 and  $(U_{i1}, U_{i2})$  under Sc3; the median centring and the temperature

0.5 give the two sources equal size on average. Under Sc3 the unmeasured confounders are  $U_j \sim \mathcal{N}(\mu_{j+4}, 1)$  and the observed proxies are  $X_{j+4} = \rho \tilde{U}_j + \sqrt{1 - \rho^2} \epsilon_j$  with  $\tilde{U}_j$  the standardized  $U_j$  and  $\epsilon_j \sim \mathcal{N}(0, 1)$ , so that  $\mathbf{U}$  drives both the assignment and the outcome and  $(X_5, X_6)$  carry information about it only through  $\rho$ . Under Sc4 and Sc5 the RWD mean is  $f_0(\mathbf{X}) + \delta_2 \mathbf{1}\{\mathbf{X} \notin \mathcal{R}\}$  with the RCT mean unchanged, so the true local discrepancy  $g = f_1 - f$  is  $-\delta_2$  outside  $\mathcal{R}$  and 0 inside it; the random-number stream is arranged so that Sc4 with  $\delta_2 = 0$  reproduces Sc1 replicate by replicate. Each replicate draws 300 RCT subjects and 300 RWD controls afresh. We compute the true estimands by standardizing to each replicate’s RCT profiles: the Gaussian ATE is  $\delta = 1$  exactly, and for survival outcomes the population median of the standardized log-normal mixture is obtained by numerical root-finding for each arm and the RMST at  $t^* = 3$  is the average over profiles of the closed-form log-normal RMST, both with  $\sigma_1$  in the control arm.

## G.2 Implementation of the comparators

We implement the comparators in Section 3.1 as follows. LM-NP, LM-CP and LM-PP are Bayesian linear regressions fitted to the RCT controls alone, to the pooled controls, and to the pooled controls with the source indicator as a covariate, and AFT-NP, AFT-CP and AFT-PP are their log-normal AFT counterparts; BART-NP, BART-CP and BART-PP are standard BART and AFT-BART [5, 24] in the same three configurations, BART-PP being the implicit borrowing of Zhou and Ji [27]. PSCL is the propensity-score composite likelihood of Chen et al. [4] with a borrowed size of 100; MAP is the meta-analytic predictive prior for the control mean, without covariate adjustment, with its heterogeneity scale calibrated to the prior ESS target; HierLM and HierAFT give the two sources distinct coefficients drawn from common means with  $\tau^2 \sim \text{Inv-Gamma}(1.5, 0.75)$ . We also include two nested LRC-BART models: C-BART fixes  $w = 1$ , the commensurate model with a single learned scale and no slab, and LRC-BART ( $w = 0$ ) fixes every leaf in the slab, the free-offset model. We implement the analysis component of covariate-adaptive hybrid borrowing (CAHB) from Jin et al. [13]. Its kernel-local control mean has a commensurate prior centred on a historical mean function, with covariate-dependent precision. Local precision estimates are thresholded and projected onto an  $L_1$  ball. We retain the fixed allocation of our simulation; the sequential allocation component of CAHB is not used.

Five transfer choices define our primary CAHB analysis. We estimate the historical mean by

Nadaraya–Watson regression on the RWD with the same kernel used for the current control mean, treating that estimate as known. We select four kernel covariates by their variable-inclusion counts in an RWD-only BART fit. We enlarge Scott’s bandwidth until the median local control sample size is 20. We rescale the precision constants to  $\gamma^2 = 3 \times 0.11/\hat{\phi}^2$  and  $\lambda_2 = 300 \log(n) \times 0.11/\hat{\phi}^2$ , where  $\hat{\phi}^2$  is the residual variance of the current-control no-borrowing smoother; the precision threshold uses its 10% quantile. Finally, we preserve the profile-specific normal marginals and induce dependence across posterior draws through the smoother weights before standardizing to the trial profiles. This last choice affects posterior SD, coverage and power, but not the point estimates or fitted surfaces. The treated mean is fitted to trial treated subjects with a flat prior. Precision estimation and projection use all trial profiles.

CAHB-p10 uses all ten covariates with the other primary choices unchanged. CAHB-lit uses historical linear regression, all ten covariates, Scott’s bandwidth without enlargement,  $\gamma^2 = 3$ ,  $\lambda_2 = 300 \log(n)$  and independent profile draws. The primary choices were assessed on diagnostic seeds 2027, 2050 and 2101, which overlap the study replicates. All variants use 1,000 draws,  $R = 200$  and data seed  $2026 + r$  for replicate  $r$ ; code is in `01-code/R/cahb.R` and the run configuration in `02-validation/cahb/run-cahb.R`. The comparison therefore assesses our transfer of the analysis, including its interval construction.

For CAHB, the local summary is the historical precision share  $1 - 1/R(x)$ , where  $R(x)$  is its information ratio, and its effective size is the sum of  $R(x) - 1$  over RCT controls. These are distinct from  $\mathcal{B}_c$  and the estimand-level ESS of LRC-BART. The literal variant has Sc1 posterior SD 12.226 and RMSE 0.950, compared with 0.542 and 0.262 for CAHB-p10 and 0.410 and 0.258 for primary CAHB. The local kernel fits are unstable or oversmoothed in this design; these results support examining tree partitions for locality, subject to the limitations of the transfer.

| Scenario | Transfer | Bias   | SD     | RMSE  | Share, $\mathcal{R}$ | Share, $\mathcal{R}^c$ |
|----------|----------|--------|--------|-------|----------------------|------------------------|
| Sc1      | CAHB     | -0.003 | 0.410  | 0.258 | 0.52                 | 0.51                   |
| Sc1      | CAHB-p10 | -0.018 | 0.542  | 0.262 | 0.50                 | 0.50                   |
| Sc1      | CAHB-lit | -0.087 | 12.226 | 0.950 | 0.01                 | 0.01                   |
| Sc4 (1)  | CAHB     | -0.208 | 0.413  | 0.333 | 0.53                 | 0.48                   |
| Sc4 (1)  | CAHB-p10 | -0.260 | 0.545  | 0.380 | 0.50                 | 0.49                   |
| Sc4 (1)  | CAHB-lit | -0.088 | 12.207 | 0.948 | 0.01                 | 0.01                   |
| Sc4 (2)  | CAHB     | -0.333 | 0.422  | 0.453 | 0.49                 | 0.41                   |
| Sc4 (2)  | CAHB-p10 | -0.265 | 0.585  | 0.533 | 0.31                 | 0.31                   |
| Sc4 (2)  | CAHB-lit | -0.089 | 12.187 | 0.947 | 0.00                 | 0.00                   |
| Sc5 (1)  | CAHB     | -0.437 | 0.419  | 0.546 | —                    | 0.47                   |
| Sc5 (1)  | CAHB-p10 | -0.290 | 0.581  | 0.549 | —                    | 0.32                   |
| Sc5 (1)  | CAHB-lit | -0.089 | 12.198 | 0.948 | —                    | 0.01                   |
| Sc5 (2)  | CAHB     | -0.067 | 0.479  | 0.400 | —                    | 0.03                   |
| Sc5 (2)  | CAHB-p10 | -0.028 | 0.648  | 0.340 | —                    | 0.00                   |
| Sc5 (2)  | CAHB-lit | -0.090 | 12.165 | 0.947 | —                    | 0.00                   |

The linear models use  $\alpha \sim \mathcal{N}(0, 5^2)$  and  $\beta \sim \mathcal{N}(0, 2^2 I)$ , BART the default prior with  $H = 50$  trees, and the residual variance of these non-CAHB methods the BART-style prior with  $\nu = 3$  and  $q = 0.9$ . The implementation details are given below. The non-CAHB samplers run 1,000 burn-in and 1,000 retained iterations. The linear, AFT and hierarchical models are fitted by exact conjugate Gibbs samplers, with right censoring handled by data augmentation of the log survival times; the HierLM sampler was checked against a Hamiltonian Monte Carlo implementation in Stan on the same data. HierLM and HierAFT use one hypermean per coefficient,  $\mu_{\beta j} \sim \mathcal{N}(0, 10^2)$ , and the group residual variances carry the BART-style prior with the scale from a group-specific least-squares or parametric AFT fit. AFT-NP, AFT-CP and AFT-PP use the same coefficient priors as the linear models. The MAP prior is built from the RWD sample mean and, with a fixed Half-Normal(0, 1) heterogeneity scale, gives an expected-local-information-ratio ESS of about 2 with a single external study of 300, so we calibrate the half-normal scale so that the ESS of the mixture-approximated MAP prior equals the target, falling back to the largest attainable ESS where the target is out of reach, with the mixture, ESS and posterior update from the RBesT package.

For PSCL, we follow Chen et al. [4]: a logistic propensity model on  $X_1$  to  $X_{10}$  over both sources, trimming of RWD subjects outside the RCT propensity range, five strata at the RCT propensity quintiles, the total borrowed size of 100 allocated across strata in proportion to the overlap coefficient of the stratum propensity distributions, composite-likelihood stratum means with

a sandwich variance, and standardization by the RCT stratum shares; its interval is Wald. BART uses the BART package with  $H = 50$ ,  $\alpha = 0.95$ ,  $\beta = 2$ ,  $k = 2$ ,  $\nu = 3$  and  $q = 0.9$ ; the treated arm always has its own ensemble on the RCT treated subjects.

The simulation summaries use the main-text definitions. Comparisons with the previous analysis require several distinctions. Previously, power was computed as the fraction of replicates with  $P(\text{ATE} > 0.5 | \text{data}) > 0.95$ , a one-sided test against 0.5, and for survival as  $P(\text{ratio} > 1 | \text{data}) > 0.95$ , although the table caption specified the 95% interval excluding the null; we use that interval-based definition. Under it, every tree model has power 1.00 at  $\delta = 1$  for Gaussian outcomes. The 90%-interval version (`pow90` in the result files) is 1.00 for the tree models and 0.90 to 0.91 for LM-NP, LM-PP and HierLM in Sc1, so neither version discriminates among the tree models.

The previous survival point estimate was the posterior median of the ratio draws; we use the posterior mean throughout, which moves the BART-NP median-ratio bias in Sc1 by about 0.05 and does not account for the remaining difference from the earlier row, which we could not reproduce from the code and the stated design.

The previous single-arm tables used a different mechanism, with a selection slope of  $-0.55$  at temperature 1.0 in place of  $-1.20$  at 0.5, different intercepts and residual scales, a dropout rate of 0.09, entry over  $(0, 0.5)$ , administrative censoring at 5.5 minus entry, an RMST horizon of 5, and a hierarchical hyperprior scale of 0.05 with covariates centred at the trial means; Section 3.4 reports the stated design.

### G.3 Configuration of the discrepancy ensemble

The configuration of  $g$  was chosen on a development run of  $R = 200$  Gaussian replicates paired with the first 200 replicates of the full study. The initial configuration,  $H_g = 10$  with  $w \sim \text{Beta}(4, 1)$ , gave a bias of 0.06 under Sc3 and the same Sc4 map as the final one. A tuning study varied  $H_g$ , the split prior, the slab scale, the leaf minimum, the prior on  $w$ , the number of  $g$  sweeps per iteration, and a warm start of  $g$  from the BART-PP source offset, and ran oracle fits with  $f$  fixed at the truth.

The configuration was informed by two observations. Underestimation of  $g$  at  $\delta_2 = 2$  persisted under every prior, under long chains, and with  $f$  known, and was not resolved by those configurations. For a moderate shift, the split-and-slab configuration incurs the tree-prior cost of one cut and the slab cost  $\log\{w/(1-w)\} + \log(\tau_1/\tau_0)$  against a likelihood gain of order  $n_\ell \delta_2^2 / (2\sigma_1^2)$ , so that fewer

trees improve the map at  $\delta_2 = 2$  and the Sc1 efficiency but raise  $\tau_1$  through the range rule and with it the moderate-shift bias. A flat prior on  $w$  reduces this slab penalty.

The final configuration,  $H_g = 5$ ,  $(\alpha_g, \beta_g) = (0.5, 3)$ , the range-rule slab, a leaf minimum of 5, one  $g$  sweep per iteration and  $w \sim \text{Beta}(1, 1)$ , lowered the Sc3 bias to 0.04 and left every other quantity within Monte Carlo error of the initial one; LRC-BART ( $H_g = 10$ ) in Web Table S1 is the initial ensemble size under the final prior on  $w$ .

The development run evaluated 25 pre-specified criteria, of which 18 pass at  $R = 500$ ; the seven that fail are the Sc1 power criterion, which is uninformative at  $\delta = 1$ , the Sc4  $\delta_2 = 1$  RMSE criterion against BART-PP, and five Sc4  $\delta_2 = 2$  criteria: the control surface RMSE in  $\mathcal{R}$  below 0.9 times BART-PP's, the all-spike map in  $\mathcal{R}$  above 0.5,  $\mathcal{B}_c$  in  $\mathcal{R}$  above 0.7, and the mean of  $g$  within 0.25 of the truth in each region, as reported in Section 3.2.

A secondary version of Sc4, Sc4' in Table 2 and Web Table S1, places the shift outside  $\mathcal{R}' = \{X_7 \leq 2\}$ , a covariate with negligible effect, so that a split of  $g$  on it is driven by the discrepancy alone. Results for the null covariate  $X_7$  are similar to those for  $X_5$ : at  $\delta_2 = 2$  the map  $\bar{\mathcal{B}}_c$  is 0.64 over the RCT control profiles in the compatible region and 0.05 outside it, and the ATE is unbiased because the errors on the two sides of the boundary have opposite signs and the estimand averages over both; at  $\delta_2 = 1$  LRC-BART (100) borrows too much outside the compatible region, with ATE bias  $-0.041$ , RMSE 0.223 against 0.207 for BART-PP, and coverage 0.89, on both  $X_5$  and  $X_7$ .

Theorem 2 collects Theorem A2(a),(b) and Theorem A3 of Web Appendix C. Part (a) describes leaf-level detection: a shift is detected only beyond a few multiples of the spike scale plus the RCT standard error of the leaf, which is why the simulation study includes shifts at and below this scale. Part (b) establishes conditional consistency of the displayed map: as the RCT controls in a region accumulate,  $\mathcal{B}_c$  converges to the truth about whether the local discrepancy is below  $c$ , irrespective of the limiting spike indicators. Part (c) is the reason we display  $\mathcal{B}_c$  and not the all-spike probability: the spike is a Gaussian component of scale  $\tau_0$ , not a point mass, so a discrepancy of that order is consistent with it and the spike indicator can remain uncertain in the limit even when the RCT data identify whether the discrepancy is below the threshold used by  $\mathcal{B}_c$ . In the two-arm sampler  $\tau_0^2$  is learned under a prior calibrated to a small scale, so the exponentially small bound in (c) is the relevant statement and abstention under a shift of several spike scales is near certain; Section 3 shows what happens at a shift near  $d_{1/2}$ . Proposition A2 extends (b) to the  $H_g$ -tree

ensemble with fixed partitions, provided the true discrepancy is representable on those partitions, with the map converging cell by cell as the RCT count in a cell grows. These results are conditional on the partitions of the  $g$  trees: we have no statement about the posterior over the partitions themselves, hence none about  $\mathcal{B}_c$  under the full posterior, since the two-ensemble decomposition with a spike-and-slab discrepancy prior lies outside the available posterior-concentration theory for BART [20]. The leaf-level results also treat  $f$  as fixed, whereas with  $f$  estimated the residual in a  $g$  leaf carries the RWD posterior error of  $f$  at the RCT profiles as an additive perturbation of  $d$ . The separation of shifted from compatible regions by  $\mathcal{B}_c$  under the full posterior is shown empirically in Section 3.

We evaluate additional surface errors over RCT control profiles and over the subsets with  $|X_5 - 2| > 0.5$ , using the first 200 paired replicates. These differ from Table 2, which uses all RCT profiles. At Sc4  $\delta_2 = 2$ , the control-profile RMSE in  $\mathcal{R}$  is 0.751 for LRC-BART and 0.704 for BART-PP, a difference of 0.047 with paired standard error 0.005. Away from the boundary it is 0.747 versus 0.723, with difference 0.024 and standard error 0.005. Boundary exclusion reduces this loss but does not remove it. At Sc4  $\delta_2 = 1$ , the far-region difference in  $\mathcal{R}$  is close to zero. Control-profile errors also place LRC-BART slightly behind BART-PP in Sc1 and Sc5, whereas the all-profile definition shows an Sc1 advantage and an Sc5 tie.

| Scenario  | Control profiles | LRC-BART | BART-PP | Difference | Paired SE |
|-----------|------------------|----------|---------|------------|-----------|
| Sc1       | $X_5 > 2$        | 0.688    | 0.682   | 0.006      | 0.005     |
| Sc1       | $X_5 \leq 2$     | 0.717    | 0.711   | 0.006      | 0.004     |
| Sc1       | $X_5 > 2.5$      | 0.701    | 0.696   | 0.005      | 0.006     |
| Sc1       | $X_5 < 1.5$      | 0.733    | 0.726   | 0.007      | 0.005     |
| Sc4 (0.5) | $X_5 > 2$        | 0.695    | 0.685   | 0.010      | 0.004     |
| Sc4 (0.5) | $X_5 \leq 2$     | 0.740    | 0.719   | 0.021      | 0.004     |
| Sc4 (0.5) | $X_5 > 2.5$      | 0.713    | 0.704   | 0.009      | 0.005     |
| Sc4 (0.5) | $X_5 < 1.5$      | 0.746    | 0.719   | 0.028      | 0.005     |
| Sc4 (1)   | $X_5 > 2$        | 0.713    | 0.705   | 0.008      | 0.004     |
| Sc4 (1)   | $X_5 \leq 2$     | 0.772    | 0.744   | 0.028      | 0.005     |
| Sc4 (1)   | $X_5 > 2.5$      | 0.731    | 0.729   | 0.002      | 0.005     |
| Sc4 (1)   | $X_5 < 1.5$      | 0.774    | 0.748   | 0.026      | 0.006     |
| Sc4 (2)   | $X_5 > 2$        | 0.751    | 0.704   | 0.047      | 0.005     |
| Sc4 (2)   | $X_5 \leq 2$     | 0.780    | 0.746   | 0.034      | 0.006     |
| Sc4 (2)   | $X_5 > 2.5$      | 0.747    | 0.723   | 0.024      | 0.005     |
| Sc4 (2)   | $X_5 < 1.5$      | 0.770    | 0.748   | 0.022      | 0.006     |
| Sc5 (2)   | $X_5 > 2$        | 0.709    | 0.688   | 0.020      | 0.004     |
| Sc5 (2)   | $X_5 \leq 2$     | 0.736    | 0.713   | 0.023      | 0.005     |
| Sc5 (2)   | $X_5 > 2.5$      | 0.722    | 0.699   | 0.023      | 0.005     |
| Sc5 (2)   | $X_5 < 1.5$      | 0.749    | 0.723   | 0.025      | 0.006     |

For the regional ESS in Table 2, we standardize the control mean to all RCT profiles in each of  $\{X_5 > 2\}$  and  $\{X_5 \leq 2\}$ . We use its RWD-only posterior variance and the posterior leaf proportions, spike indicators and scales to evaluate the realized ESS within each region. In Sc1 and Sc5 the partition is retained for comparison; it does not define a compatible region in Sc5. The prior pointwise ESS, about 2.1–2.3 across regions and scenarios, describes external support before the discrepancy is learned and does not distinguish compatible from conflicting regions.

#### G.4 Full simulation tables

Table S1: Gaussian outcomes, two-arm design, all scenarios and methods,  $R = 500$ . Columns as in Table 1, plus bias and RMSE of the RCT-standardized treated mean. MAP ( $N$ ) is the meta-analytic predictive prior calibrated to prior ESS  $N$ ; LRC-BART ( $N$ ) targets prior ESS  $N$  and LRC-BART ( $p$ ) the fraction  $p$  of the ESS ceiling; LRC-BART ( $w = 0$ ) fixes every leaf in the slab; LRC-BART ( $H_g = 10$ ) doubles the discrepancy ensemble.

| Scenario | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|----------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|          |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
| Sc1      | LM-NP                   | −0.002 | 0.341 | 0.342 | 0.94 | 0.83 | −0.001       | 0.145 | 0.001        | 0.250 |
|          | LM-CP                   | 0.009  | 0.243 | 0.236 | 0.96 | 0.99 | −0.000       | 0.145 | −0.009       | 0.175 |
|          | LM-PP                   | −0.002 | 0.347 | 0.337 | 0.94 | 0.85 | −0.000       | 0.145 | 0.001        | 0.243 |
|          | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|          | BART-CP                 | 0.008  | 0.145 | 0.142 | 0.96 | 1.00 | 0.005        | 0.109 | −0.003       | 0.096 |
|          | BART-PP                 | 0.006  | 0.207 | 0.205 | 0.94 | 1.00 | 0.005        | 0.110 | −0.001       | 0.170 |
|          | PSCL                    | 0.005  | 0.269 | 0.261 | 0.95 | 0.97 | 0.000        | 0.148 | −0.005       | 0.171 |
|          | MAP (100)               | 0.006  | 0.292 | 0.265 | 0.97 | 0.95 | 0.000        | 0.146 | −0.005       | 0.174 |
|          | MAP (75)                | 0.004  | 0.301 | 0.274 | 0.97 | 0.94 | −0.000       | 0.146 | −0.004       | 0.178 |
|          | MAP (50)                | 0.003  | 0.311 | 0.285 | 0.97 | 0.92 | −0.000       | 0.146 | −0.004       | 0.186 |
|          | HierLM                  | −0.003 | 0.338 | 0.341 | 0.93 | 0.85 | −0.001       | 0.145 | 0.003        | 0.249 |
|          | C-BART                  | 0.007  | 0.155 | 0.150 | 0.95 | 1.00 | 0.005        | 0.108 | −0.002       | 0.106 |
|          | LRC-BART ( $w = 0$ )    | 0.009  | 0.189 | 0.205 | 0.92 | 1.00 | 0.005        | 0.110 | −0.004       | 0.169 |
|          | LRC-BART (100)          | 0.009  | 0.171 | 0.172 | 0.94 | 1.00 | 0.005        | 0.109 | −0.004       | 0.134 |
|          | LRC-BART (75)           | 0.008  | 0.174 | 0.176 | 0.94 | 1.00 | 0.005        | 0.110 | −0.003       | 0.136 |
|          | LRC-BART (50)           | 0.009  | 0.177 | 0.181 | 0.94 | 1.00 | 0.005        | 0.110 | −0.004       | 0.142 |
|          | LRC-BART (0.9)          | 0.011  | 0.171 | 0.171 | 0.95 | 1.00 | 0.005        | 0.109 | −0.005       | 0.132 |
|          | LRC-BART (0.5)          | 0.009  | 0.175 | 0.178 | 0.93 | 1.00 | 0.005        | 0.109 | −0.004       | 0.139 |
|          | LRC-BART (0.25)         | 0.008  | 0.180 | 0.184 | 0.94 | 1.00 | 0.005        | 0.109 | −0.002       | 0.147 |
|          | LRC-BART ( $H_g = 10$ ) | 0.009  | 0.171 | 0.169 | 0.94 | 1.00 | 0.005        | 0.109 | −0.004       | 0.131 |
| Sc2      | LM-NP                   | −0.016 | 0.320 | 0.324 | 0.95 | 0.87 | −0.000       | 0.140 | 0.016        | 0.241 |
|          | LM-CP                   | 0.379  | 0.286 | 0.487 | 0.73 | 0.99 | 0.000        | 0.140 | −0.379       | 0.445 |
|          | LM-PP                   | 0.010  | 0.340 | 0.331 | 0.95 | 0.84 | 0.000        | 0.140 | −0.010       | 0.248 |
|          | BART-NP                 | −0.008 | 0.201 | 0.219 | 0.93 | 1.00 | 0.001        | 0.115 | 0.009        | 0.180 |
|          | BART-CP                 | 0.058  | 0.177 | 0.195 | 0.91 | 1.00 | 0.002        | 0.115 | −0.056       | 0.151 |
|          | BART-PP                 | 0.007  | 0.208 | 0.204 | 0.96 | 1.00 | 0.002        | 0.115 | −0.005       | 0.163 |
|          | PSCL                    | 0.024  | 0.321 | 0.292 | 0.97 | 0.91 | −0.005       | 0.141 | −0.029       | 0.205 |
|          | MAP (100)               | 0.332  | 0.360 | 0.491 | 0.83 | 0.97 | −0.005       | 0.149 | −0.337       | 0.424 |
|          | MAP (75)                | 0.250  | 0.363 | 0.445 | 0.89 | 0.93 | −0.006       | 0.149 | −0.256       | 0.368 |
|          | MAP (50)                | 0.177  | 0.365 | 0.412 | 0.92 | 0.88 | −0.006       | 0.149 | −0.183       | 0.324 |
|          | HierLM                  | −0.011 | 0.318 | 0.321 | 0.95 | 0.88 | −0.000       | 0.140 | 0.011        | 0.238 |
|          | C-BART                  | 0.042  | 0.168 | 0.186 | 0.91 | 1.00 | 0.001        | 0.114 | −0.041       | 0.141 |
|          | LRC-BART ( $w = 0$ )    | 0.008  | 0.190 | 0.204 | 0.94 | 1.00 | 0.002        | 0.114 | −0.005       | 0.165 |
|          | LRC-BART (100)          | 0.023  | 0.181 | 0.192 | 0.94 | 1.00 | 0.002        | 0.114 | −0.021       | 0.148 |
|          | LRC-BART (75)           | 0.022  | 0.182 | 0.192 | 0.93 | 1.00 | 0.002        | 0.114 | −0.021       | 0.150 |
|          | LRC-BART (50)           | 0.023  | 0.181 | 0.193 | 0.93 | 1.00 | 0.002        | 0.115 | −0.021       | 0.149 |
|          | LRC-BART (0.9)          | 0.019  | 0.183 | 0.193 | 0.93 | 1.00 | 0.002        | 0.114 | −0.017       | 0.151 |
|          | LRC-BART (0.5)          | 0.016  | 0.186 | 0.197 | 0.93 | 1.00 | 0.002        | 0.114 | −0.013       | 0.156 |
|          | LRC-BART (0.25)         | 0.012  | 0.188 | 0.200 | 0.94 | 1.00 | 0.002        | 0.114 | −0.010       | 0.160 |
|          | LRC-BART ( $H_g = 10$ ) | 0.022  | 0.181 | 0.190 | 0.93 | 1.00 | 0.001        | 0.114 | −0.021       | 0.147 |
|          | LM-NP                   | 0.003  | 0.359 | 0.352 | 0.95 | 0.81 | 0.015        | 0.151 | 0.011        | 0.255 |

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Web Table S1, continued

| Scenario        | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                 |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                 | LM-CP                   | 0.888  | 0.264 | 0.927 | 0.10 | 1.00 | 0.015        | 0.150 | -0.873       | 0.893 |
|                 | LM-PP                   | 0.041  | 0.353 | 0.342 | 0.95 | 0.85 | 0.015        | 0.150 | -0.026       | 0.240 |
|                 | BART-NP                 | -0.004 | 0.284 | 0.282 | 0.94 | 0.94 | 0.020        | 0.131 | 0.024        | 0.216 |
|                 | BART-CP                 | 0.846  | 0.206 | 0.871 | 0.01 | 1.00 | 0.020        | 0.131 | -0.825       | 0.839 |
|                 | BART-PP                 | 0.027  | 0.271 | 0.270 | 0.93 | 0.96 | 0.020        | 0.131 | -0.008       | 0.200 |
|                 | PSCL                    | 0.675  | 0.277 | 0.728 | 0.31 | 1.00 | 0.009        | 0.148 | -0.665       | 0.690 |
|                 | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                 | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                 | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                 | HierLM                  | 0.033  | 0.356 | 0.348 | 0.95 | 0.84 | 0.015        | 0.151 | -0.018       | 0.251 |
|                 | C-BART                  | 0.280  | 0.292 | 0.431 | 0.81 | 0.99 | 0.020        | 0.131 | -0.259       | 0.375 |
|                 | LRC-BART ( $w = 0$ )    | 0.024  | 0.272 | 0.271 | 0.93 | 0.96 | 0.020        | 0.132 | -0.003       | 0.201 |
|                 | LRC-BART (100)          | 0.042  | 0.276 | 0.278 | 0.93 | 0.96 | 0.021        | 0.131 | -0.021       | 0.209 |
|                 | LRC-BART (75)           | 0.043  | 0.276 | 0.278 | 0.93 | 0.96 | 0.021        | 0.131 | -0.022       | 0.208 |
|                 | LRC-BART (50)           | 0.045  | 0.276 | 0.278 | 0.93 | 0.97 | 0.020        | 0.131 | -0.025       | 0.208 |
|                 | LRC-BART (0.9)          | 0.039  | 0.275 | 0.276 | 0.93 | 0.96 | 0.020        | 0.132 | -0.019       | 0.206 |
|                 | LRC-BART (0.5)          | 0.042  | 0.277 | 0.277 | 0.93 | 0.96 | 0.020        | 0.131 | -0.022       | 0.208 |
|                 | LRC-BART (0.25)         | 0.047  | 0.276 | 0.279 | 0.92 | 0.96 | 0.020        | 0.131 | -0.026       | 0.209 |
|                 | LRC-BART ( $H_g = 10$ ) | 0.054  | 0.276 | 0.280 | 0.93 | 0.96 | 0.021        | 0.132 | -0.034       | 0.210 |
| Sc3, $\rho = 0$ | LM-NP                   | 0.005  | 0.366 | 0.355 | 0.96 | 0.80 | 0.016        | 0.152 | 0.011        | 0.256 |
|                 | LM-CP                   | 1.037  | 0.256 | 1.067 | 0.03 | 1.00 | 0.017        | 0.152 | -1.020       | 1.035 |
|                 | LM-PP                   | 0.043  | 0.356 | 0.346 | 0.95 | 0.85 | 0.016        | 0.152 | -0.027       | 0.241 |
|                 | BART-NP                 | -0.002 | 0.292 | 0.289 | 0.94 | 0.92 | 0.022        | 0.134 | 0.024        | 0.219 |
|                 | BART-CP                 | 0.984  | 0.202 | 1.003 | 0.00 | 1.00 | 0.022        | 0.134 | -0.962       | 0.971 |
|                 | BART-PP                 | 0.029  | 0.275 | 0.273 | 0.94 | 0.96 | 0.023        | 0.135 | -0.007       | 0.197 |
|                 | PSCL                    | 0.704  | 0.269 | 0.753 | 0.26 | 1.00 | 0.012        | 0.149 | -0.693       | 0.713 |
|                 | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                 | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                 | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                 | HierLM                  | 0.039  | 0.362 | 0.354 | 0.96 | 0.82 | 0.016        | 0.152 | -0.023       | 0.253 |
|                 | C-BART                  | 0.226  | 0.300 | 0.380 | 0.87 | 0.98 | 0.023        | 0.134 | -0.204       | 0.311 |
|                 | LRC-BART ( $w = 0$ )    | 0.023  | 0.280 | 0.275 | 0.94 | 0.96 | 0.023        | 0.134 | -0.000       | 0.201 |
|                 | LRC-BART (100)          | 0.041  | 0.284 | 0.280 | 0.94 | 0.96 | 0.023        | 0.134 | -0.018       | 0.206 |
|                 | LRC-BART (75)           | 0.042  | 0.283 | 0.281 | 0.94 | 0.96 | 0.023        | 0.134 | -0.019       | 0.207 |
|                 | LRC-BART (50)           | 0.045  | 0.283 | 0.282 | 0.94 | 0.96 | 0.023        | 0.134 | -0.022       | 0.208 |
|                 | LRC-BART (0.9)          | 0.037  | 0.282 | 0.277 | 0.94 | 0.96 | 0.023        | 0.134 | -0.015       | 0.203 |
|                 | LRC-BART (0.5)          | 0.041  | 0.282 | 0.280 | 0.94 | 0.96 | 0.023        | 0.134 | -0.018       | 0.205 |
|                 | LRC-BART (0.25)         | 0.045  | 0.283 | 0.279 | 0.94 | 0.97 | 0.023        | 0.134 | -0.022       | 0.205 |
|                 | LRC-BART ( $H_g = 10$ ) | 0.055  | 0.283 | 0.282 | 0.94 | 0.97 | 0.023        | 0.134 | -0.032       | 0.206 |
|                 | LM-NP                   | 0.008  | 0.360 | 0.350 | 0.96 | 0.80 | 0.017        | 0.151 | 0.009        | 0.252 |
|                 | LM-CP                   | 0.899  | 0.265 | 0.937 | 0.08 | 1.00 | 0.017        | 0.151 | -0.882       | 0.901 |
|                 | LM-PP                   | 0.043  | 0.353 | 0.343 | 0.95 | 0.84 | 0.017        | 0.151 | -0.027       | 0.241 |
|                 | BART-NP                 | 0.001  | 0.284 | 0.280 | 0.95 | 0.94 | 0.021        | 0.132 | 0.020        | 0.213 |
|                 | BART-CP                 | 0.847  | 0.205 | 0.872 | 0.01 | 1.00 | 0.021        | 0.133 | -0.826       | 0.838 |
|                 | BART-PP                 | 0.029  | 0.271 | 0.267 | 0.94 | 0.96 | 0.021        | 0.132 | -0.008       | 0.197 |
|                 | PSCL                    | 0.678  | 0.277 | 0.731 | 0.30 | 1.00 | 0.010        | 0.149 | -0.668       | 0.692 |

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Web Table S1, continued

| Scenario                      | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-------------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                               |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                               | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                               | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                               | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                               | HierLM                  | 0.038  | 0.356 | 0.349 | 0.96 | 0.82 | 0.017        | 0.151 | -0.021       | 0.250 |
|                               | C-BART                  | 0.277  | 0.292 | 0.430 | 0.82 | 0.99 | 0.021        | 0.133 | -0.256       | 0.371 |
|                               | LRC-BART ( $w = 0$ )    | 0.021  | 0.273 | 0.266 | 0.95 | 0.96 | 0.021        | 0.133 | -0.001       | 0.196 |
|                               | LRC-BART (100)          | 0.040  | 0.276 | 0.275 | 0.94 | 0.96 | 0.021        | 0.132 | -0.020       | 0.204 |
|                               | LRC-BART (75)           | 0.042  | 0.276 | 0.274 | 0.94 | 0.97 | 0.021        | 0.132 | -0.021       | 0.203 |
|                               | LRC-BART (50)           | 0.044  | 0.277 | 0.274 | 0.95 | 0.97 | 0.021        | 0.132 | -0.023       | 0.204 |
|                               | LRC-BART (0.9)          | 0.039  | 0.276 | 0.273 | 0.95 | 0.97 | 0.021        | 0.133 | -0.018       | 0.202 |
|                               | LRC-BART (0.5)          | 0.043  | 0.276 | 0.274 | 0.95 | 0.97 | 0.021        | 0.132 | -0.021       | 0.203 |
|                               | LRC-BART (0.25)         | 0.046  | 0.277 | 0.275 | 0.94 | 0.97 | 0.021        | 0.132 | -0.025       | 0.205 |
|                               | LRC-BART ( $H_g = 10$ ) | 0.053  | 0.277 | 0.276 | 0.94 | 0.97 | 0.021        | 0.132 | -0.032       | 0.206 |
| Sc4, $X_5$ , $\delta_2 = 0.5$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                               | LM-CP                   | -0.176 | 0.243 | 0.295 | 0.90 | 0.92 | -0.000       | 0.145 | 0.176        | 0.249 |
|                               | LM-PP                   | -0.007 | 0.348 | 0.338 | 0.94 | 0.83 | -0.000       | 0.145 | 0.006        | 0.243 |
|                               | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                               | BART-CP                 | -0.164 | 0.145 | 0.217 | 0.78 | 1.00 | 0.005        | 0.109 | 0.169        | 0.195 |
|                               | BART-PP                 | 0.000  | 0.208 | 0.203 | 0.95 | 1.00 | 0.005        | 0.110 | 0.005        | 0.168 |
|                               | PSCL                    | -0.120 | 0.270 | 0.287 | 0.95 | 0.92 | 0.000        | 0.148 | 0.121        | 0.209 |
|                               | MAP (100)               | -0.115 | 0.296 | 0.294 | 0.95 | 0.88 | 0.000        | 0.146 | 0.116        | 0.212 |
|                               | MAP (75)                | -0.101 | 0.306 | 0.297 | 0.95 | 0.88 | -0.000       | 0.146 | 0.100        | 0.209 |
|                               | MAP (50)                | -0.082 | 0.316 | 0.303 | 0.96 | 0.88 | -0.000       | 0.146 | 0.082        | 0.210 |
|                               | HierLM                  | -0.014 | 0.338 | 0.342 | 0.94 | 0.84 | -0.001       | 0.145 | 0.014        | 0.249 |
|                               | C-BART                  | -0.109 | 0.159 | 0.190 | 0.90 | 1.00 | 0.005        | 0.108 | 0.115        | 0.162 |
|                               | LRC-BART ( $w = 0$ )    | 0.007  | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | -0.001       | 0.169 |
|                               | LRC-BART (100)          | -0.043 | 0.180 | 0.200 | 0.92 | 1.00 | 0.005        | 0.109 | 0.048        | 0.168 |
|                               | LRC-BART (75)           | -0.040 | 0.181 | 0.197 | 0.93 | 1.00 | 0.005        | 0.110 | 0.045        | 0.165 |
|                               | LRC-BART (50)           | -0.031 | 0.182 | 0.194 | 0.94 | 1.00 | 0.005        | 0.110 | 0.036        | 0.160 |
|                               | LRC-BART (0.9)          | -0.043 | 0.180 | 0.200 | 0.93 | 1.00 | 0.005        | 0.109 | 0.048        | 0.169 |
|                               | LRC-BART (0.5)          | -0.035 | 0.182 | 0.196 | 0.93 | 1.00 | 0.005        | 0.109 | 0.040        | 0.163 |
|                               | LRC-BART (0.25)         | -0.022 | 0.184 | 0.195 | 0.93 | 1.00 | 0.005        | 0.109 | 0.028        | 0.161 |
|                               | LRC-BART ( $H_g = 10$ ) | -0.048 | 0.179 | 0.198 | 0.91 | 1.00 | 0.005        | 0.109 | 0.053        | 0.167 |
| Sc4, $X_5$ , $\delta_2 = 1$   | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                               | LM-CP                   | -0.361 | 0.244 | 0.433 | 0.67 | 0.74 | -0.000       | 0.145 | 0.361        | 0.403 |
|                               | LM-PP                   | -0.012 | 0.349 | 0.339 | 0.95 | 0.83 | -0.000       | 0.145 | 0.011        | 0.245 |
|                               | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                               | BART-CP                 | -0.338 | 0.146 | 0.368 | 0.35 | 1.00 | 0.005        | 0.109 | 0.343        | 0.357 |
|                               | BART-PP                 | -0.002 | 0.209 | 0.207 | 0.95 | 1.00 | 0.005        | 0.110 | 0.007        | 0.173 |
|                               | PSCL                    | -0.246 | 0.271 | 0.358 | 0.87 | 0.80 | 0.000        | 0.148 | 0.247        | 0.300 |
|                               | MAP (100)               | -0.224 | 0.306 | 0.361 | 0.91 | 0.75 | 0.000        | 0.146 | 0.224        | 0.292 |
|                               | MAP (75)                | -0.190 | 0.317 | 0.352 | 0.92 | 0.75 | -0.000       | 0.146 | 0.190        | 0.274 |
|                               | MAP (50)                | -0.152 | 0.328 | 0.346 | 0.94 | 0.77 | -0.000       | 0.146 | 0.152        | 0.260 |
|                               | HierLM                  | -0.025 | 0.338 | 0.343 | 0.94 | 0.83 | -0.001       | 0.145 | 0.024        | 0.250 |
|                               | C-BART                  | -0.207 | 0.173 | 0.273 | 0.73 | 1.00 | 0.005        | 0.108 | 0.212        | 0.258 |
|                               | LRC-BART ( $w = 0$ )    | 0.006  | 0.191 | 0.204 | 0.93 | 1.00 | 0.005        | 0.110 | -0.000       | 0.171 |

continued on the next page

Web Table S1, continued

| Scenario                    | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                             |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                             | LRC-BART (100)          | -0.041 | 0.195 | 0.223 | 0.89 | 1.00 | 0.005        | 0.109 | 0.046        | 0.194 |
|                             | LRC-BART (75)           | -0.041 | 0.194 | 0.221 | 0.90 | 1.00 | 0.005        | 0.110 | 0.046        | 0.191 |
|                             | LRC-BART (50)           | -0.036 | 0.193 | 0.213 | 0.90 | 1.00 | 0.005        | 0.110 | 0.042        | 0.184 |
|                             | LRC-BART (0.9)          | -0.041 | 0.196 | 0.224 | 0.90 | 1.00 | 0.005        | 0.109 | 0.047        | 0.194 |
|                             | LRC-BART (0.5)          | -0.037 | 0.194 | 0.218 | 0.90 | 1.00 | 0.005        | 0.109 | 0.042        | 0.187 |
|                             | LRC-BART (0.25)         | -0.032 | 0.192 | 0.208 | 0.92 | 1.00 | 0.005        | 0.109 | 0.037        | 0.178 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.046 | 0.195 | 0.222 | 0.89 | 1.00 | 0.005        | 0.109 | 0.051        | 0.195 |
| Sc4, $X_5$ , $\delta_2 = 2$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.732 | 0.247 | 0.771 | 0.16 | 0.21 | -0.000       | 0.145 | 0.732        | 0.755 |
|                             | LM-PP                   | -0.023 | 0.354 | 0.343 | 0.95 | 0.81 | -0.000       | 0.145 | 0.022        | 0.249 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.685 | 0.149 | 0.701 | 0.00 | 0.56 | 0.005        | 0.109 | 0.690        | 0.698 |
|                             | BART-PP                 | -0.006 | 0.209 | 0.205 | 0.96 | 1.00 | 0.005        | 0.110 | 0.011        | 0.170 |
|                             | PSCL                    | -0.498 | 0.273 | 0.562 | 0.54 | 0.44 | 0.000        | 0.148 | 0.498        | 0.528 |
|                             | MAP (100)               | -0.377 | 0.333 | 0.495 | 0.78 | 0.46 | 0.000        | 0.146 | 0.378        | 0.437 |
|                             | MAP (75)                | -0.296 | 0.344 | 0.451 | 0.84 | 0.54 | -0.000       | 0.146 | 0.296        | 0.380 |
|                             | MAP (50)                | -0.219 | 0.352 | 0.415 | 0.92 | 0.60 | -0.000       | 0.146 | 0.219        | 0.333 |
|                             | HierLM                  | -0.044 | 0.339 | 0.345 | 0.94 | 0.81 | -0.001       | 0.145 | 0.043        | 0.253 |
|                             | C-BART                  | -0.227 | 0.197 | 0.391 | 0.66 | 0.90 | 0.005        | 0.108 | 0.232        | 0.383 |
|                             | LRC-BART ( $w = 0$ )    | 0.004  | 0.192 | 0.207 | 0.92 | 1.00 | 0.005        | 0.110 | 0.001        | 0.173 |
|                             | LRC-BART (100)          | 0.000  | 0.189 | 0.202 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.168 |
|                             | LRC-BART (75)           | 0.001  | 0.190 | 0.201 | 0.93 | 1.00 | 0.005        | 0.110 | 0.004        | 0.167 |
|                             | LRC-BART (50)           | -0.002 | 0.190 | 0.201 | 0.92 | 1.00 | 0.005        | 0.110 | 0.008        | 0.168 |
|                             | LRC-BART (0.9)          | -0.000 | 0.189 | 0.202 | 0.93 | 1.00 | 0.005        | 0.109 | 0.006        | 0.170 |
|                             | LRC-BART (0.5)          | -0.001 | 0.190 | 0.202 | 0.92 | 1.00 | 0.005        | 0.109 | 0.006        | 0.168 |
|                             | LRC-BART (0.25)         | -0.001 | 0.191 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.007        | 0.168 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.004 | 0.188 | 0.199 | 0.93 | 1.00 | 0.005        | 0.109 | 0.009        | 0.166 |
| Sc4, $X_7$ , $\delta_2 = 1$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.363 | 0.244 | 0.434 | 0.66 | 0.72 | -0.000       | 0.145 | 0.363        | 0.404 |
|                             | LM-PP                   | -0.012 | 0.348 | 0.337 | 0.94 | 0.83 | -0.000       | 0.145 | 0.012        | 0.244 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.340 | 0.146 | 0.369 | 0.35 | 1.00 | 0.005        | 0.109 | 0.345        | 0.359 |
|                             | BART-PP                 | -0.001 | 0.209 | 0.206 | 0.96 | 1.00 | 0.005        | 0.110 | 0.006        | 0.172 |
|                             | PSCL                    | -0.246 | 0.269 | 0.358 | 0.87 | 0.81 | 0.000        | 0.148 | 0.246        | 0.301 |
|                             | MAP (100)               | -0.222 | 0.308 | 0.360 | 0.91 | 0.75 | 0.000        | 0.146 | 0.222        | 0.291 |
|                             | MAP (75)                | -0.189 | 0.318 | 0.352 | 0.92 | 0.74 | -0.000       | 0.146 | 0.189        | 0.274 |
|                             | MAP (50)                | -0.152 | 0.328 | 0.347 | 0.94 | 0.77 | -0.000       | 0.146 | 0.151        | 0.260 |
|                             | HierLM                  | 0.001  | 0.338 | 0.341 | 0.93 | 0.85 | -0.001       | 0.145 | -0.001       | 0.249 |
|                             | C-BART                  | -0.206 | 0.173 | 0.276 | 0.73 | 1.00 | 0.005        | 0.108 | 0.211        | 0.258 |
|                             | LRC-BART ( $w = 0$ )    | 0.009  | 0.191 | 0.206 | 0.92 | 1.00 | 0.005        | 0.110 | -0.004       | 0.172 |
|                             | LRC-BART (100)          | -0.038 | 0.195 | 0.224 | 0.90 | 1.00 | 0.005        | 0.109 | 0.043        | 0.196 |
|                             | LRC-BART (75)           | -0.035 | 0.194 | 0.220 | 0.91 | 1.00 | 0.005        | 0.110 | 0.041        | 0.190 |
|                             | LRC-BART (50)           | -0.031 | 0.193 | 0.217 | 0.90 | 1.00 | 0.005        | 0.110 | 0.037        | 0.187 |
|                             | LRC-BART (0.9)          | -0.040 | 0.195 | 0.223 | 0.90 | 1.00 | 0.005        | 0.109 | 0.045        | 0.194 |
|                             | LRC-BART (0.5)          | -0.034 | 0.193 | 0.218 | 0.91 | 1.00 | 0.005        | 0.109 | 0.039        | 0.188 |
|                             | LRC-BART (0.25)         | -0.025 | 0.191 | 0.212 | 0.91 | 1.00 | 0.005        | 0.109 | 0.031        | 0.180 |

continued on the next page

Web Table S1, continued

| Scenario                    | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                             |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
| Sc4, $X_7$ , $\delta_2 = 2$ | LRC-BART ( $H_g = 10$ ) | -0.044 | 0.195 | 0.221 | 0.89 | 1.00 | 0.005        | 0.109 | 0.049        | 0.192 |
|                             | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.735 | 0.246 | 0.773 | 0.16 | 0.20 | -0.000       | 0.145 | 0.735        | 0.758 |
|                             | LM-PP                   | -0.023 | 0.352 | 0.339 | 0.95 | 0.81 | -0.000       | 0.145 | 0.023        | 0.247 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.685 | 0.148 | 0.701 | 0.00 | 0.54 | 0.005        | 0.109 | 0.690        | 0.698 |
|                             | BART-PP                 | -0.002 | 0.209 | 0.206 | 0.95 | 1.00 | 0.005        | 0.110 | 0.007        | 0.173 |
|                             | PSCL                    | -0.498 | 0.271 | 0.562 | 0.53 | 0.45 | 0.000        | 0.148 | 0.498        | 0.528 |
|                             | MAP (100)               | -0.358 | 0.338 | 0.486 | 0.80 | 0.48 | 0.000        | 0.146 | 0.358        | 0.424 |
|                             | MAP (75)                | -0.284 | 0.347 | 0.446 | 0.88 | 0.54 | -0.000       | 0.146 | 0.283        | 0.374 |
|                             | MAP (50)                | -0.212 | 0.353 | 0.413 | 0.92 | 0.59 | -0.000       | 0.146 | 0.211        | 0.331 |
|                             | HierLM                  | 0.005  | 0.338 | 0.341 | 0.93 | 0.85 | -0.001       | 0.145 | -0.005       | 0.249 |
|                             | C-BART                  | -0.218 | 0.199 | 0.383 | 0.69 | 0.91 | 0.005        | 0.108 | 0.223        | 0.374 |
|                             | LRC-BART ( $w = 0$ )    | 0.007  | 0.192 | 0.209 | 0.92 | 1.00 | 0.005        | 0.110 | -0.002       | 0.176 |
|                             | LRC-BART (100)          | 0.001  | 0.189 | 0.203 | 0.93 | 1.00 | 0.005        | 0.109 | 0.004        | 0.171 |
|                             | LRC-BART (75)           | 0.004  | 0.189 | 0.202 | 0.93 | 1.00 | 0.005        | 0.110 | 0.001        | 0.169 |
|                             | LRC-BART (50)           | 0.002  | 0.190 | 0.205 | 0.93 | 1.00 | 0.005        | 0.110 | 0.003        | 0.173 |
|                             | LRC-BART (0.9)          | 0.001  | 0.189 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.004        | 0.171 |
|                             | LRC-BART (0.5)          | 0.003  | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.109 | 0.002        | 0.169 |
|                             | LRC-BART (0.25)         | 0.002  | 0.190 | 0.204 | 0.92 | 1.00 | 0.005        | 0.109 | 0.004        | 0.172 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.002 | 0.189 | 0.204 | 0.92 | 1.00 | 0.005        | 0.109 | 0.007        | 0.170 |
| Sc5, $\delta_2 = 1$         | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.734 | 0.244 | 0.771 | 0.15 | 0.20 | -0.000       | 0.145 | 0.734        | 0.755 |
|                             | LM-PP                   | -0.022 | 0.347 | 0.338 | 0.95 | 0.82 | -0.000       | 0.145 | 0.022        | 0.244 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.687 | 0.147 | 0.702 | 0.00 | 0.55 | 0.005        | 0.109 | 0.692        | 0.699 |
|                             | BART-PP                 | -0.004 | 0.208 | 0.205 | 0.95 | 1.00 | 0.005        | 0.110 | 0.010        | 0.171 |
|                             | PSCL                    | -0.498 | 0.269 | 0.562 | 0.53 | 0.45 | 0.000        | 0.148 | 0.498        | 0.527 |
|                             | MAP (100)               | -0.345 | 0.341 | 0.480 | 0.82 | 0.49 | 0.000        | 0.146 | 0.345        | 0.417 |
|                             | MAP (75)                | -0.275 | 0.349 | 0.444 | 0.88 | 0.54 | -0.000       | 0.146 | 0.275        | 0.370 |
|                             | MAP (50)                | -0.207 | 0.355 | 0.412 | 0.92 | 0.60 | -0.000       | 0.146 | 0.206        | 0.329 |
|                             | HierLM                  | -0.021 | 0.338 | 0.342 | 0.94 | 0.84 | -0.001       | 0.145 | 0.021        | 0.249 |
|                             | C-BART                  | -0.244 | 0.198 | 0.373 | 0.68 | 0.94 | 0.005        | 0.108 | 0.249        | 0.362 |
|                             | LRC-BART ( $w = 0$ )    | 0.007  | 0.190 | 0.205 | 0.92 | 1.00 | 0.005        | 0.110 | -0.002       | 0.170 |
|                             | LRC-BART (100)          | -0.005 | 0.192 | 0.210 | 0.91 | 1.00 | 0.005        | 0.109 | 0.010        | 0.176 |
|                             | LRC-BART (75)           | -0.005 | 0.192 | 0.210 | 0.90 | 1.00 | 0.005        | 0.110 | 0.010        | 0.177 |
|                             | LRC-BART (50)           | -0.008 | 0.192 | 0.209 | 0.92 | 1.00 | 0.005        | 0.110 | 0.013        | 0.176 |
|                             | LRC-BART (0.9)          | -0.002 | 0.192 | 0.208 | 0.92 | 1.00 | 0.005        | 0.109 | 0.007        | 0.176 |
|                             | LRC-BART (0.5)          | -0.006 | 0.192 | 0.211 | 0.92 | 1.00 | 0.005        | 0.109 | 0.011        | 0.178 |
|                             | LRC-BART (0.25)         | -0.011 | 0.192 | 0.211 | 0.92 | 1.00 | 0.005        | 0.109 | 0.016        | 0.179 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.009 | 0.191 | 0.207 | 0.92 | 1.00 | 0.005        | 0.109 | 0.014        | 0.176 |
|                             | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -1.478 | 0.247 | 1.497 | 0.00 | 0.51 | -0.000       | 0.145 | 1.478        | 1.489 |
|                             | LM-PP                   | -0.042 | 0.347 | 0.339 | 0.94 | 0.80 | -0.000       | 0.145 | 0.042        | 0.246 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -1.390 | 0.152 | 1.398 | 0.00 | 0.72 | 0.005        | 0.109 | 1.395        | 1.399 |

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Web Table S1, continued

| Scenario | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|----------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|          |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|          | BART-PP                 | -0.012 | 0.208 | 0.205 | 0.95 | 1.00 | 0.005        | 0.110 | 0.017        | 0.171 |
|          | PSCL                    | -1.002 | 0.269 | 1.035 | 0.04 | 0.04 | 0.000        | 0.148 | 1.002        | 1.017 |
|          | MAP (100)               | -0.395 | 0.348 | 0.525 | 0.78 | 0.41 | 0.000        | 0.146 | 0.395        | 0.465 |
|          | MAP (75)                | -0.279 | 0.355 | 0.452 | 0.89 | 0.50 | -0.000       | 0.146 | 0.278        | 0.377 |
|          | MAP (50)                | -0.187 | 0.357 | 0.406 | 0.93 | 0.60 | -0.000       | 0.146 | 0.186        | 0.319 |
|          | HierLM                  | -0.038 | 0.338 | 0.344 | 0.94 | 0.82 | -0.001       | 0.145 | 0.037        | 0.251 |
|          | C-BART                  | -0.316 | 0.194 | 0.650 | 0.76 | 0.93 | 0.005        | 0.108 | 0.322        | 0.648 |
|          | LRC-BART ( $w = 0$ )    | 0.006  | 0.189 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | -0.001       | 0.169 |
|          | LRC-BART (100)          | -0.001 | 0.190 | 0.203 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.171 |
|          | LRC-BART (75)           | -0.001 | 0.190 | 0.205 | 0.93 | 1.00 | 0.005        | 0.110 | 0.006        | 0.172 |
|          | LRC-BART (50)           | -0.001 | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | 0.006        | 0.169 |
|          | LRC-BART (0.9)          | -0.001 | 0.190 | 0.205 | 0.92 | 1.00 | 0.005        | 0.109 | 0.006        | 0.171 |
|          | LRC-BART (0.5)          | -0.001 | 0.190 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.006        | 0.171 |
|          | LRC-BART (0.25)         | 0.000  | 0.189 | 0.203 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.169 |
|          | LRC-BART ( $H_g = 10$ ) | -0.007 | 0.190 | 0.201 | 0.92 | 1.00 | 0.005        | 0.109 | 0.012        | 0.169 |

Table S2: Survival outcomes, two-arm design, all scenarios and methods,  $R = 500$ . Columns as in Table 3, with the mean posterior SD and the bias of the standardized control median on the log scale (Ctrl bias).

| Scenario | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|----------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|          |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
| Sc1      | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|          | AFT-CP               | 0.165        | 0.427 | 0.428 | 0.95 | 0.29 | -0.007    | -0.012                | 0.094 | 0.089 | 0.95 | 0.22 |
|          | AFT-PP               | 0.261        | 0.719 | 0.737 | 0.95 | 0.12 | 0.052     | -0.009                | 0.144 | 0.142 | 0.94 | 0.10 |
|          | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|          | BART-CP              | 0.064        | 0.238 | 0.232 | 0.96 | 0.63 | -0.019    | -0.011                | 0.074 | 0.072 | 0.95 | 0.35 |
|          | BART-PP              | 0.032        | 0.350 | 0.345 | 0.95 | 0.25 | 0.016     | -0.023                | 0.103 | 0.106 | 0.94 | 0.17 |
|          | HierAFT              | 0.246        | 0.701 | 0.733 | 0.94 | 0.12 | 0.060     | -0.008                | 0.148 | 0.152 | 0.92 | 0.11 |
|          | C-BART               | 0.034        | 0.268 | 0.248 | 0.97 | 0.42 | -0.007    | 0.003                 | 0.085 | 0.079 | 0.95 | 0.32 |
|          | LRC-BART ( $w = 0$ ) | 0.050        | 0.329 | 0.351 | 0.94 | 0.33 | 0.006     | -0.001                | 0.101 | 0.108 | 0.94 | 0.25 |
|          | LRC-BART (100)       | 0.045        | 0.305 | 0.312 | 0.95 | 0.34 | -0.001    | 0.003                 | 0.095 | 0.097 | 0.94 | 0.28 |
|          | LRC-BART (75)        | 0.048        | 0.308 | 0.312 | 0.96 | 0.36 | -0.001    | 0.003                 | 0.096 | 0.097 | 0.95 | 0.28 |
|          | LRC-BART (50)        | 0.047        | 0.312 | 0.318 | 0.96 | 0.35 | 0.001     | 0.003                 | 0.097 | 0.099 | 0.95 | 0.27 |
|          | LRC-BART (0.9)       | 0.042        | 0.298 | 0.304 | 0.96 | 0.38 | -0.001    | 0.002                 | 0.093 | 0.095 | 0.95 | 0.30 |
|          | LRC-BART (0.5)       | 0.048        | 0.305 | 0.309 | 0.96 | 0.35 | -0.002    | 0.004                 | 0.095 | 0.097 | 0.94 | 0.28 |
|          | LRC-BART (0.25)      | 0.047        | 0.311 | 0.320 | 0.95 | 0.35 | 0.001     | 0.003                 | 0.097 | 0.099 | 0.94 | 0.28 |
| Sc2      | AFT-NP               | 0.175        | 0.672 | 0.672 | 0.96 | 0.12 | 0.102     | -0.024                | 0.131 | 0.131 | 0.93 | 0.09 |
|          | AFT-CP               | 1.519        | 0.996 | 1.831 | 0.55 | 0.83 | -0.207    | 0.261                 | 0.153 | 0.304 | 0.57 | 0.82 |
|          | AFT-PP               | 0.389        | 0.763 | 0.818 | 0.95 | 0.19 | 0.020     | 0.050                 | 0.149 | 0.154 | 0.95 | 0.20 |
|          | BART-NP              | 0.251        | 0.427 | 0.495 | 0.91 | 0.43 | -0.042    | 0.110                 | 0.137 | 0.177 | 0.88 | 0.45 |
|          | BART-CP              | 0.194        | 0.330 | 0.383 | 0.93 | 0.56 | -0.048    | 0.037                 | 0.092 | 0.100 | 0.94 | 0.43 |

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Sc2

Web Table S2, continued

| Scenario           | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|--------------------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|                    |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
| Sc3, $\rho = -0.5$ | BART-PP              | 0.036        | 0.360 | 0.335 | 0.95 | 0.24 | 0.017     | -0.002                | 0.101 | 0.097 | 0.95 | 0.21 |
|                    | HierAFT              | 0.233        | 0.679 | 0.695 | 0.97 | 0.14 | 0.069     | 0.001                 | 0.136 | 0.134 | 0.95 | 0.12 |
|                    | C-BART               | 0.138        | 0.300 | 0.331 | 0.93 | 0.55 | -0.041    | 0.043                 | 0.090 | 0.104 | 0.92 | 0.49 |
|                    | LRC-BART ( $w = 0$ ) | 0.028        | 0.328 | 0.332 | 0.93 | 0.29 | 0.009     | 0.004                 | 0.096 | 0.099 | 0.94 | 0.24 |
|                    | LRC-BART (100)       | 0.061        | 0.322 | 0.322 | 0.94 | 0.34 | -0.007    | 0.018                 | 0.096 | 0.100 | 0.93 | 0.31 |
|                    | LRC-BART (75)        | 0.065        | 0.322 | 0.323 | 0.94 | 0.34 | -0.008    | 0.019                 | 0.096 | 0.100 | 0.93 | 0.32 |
|                    | LRC-BART (50)        | 0.061        | 0.322 | 0.322 | 0.94 | 0.34 | -0.007    | 0.018                 | 0.095 | 0.100 | 0.94 | 0.31 |
|                    | LRC-BART (0.9)       | 0.058        | 0.324 | 0.325 | 0.94 | 0.33 | -0.006    | 0.017                 | 0.096 | 0.100 | 0.94 | 0.30 |
|                    | LRC-BART (0.5)       | 0.051        | 0.326 | 0.322 | 0.95 | 0.31 | -0.002    | 0.015                 | 0.096 | 0.099 | 0.94 | 0.29 |
|                    | LRC-BART (0.25)      | 0.042        | 0.327 | 0.327 | 0.94 | 0.31 | 0.002     | 0.011                 | 0.096 | 0.100 | 0.94 | 0.28 |
|                    | AFT-NP               | 0.222        | 0.757 | 0.806 | 0.95 | 0.10 | 0.126     | -0.025                | 0.140 | 0.142 | 0.93 | 0.08 |
|                    | AFT-CP               | 3.017        | 1.311 | 3.271 | 0.03 | 1.00 | -0.320    | 0.483                 | 0.159 | 0.507 | 0.04 | 1.00 |
|                    | AFT-PP               | 0.438        | 0.802 | 0.909 | 0.95 | 0.18 | 0.009     | 0.054                 | 0.153 | 0.157 | 0.95 | 0.19 |
|                    | BART-NP              | 0.188        | 0.439 | 0.555 | 0.91 | 0.32 | -0.041    | 0.064                 | 0.150 | 0.172 | 0.92 | 0.24 |
|                    | BART-CP              | 1.953        | 0.596 | 2.063 | 0.00 | 1.00 | -0.302    | 0.499                 | 0.145 | 0.523 | 0.02 | 1.00 |
| Sc3, $\rho = 0$    | BART-PP              | 0.047        | 0.387 | 0.441 | 0.92 | 0.23 | -0.007    | -0.000                | 0.124 | 0.134 | 0.92 | 0.15 |
|                    | HierAFT              | 0.321        | 0.778 | 0.857 | 0.95 | 0.12 | 0.070     | 0.005                 | 0.146 | 0.143 | 0.95 | 0.10 |
|                    | C-BART               | 0.627        | 0.668 | 0.948 | 0.84 | 0.45 | -0.094    | 0.156                 | 0.158 | 0.236 | 0.81 | 0.43 |
|                    | LRC-BART ( $w = 0$ ) | 0.094        | 0.477 | 0.494 | 0.95 | 0.17 | 0.034     | 0.015                 | 0.131 | 0.135 | 0.94 | 0.16 |
|                    | LRC-BART (100)       | 0.147        | 0.502 | 0.539 | 0.95 | 0.18 | 0.018     | 0.030                 | 0.136 | 0.144 | 0.94 | 0.19 |
|                    | LRC-BART (75)        | 0.147        | 0.501 | 0.531 | 0.94 | 0.19 | 0.017     | 0.031                 | 0.136 | 0.142 | 0.94 | 0.19 |
|                    | LRC-BART (50)        | 0.150        | 0.499 | 0.534 | 0.95 | 0.19 | 0.016     | 0.032                 | 0.135 | 0.143 | 0.94 | 0.19 |
|                    | LRC-BART (0.9)       | 0.145        | 0.501 | 0.539 | 0.94 | 0.19 | 0.018     | 0.030                 | 0.135 | 0.143 | 0.94 | 0.20 |
|                    | LRC-BART (0.5)       | 0.147        | 0.501 | 0.540 | 0.95 | 0.17 | 0.018     | 0.031                 | 0.135 | 0.144 | 0.94 | 0.19 |
|                    | LRC-BART (0.25)      | 0.151        | 0.501 | 0.539 | 0.95 | 0.19 | 0.016     | 0.032                 | 0.136 | 0.144 | 0.94 | 0.19 |
|                    | AFT-NP               | 0.231        | 0.771 | 0.832 | 0.94 | 0.10 | 0.128     | -0.025                | 0.141 | 0.145 | 0.92 | 0.09 |
|                    | AFT-CP               | 3.527        | 1.395 | 3.769 | 0.01 | 1.00 | -0.342    | 0.553                 | 0.158 | 0.573 | 0.01 | 1.00 |
|                    | AFT-PP               | 0.440        | 0.806 | 0.919 | 0.94 | 0.19 | 0.008     | 0.054                 | 0.153 | 0.158 | 0.95 | 0.19 |
|                    | BART-NP              | 0.186        | 0.443 | 0.571 | 0.90 | 0.30 | -0.041    | 0.058                 | 0.152 | 0.173 | 0.92 | 0.21 |
|                    | BART-CP              | 2.308        | 0.631 | 2.412 | 0.00 | 1.00 | -0.323    | 0.588                 | 0.148 | 0.607 | 0.01 | 1.00 |
| Sc3, $\rho = 0.5$  | BART-PP              | 0.041        | 0.387 | 0.451 | 0.91 | 0.20 | -0.006    | -0.005                | 0.124 | 0.137 | 0.92 | 0.15 |
|                    | HierAFT              | 0.337        | 0.795 | 0.890 | 0.95 | 0.13 | 0.070     | 0.006                 | 0.147 | 0.147 | 0.95 | 0.11 |
|                    | C-BART               | 0.576        | 0.670 | 0.909 | 0.88 | 0.40 | -0.081    | 0.139                 | 0.159 | 0.222 | 0.85 | 0.38 |
|                    | LRC-BART ( $w = 0$ ) | 0.096        | 0.489 | 0.510 | 0.95 | 0.15 | 0.038     | 0.014                 | 0.133 | 0.137 | 0.94 | 0.17 |
|                    | LRC-BART (100)       | 0.155        | 0.516 | 0.558 | 0.94 | 0.17 | 0.020     | 0.031                 | 0.138 | 0.146 | 0.94 | 0.18 |
|                    | LRC-BART (75)        | 0.156        | 0.516 | 0.553 | 0.94 | 0.17 | 0.019     | 0.031                 | 0.138 | 0.145 | 0.93 | 0.18 |
|                    | LRC-BART (50)        | 0.156        | 0.514 | 0.554 | 0.94 | 0.18 | 0.019     | 0.031                 | 0.137 | 0.145 | 0.93 | 0.19 |
|                    | LRC-BART (0.9)       | 0.148        | 0.511 | 0.554 | 0.94 | 0.17 | 0.021     | 0.029                 | 0.136 | 0.144 | 0.93 | 0.18 |
|                    | LRC-BART (0.5)       | 0.151        | 0.515 | 0.552 | 0.95 | 0.18 | 0.020     | 0.030                 | 0.137 | 0.145 | 0.94 | 0.19 |
|                    | LRC-BART (0.25)      | 0.156        | 0.515 | 0.557 | 0.94 | 0.18 | 0.019     | 0.031                 | 0.138 | 0.146 | 0.93 | 0.18 |
|                    | AFT-NP               | 0.226        | 0.760 | 0.819 | 0.94 | 0.11 | 0.127     | -0.024                | 0.140 | 0.144 | 0.93 | 0.09 |
|                    | AFT-CP               | 3.038        | 1.318 | 3.285 | 0.03 | 1.00 | -0.321    | 0.486                 | 0.160 | 0.509 | 0.04 | 1.00 |
|                    | AFT-PP               | 0.441        | 0.803 | 0.914 | 0.95 | 0.18 | 0.009     | 0.054                 | 0.153 | 0.157 | 0.94 | 0.19 |
|                    | BART-NP              | 0.195        | 0.440 | 0.574 | 0.90 | 0.31 | -0.043    | 0.066                 | 0.151 | 0.176 | 0.92 | 0.23 |
|                    | BART-CP              | 1.957        | 0.597 | 2.069 | 0.01 | 1.00 | -0.302    | 0.499                 | 0.145 | 0.522 | 0.02 | 1.00 |
|                    | BART-PP              | 0.046        | 0.387 | 0.445 | 0.91 | 0.21 | -0.007    | -0.000                | 0.124 | 0.134 | 0.92 | 0.15 |

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Sc3,  $\rho = 0.5$

Web Table S2, continued

| Scenario                      | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|-------------------------------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|                               |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
|                               | HierAFT              | 0.327        | 0.781 | 0.872 | 0.95 | 0.14 | 0.071     | 0.006                 | 0.146 | 0.146 | 0.94 | 0.12 |
|                               | C-BART               | 0.617        | 0.662 | 0.948 | 0.85 | 0.44 | -0.091    | 0.154                 | 0.158 | 0.236 | 0.82 | 0.42 |
|                               | LRC-BART ( $w = 0$ ) | 0.096        | 0.476 | 0.503 | 0.94 | 0.16 | 0.033     | 0.016                 | 0.131 | 0.137 | 0.93 | 0.17 |
|                               | LRC-BART (100)       | 0.152        | 0.501 | 0.552 | 0.93 | 0.19 | 0.016     | 0.032                 | 0.136 | 0.146 | 0.92 | 0.20 |
|                               | LRC-BART (75)        | 0.153        | 0.502 | 0.551 | 0.94 | 0.19 | 0.016     | 0.033                 | 0.136 | 0.146 | 0.93 | 0.19 |
|                               | LRC-BART (50)        | 0.156        | 0.502 | 0.545 | 0.94 | 0.20 | 0.014     | 0.034                 | 0.136 | 0.144 | 0.93 | 0.20 |
|                               | LRC-BART (0.9)       | 0.146        | 0.501 | 0.549 | 0.94 | 0.18 | 0.018     | 0.031                 | 0.136 | 0.146 | 0.94 | 0.19 |
|                               | LRC-BART (0.5)       | 0.150        | 0.502 | 0.552 | 0.93 | 0.18 | 0.017     | 0.031                 | 0.136 | 0.146 | 0.94 | 0.19 |
|                               | LRC-BART (0.25)      | 0.156        | 0.501 | 0.548 | 0.94 | 0.20 | 0.014     | 0.034                 | 0.136 | 0.146 | 0.93 | 0.21 |
| Sc4, $X_5$ , $\delta_2 = 0.5$ | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                               | AFT-CP               | -0.123       | 0.351 | 0.347 | 0.92 | 0.09 | 0.083     | -0.083                | 0.086 | 0.116 | 0.85 | 0.06 |
|                               | AFT-PP               | 0.228        | 0.714 | 0.712 | 0.95 | 0.10 | 0.062     | -0.023                | 0.141 | 0.139 | 0.94 | 0.08 |
|                               | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                               | BART-CP              | -0.168       | 0.205 | 0.252 | 0.86 | 0.17 | 0.053     | -0.088                | 0.068 | 0.109 | 0.74 | 0.06 |
|                               | BART-PP              | 0.011        | 0.354 | 0.339 | 0.96 | 0.21 | 0.023     | -0.040                | 0.102 | 0.107 | 0.94 | 0.12 |
|                               | HierAFT              | 0.219        | 0.690 | 0.713 | 0.95 | 0.11 | 0.068     | -0.015                | 0.147 | 0.150 | 0.92 | 0.10 |
|                               | C-BART               | -0.094       | 0.253 | 0.247 | 0.94 | 0.21 | 0.033     | -0.048                | 0.081 | 0.089 | 0.92 | 0.13 |
|                               | LRC-BART ( $w = 0$ ) | 0.047        | 0.329 | 0.354 | 0.94 | 0.32 | 0.007     | -0.011                | 0.099 | 0.107 | 0.93 | 0.24 |
|                               | LRC-BART (100)       | 0.001        | 0.308 | 0.317 | 0.95 | 0.29 | 0.014     | -0.022                | 0.094 | 0.100 | 0.93 | 0.19 |
|                               | LRC-BART (75)        | 0.005        | 0.308 | 0.317 | 0.95 | 0.29 | 0.014     | -0.021                | 0.094 | 0.100 | 0.93 | 0.20 |
|                               | LRC-BART (50)        | 0.014        | 0.312 | 0.322 | 0.95 | 0.30 | 0.011     | -0.018                | 0.095 | 0.099 | 0.93 | 0.22 |
|                               | LRC-BART (0.9)       | -0.004       | 0.305 | 0.313 | 0.95 | 0.28 | 0.015     | -0.023                | 0.093 | 0.099 | 0.92 | 0.19 |
|                               | LRC-BART (0.5)       | 0.004        | 0.307 | 0.315 | 0.95 | 0.28 | 0.013     | -0.021                | 0.094 | 0.099 | 0.93 | 0.20 |
|                               | LRC-BART (0.25)      | 0.015        | 0.312 | 0.324 | 0.95 | 0.30 | 0.011     | -0.018                | 0.095 | 0.100 | 0.93 | 0.20 |
| Sc4, $X_5$ , $\delta_2 = 1$   | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                               | AFT-CP               | -0.357       | 0.289 | 0.446 | 0.75 | 0.05 | 0.193     | -0.144                | 0.079 | 0.163 | 0.57 | 0.05 |
|                               | AFT-PP               | 0.191        | 0.703 | 0.691 | 0.95 | 0.10 | 0.075     | -0.039                | 0.137 | 0.139 | 0.92 | 0.07 |
|                               | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                               | BART-CP              | -0.367       | 0.176 | 0.401 | 0.50 | 0.04 | 0.142     | -0.156                | 0.062 | 0.168 | 0.32 | 0.07 |
|                               | BART-PP              | -0.012       | 0.361 | 0.339 | 0.97 | 0.17 | 0.033     | -0.059                | 0.101 | 0.113 | 0.91 | 0.07 |
|                               | HierAFT              | 0.193        | 0.681 | 0.700 | 0.94 | 0.10 | 0.078     | -0.023                | 0.145 | 0.149 | 0.91 | 0.09 |
|                               | C-BART               | -0.202       | 0.251 | 0.305 | 0.84 | 0.08 | 0.076     | -0.093                | 0.080 | 0.120 | 0.79 | 0.04 |
|                               | LRC-BART ( $w = 0$ ) | 0.040        | 0.331 | 0.353 | 0.94 | 0.31 | 0.010     | -0.023                | 0.097 | 0.107 | 0.92 | 0.19 |
|                               | LRC-BART (100)       | -0.011       | 0.321 | 0.340 | 0.93 | 0.24 | 0.024     | -0.037                | 0.096 | 0.108 | 0.90 | 0.16 |
|                               | LRC-BART (75)        | -0.007       | 0.322 | 0.337 | 0.93 | 0.24 | 0.022     | -0.036                | 0.096 | 0.107 | 0.90 | 0.17 |
|                               | LRC-BART (50)        | -0.001       | 0.322 | 0.333 | 0.94 | 0.24 | 0.020     | -0.034                | 0.096 | 0.106 | 0.91 | 0.17 |
|                               | LRC-BART (0.9)       | -0.018       | 0.321 | 0.342 | 0.93 | 0.24 | 0.027     | -0.040                | 0.096 | 0.110 | 0.90 | 0.16 |
|                               | LRC-BART (0.5)       | -0.013       | 0.322 | 0.338 | 0.94 | 0.24 | 0.024     | -0.038                | 0.096 | 0.108 | 0.91 | 0.17 |
|                               | LRC-BART (0.25)      | 0.001        | 0.321 | 0.338 | 0.93 | 0.26 | 0.020     | -0.034                | 0.095 | 0.106 | 0.91 | 0.17 |
| Sc5, $\delta_2 = 1$           | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                               | AFT-CP               | -0.720       | 0.191 | 0.741 | 0.18 | 0.39 | 0.510     | -0.244                | 0.069 | 0.253 | 0.09 | 0.40 |
|                               | AFT-PP               | 0.193        | 0.702 | 0.695 | 0.94 | 0.09 | 0.075     | -0.029                | 0.141 | 0.141 | 0.93 | 0.08 |
|                               | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                               | BART-CP              | -0.703       | 0.119 | 0.712 | 0.00 | 0.64 | 0.402     | -0.256                | 0.054 | 0.262 | 0.01 | 0.65 |
|                               | BART-PP              | -0.029       | 0.352 | 0.333 | 0.95 | 0.17 | 0.037     | -0.041                | 0.107 | 0.110 | 0.93 | 0.11 |
|                               | HierAFT              | 0.204        | 0.683 | 0.701 | 0.94 | 0.10 | 0.073     | -0.018                | 0.146 | 0.150 | 0.92 | 0.09 |

Sc5,  $\delta_2 = 1$ 

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Web Table S2, continued

| Scenario | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|----------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|          |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
|          | C-BART               | -0.233       | 0.288 | 0.405 | 0.77 | 0.13 | 0.120     | -0.085                | 0.099 | 0.143 | 0.77 | 0.11 |
|          | LRC-BART ( $w = 0$ ) | 0.053        | 0.329 | 0.353 | 0.94 | 0.32 | 0.005     | 0.006                 | 0.103 | 0.110 | 0.93 | 0.26 |
|          | LRC-BART (100)       | 0.032        | 0.328 | 0.353 | 0.93 | 0.30 | 0.013     | 0.001                 | 0.104 | 0.112 | 0.93 | 0.26 |
|          | LRC-BART (75)        | 0.030        | 0.328 | 0.348 | 0.94 | 0.29 | 0.013     | 0.000                 | 0.104 | 0.111 | 0.93 | 0.25 |
|          | LRC-BART (50)        | 0.026        | 0.326 | 0.350 | 0.94 | 0.30 | 0.014     | -0.001                | 0.104 | 0.111 | 0.93 | 0.24 |
|          | LRC-BART (0.9)       | 0.029        | 0.327 | 0.350 | 0.94 | 0.31 | 0.014     | 0.001                 | 0.104 | 0.111 | 0.94 | 0.26 |
|          | LRC-BART (0.5)       | 0.028        | 0.326 | 0.349 | 0.94 | 0.30 | 0.013     | 0.000                 | 0.104 | 0.111 | 0.93 | 0.25 |
|          | LRC-BART (0.25)      | 0.028        | 0.325 | 0.351 | 0.94 | 0.30 | 0.013     | 0.000                 | 0.103 | 0.112 | 0.93 | 0.25 |

## G.5 Application

The merged dataset matches the baseline counts. We harmonize the covariate labels before constructing the model matrix. The non-Hispanic level of ethnicity is labelled differently in the two sources, so a dummy coding taken as recorded produces two non-Hispanic indicators, one per source, and every control model trained on UCMM and evaluated at the trial profiles then predicts the 29 non-Hispanic trial patients from the 13-patient Hispanic stratum of the registry; race as recorded has four levels with Asian as the reference level, although the table pools it into Other/Unknown. We harmonize both, to seven columns (age, male, Black, Other/Unknown, Hispanic, high-risk cytogenetics, ASCT), fit every model of Section 4.3 under the harmonized coding, and keep the original coding as a sensitivity analysis (Web Appendix G.5). The calibration in Section 2.4 also shows the effect of coding: under the original coding the ESS ceiling at the trial profiles is 12.6 for PFS and 14.6 for OS, so a target of 30 is capped, whereas under the harmonized coding it is 63.7 and 36.7.

We construct the registry cohort from 797 patients by excluding 7 with inconsistent survival times, those without a recorded induction regimen, 49 transplanted on or before induction, and 1 on elotuzumab with lenalidomide and dexamethasone. Under the original covariate coding the reference analyses of the earlier version give the following comparisons under the same scripts (posterior medians: complete-pooling AFT 1.03 and 0.90, hierarchical AFT 1.24 and 1.25, AFT-BART 1.06 and 0.91 for PFS and OS, against 1.03, 0.90, 1.21, 1.18, 1.06 and 0.90), which permits a coding

Table S3: The ESS ladder, Gaussian outcomes,  $R = 500$ : absolute targets  $N_{\text{target}} \in \{50, 75, 100\}$  and fractional targets 0.25, 0.5, 0.9 of the ceiling  $\sigma_1^2/V_\mu^f$ . Prior ESS is  $\text{ESS}_0$  at the calibrated scale, realized ESS its posterior counterpart, ceiling the mean ceiling, and capped the fraction of replicates in which the target was at or above the ceiling and the calibration returned the point-mass limit. Prior ESS is a block average and can exceed the whole-chain ceiling by a few percent when capped (Section 3.1).

| Scenario                 | Target          | Bias   | RMSE  | Cov  | Prior ESS | Realized ESS | Ceiling | Capped |
|--------------------------|-----------------|--------|-------|------|-----------|--------------|---------|--------|
| Sc1                      | LRC-BART (50)   | 0.009  | 0.181 | 0.94 | 42.7      | 28.5         | 128     | 0.00   |
|                          | LRC-BART (75)   | 0.008  | 0.176 | 0.94 | 62.1      | 40.6         | 127     | 0.04   |
|                          | LRC-BART (100)  | 0.009  | 0.172 | 0.94 | 82.1      | 52.5         | 127     | 0.20   |
|                          | LRC-BART (0.25) | 0.008  | 0.184 | 0.94 | 28.0      | 18.7         | 127     | 0.00   |
|                          | LRC-BART (0.5)  | 0.009  | 0.178 | 0.93 | 53.6      | 34.8         | 128     | 0.00   |
|                          | LRC-BART (0.9)  | 0.011  | 0.171 | 0.95 | 90.0      | 55.7         | 127     | 0.00   |
| Sc2                      | LRC-BART (50)   | 0.023  | 0.193 | 0.93 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (75)   | 0.022  | 0.192 | 0.93 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (100)  | 0.023  | 0.192 | 0.94 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (0.25) | 0.012  | 0.200 | 0.94 | 5.1       | 3.3          | 23      | 0.00   |
|                          | LRC-BART (0.5)  | 0.016  | 0.197 | 0.93 | 9.7       | 6.0          | 23      | 0.00   |
|                          | LRC-BART (0.9)  | 0.019  | 0.193 | 0.93 | 16.5      | 9.6          | 23      | 0.00   |
| Sc3, $\rho = 0$          | LRC-BART (50)   | 0.045  | 0.282 | 0.94 | 44.2      | 1.9          | 224     | 0.00   |
|                          | LRC-BART (75)   | 0.042  | 0.281 | 0.94 | 64.7      | 1.7          | 224     | 0.00   |
|                          | LRC-BART (100)  | 0.041  | 0.280 | 0.94 | 84.5      | 1.6          | 224     | 0.00   |
|                          | LRC-BART (0.25) | 0.045  | 0.279 | 0.94 | 49.4      | 1.9          | 224     | 0.00   |
|                          | LRC-BART (0.5)  | 0.041  | 0.280 | 0.94 | 94.1      | 1.6          | 224     | 0.00   |
|                          | LRC-BART (0.9)  | 0.037  | 0.277 | 0.94 | 160.7     | 1.3          | 224     | 0.00   |
| Sc4, $X_5, \delta_2 = 1$ | LRC-BART (50)   | -0.036 | 0.213 | 0.90 | 42.5      | 13.1         | 126     | 0.00   |
|                          | LRC-BART (75)   | -0.041 | 0.221 | 0.90 | 62.0      | 17.1         | 125     | 0.05   |
|                          | LRC-BART (100)  | -0.041 | 0.223 | 0.89 | 82.0      | 20.8         | 125     | 0.22   |
|                          | LRC-BART (0.25) | -0.032 | 0.208 | 0.92 | 27.6      | 9.7          | 126     | 0.00   |
|                          | LRC-BART (0.5)  | -0.037 | 0.218 | 0.90 | 52.5      | 14.8         | 125     | 0.00   |
|                          | LRC-BART (0.9)  | -0.041 | 0.224 | 0.90 | 89.4      | 21.1         | 126     | 0.00   |
| Sc4, $X_5, \delta_2 = 2$ | LRC-BART (50)   | -0.002 | 0.201 | 0.92 | 42.4      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (75)   | 0.001  | 0.201 | 0.93 | 61.8      | 0.8          | 125     | 0.04   |
|                          | LRC-BART (100)  | 0.000  | 0.202 | 0.92 | 83.1      | 0.8          | 124     | 0.27   |
|                          | LRC-BART (0.25) | -0.001 | 0.204 | 0.93 | 27.4      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (0.5)  | -0.001 | 0.202 | 0.92 | 52.2      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (0.9)  | -0.000 | 0.202 | 0.93 | 88.5      | 0.8          | 124     | 0.00   |
| Sc5, $\delta_2 = 2$      | LRC-BART (50)   | -0.001 | 0.203 | 0.93 | 42.7      | 0.3          | 128     | 0.00   |
|                          | LRC-BART (75)   | -0.001 | 0.205 | 0.93 | 62.1      | 0.3          | 127     | 0.04   |
|                          | LRC-BART (100)  | -0.001 | 0.203 | 0.92 | 82.1      | 0.2          | 127     | 0.20   |
|                          | LRC-BART (0.25) | 0.000  | 0.203 | 0.92 | 28.0      | 0.3          | 127     | 0.00   |
|                          | LRC-BART (0.5)  | -0.001 | 0.204 | 0.93 | 53.6      | 0.3          | 128     | 0.00   |
|                          | LRC-BART (0.9)  | -0.001 | 0.205 | 0.92 | 90.0      | 0.3          | 127     | 0.00   |

Table S4: Regional and global shifts, survival outcomes (median ratio),  $R = 500$ : columns as in Table 2, with the surface RMSE, the map threshold  $c = 0.5$  and  $g$  on the log-time scale.

| Scenario  | Method               | ATE    |      | Surface RMSE  |                 | $\bar{B}_c$   |                 | Mean $\hat{g}$ |                 | ESS  |
|-----------|----------------------|--------|------|---------------|-----------------|---------------|-----------------|----------------|-----------------|------|
|           |                      | Bias   | Cov  | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$  | $\mathcal{R}^c$ |      |
| Sc4 (0.5) | BART-PP              | 0.011  | 0.96 | 0.783         | 0.838           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.094 | 0.94 | 0.747         | 0.805           | 0.96          | 0.96            | −0.08          | −0.08           | 44.6 |
|           | LRC-BART ( $w = 0$ ) | 0.047  | 0.94 | 0.784         | 0.820           | 0.65          | 0.64            | −0.18          | −0.24           | 0.2  |
|           | LRC-BART (100)       | 0.001  | 0.95 | 0.764         | 0.809           | 0.79          | 0.78            | −0.15          | −0.18           | 19.2 |
| Sc4 (1)   | BART-PP              | −0.012 | 0.97 | 0.803         | 0.888           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.202 | 0.84 | 0.762         | 0.920           | 0.91          | 0.90            | −0.16          | −0.17           | 44.5 |
|           | LRC-BART ( $w = 0$ ) | 0.040  | 0.94 | 0.814         | 0.861           | 0.52          | 0.46            | −0.33          | −0.53           | 0.2  |
|           | LRC-BART (100)       | −0.011 | 0.93 | 0.792         | 0.868           | 0.64          | 0.58            | −0.30          | −0.44           | 12.8 |
| Sc5 (1)   | BART-PP              | −0.029 | 0.95 | —             | 0.827           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.233 | 0.77 | —             | 0.843           | —             | 0.43            | —              | −0.62           | 25.9 |
|           | LRC-BART ( $w = 0$ ) | 0.053  | 0.94 | —             | 0.810           | —             | 0.15            | —              | −0.96           | 0.3  |
|           | LRC-BART (100)       | 0.032  | 0.93 | —             | 0.807           | —             | 0.16            | —              | −0.94           | 2.0  |

Table S5: Baseline characteristics of the merged analytic cohort: median (range) for age,  $n$  (%) otherwise. Survival is measured from the first day of treatment (EloKRd) or the start of induction (UCMM); RMST at 5 years is the Kaplan–Meier estimate.

| Characteristic         | EloKRd ( $n = 30$ ) | UCMM ( $n = 253$ ) |
|------------------------|---------------------|--------------------|
| Age, years             | 62.8 (43–81)        | 61.3 (32–97)       |
| Male                   | 21 (70%)            | 130 (51%)          |
| Race: White            | 23 (77%)            | 164 (65%)          |
| Race: Black            | 4 (13%)             | 76 (30%)           |
| Race: Other/Unknown    | 3 (10%)             | 13 (5%)            |
| Hispanic or Latino     | 1 (3%)              | 13 (5%)            |
| High-risk cytogenetics | 15 (50%)            | 58 (23%)           |
| ASCT after induction   | 9 (30%)             | 203 (80%)          |
| PFS events             | 13 (43%)            | 159 (63%)          |
| Median PFS, years      | not reached         | 5.09               |
| PFS RMST at 5 years    | 3.71                | 3.65               |
| OS events              | 10 (33%)            | 60 (24%)           |
| Median OS, years       | not reached         | not reached        |
| OS RMST at 5 years     | 4.12                | 4.57               |

sensitivity comparison of the reference rows in Table 5; the MAP-AFT-BART rows cannot be rerun because their sampler is not in the repository.

The posterior of the RMST ratio under the three absolute targets (Web Figure S2) is unimodal with a largest observed draw of 2.0 for OS and one draw reaching 3.1 among 8,000 for PFS.

The covariate coding behind the model-based analyses of the earlier version does not match the baseline table, in the two respects described in Section 4.1: the non-Hispanic level of ethnicity was labelled differently in the two sources, so the dummy coding produced two non-Hispanic indicators, one per source, and race kept a four-level coding with Asian as the reference level although the table pools it into Other/Unknown. The reference analyses of Table 5 are rerun under the harmonized coding; the hierarchical AFT row keeps the prior scale used in the earlier analysis. The lower block of Table 5 reports the previous analysis. Its MAP-AFT-BART rows had 97.5% limits of 16 to 33 for PFS and 4.8 to 7.2 for OS, attributed at the time to an identification limit of a 30-patient single-arm trial. The corresponding limits in our analysis are 1.23 to 1.40 and 1.04 to 1.11, and over 8,000 draws the PFS ratio at  $N_{\text{target}} = 30$  has a 99.9th percentile of 1.6 and a maximum of 3.1. Under the earlier prior, a leaf-level discrepancy of unbounded scale could send the control RMST toward zero on rare draws; with the discrepancy carried by a five-tree ensemble of pointwise variance  $H_g \tau_0^2$ , about 0.07 on the log-time scale at  $N_{\text{target}} = 30$ , the observed tails are shorter, and the calibration keeps the point estimate within 0.02 across the ladder. The point estimates move from 1.08 to 1.10 down to 0.97 to 0.99 for PFS and from 0.97 to 1.00 down to 0.90 to 0.92 for OS; the original-coding sensitivity row gives 1.04 for PFS, so about 0.06 of the PFS change is the coding and the rest is the model.

## G.6 The prior cost of a regional shift in a shared-leaf design

Section 2.2 states that the decomposition into a pooled surface  $f$  and a discrepancy ensemble  $g$  is what lets a regional shift be detected, and we give the calculation here because the alternative, one ensemble with two means per leaf coupled through a common variance, looks equally natural. Suppose the RWD surface is shifted by  $\delta$  on a region. In the shared-leaf design with  $H$  trees and a commensurate scale  $\tau^2$ , the cheapest fit spreads  $\delta/H$  across every tree under the spike, at prior cost  $\delta^2/(2H\tau^2)$ ; routing the shift through  $m$  slab leaves of variance  $\lambda^2$  instead costs  $m c + \delta^2/(2m\lambda^2)$  with  $c$  the log prior odds plus the log ratio of the two scales, and the slab route is cheaper only if

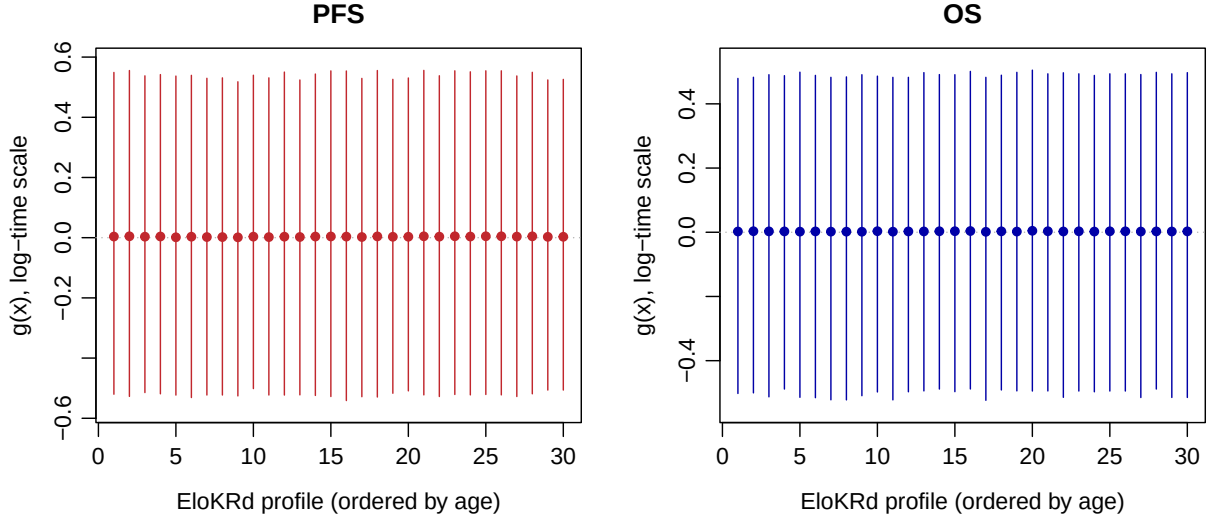


Figure S1: Posterior mean and 95% interval of the discrepancy  $g(x)$  on the log-time scale at the 30 EloKRd profiles, ordered by age, at  $N_{\text{target}} = 30$  for PFS (red) and OS (blue). In single-arm mode  $g$  is a prior draw, so the intervals are centred at zero with SD 0.26 everywhere; discrepancy uncertainty is therefore the same at every profile. Variation in external support is summarized by the local ESS map in Figure 2.

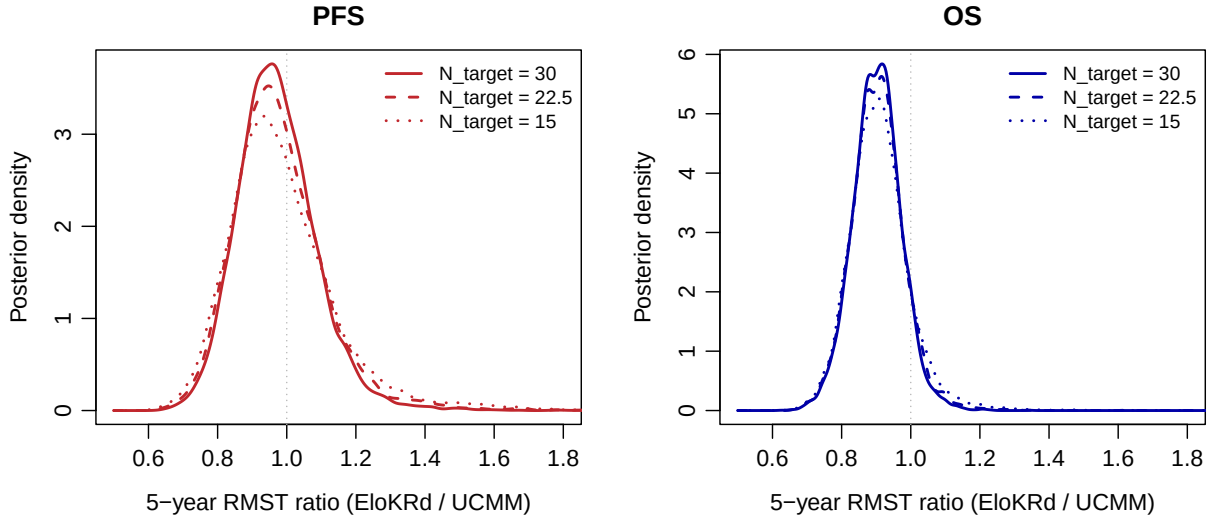


Figure S2: Posterior density of the standardized 5-year RMST ratio for PFS and OS under LRC-BART at  $N_{\text{target}} = 30, 22.5$  and 15 (harmonized coding,  $H_f = 10$ ,  $w = 1$ ).

$\delta > 2H\tau^2\sqrt{2c}/\lambda$ . At  $H = 50$  and the scales of our simulation study this threshold is between 17 and 86 on an outcome with range 10, so no shift of clinical size flips an indicator and the borrowing map stays flat. In the decomposition (1) the shift lands in one or two leaves of one  $g$  tree after a single split on the region boundary, which the tree prior allows at ordinary cost, and the slab wins as soon as  $\frac{\delta^2}{2}\{(H_g\tau_0^2)^{-1} - \tau_1^{-2}\}$  exceeds the slab's prior cost  $\log \frac{w}{1-w} + \log \frac{\tau_1}{\tau_0}$ . The design keeps  $H_g$  small on purpose: with few trees and a small  $\tau_0$ , the spike cannot hide a shift by spreading it. Letting  $f$  absorb part of the shift misfits both sources and pays in the likelihood in proportion to  $n_1 + n_2$ .

## G.7 The single-arm simulation in detail

We provide further details of the single-arm study in Section 3.4. The trial contributes  $n_T$  treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and the estimands are standardized to the trial profiles. The comparators reduce to the complete-pooling models, the hierarchical models, BART-CP, and LRC-BART in the single-arm mode of Section 2.6 with  $w = 1$ , at the absolute target 100, at 0.9 of the ceiling, and with the spike scale at the grid minimum, which is near-complete pooling of the leaf means;  $w = 0.9$  is the slab sensitivity. Sc3 is not identifiable without an internal control and is omitted. Table 4 and Web Table S6 report the results.

Under Sc1, BART-CP, the pooled LRC-BART and the target-100 LRC-BART have Gaussian RMSE 0.16 with coverage 0.94 to 1.00, and at  $n_T = 30$  the survival RMST ratio has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Under Sc2, all methods have substantial bias. With the source assignment of Sc2 and no RCT controls, the RWD is the only information about the control surface at trial profiles that lie where RWD support is limited, and every method is far more biased than in the two-arm study: on the Gaussian ATE, LM-CP has bias 1.68, BART-CP 0.41 with coverage 0.75, and LRC-BART 0.42, and the survival median ratio at  $n_T = 30$  has biases of 0.8 to 1.3 for the tree models. Single-arm borrowing under measured confounding is a harder problem than the two-arm study, because nothing in the data can reveal a discrepancy at profiles the RWD does not cover; the rows reported for this scenario in the earlier version came from a different mechanism (Web Appendix G.2).

For LRC-BART, uncertainty depends on whether the target is capped. LRC-BART (100) gives the same point estimate as the pooled model in every design, within 0.005, and the same posterior

wherever the target is capped: the single-arm ceiling is 27 to 34 in Sc2 and 75 at  $n_T = 30$  in Sc1, so the absolute target is capped there, and where it is not, about 160 in Sc1 at  $n_T = 200$ , the prior draw of  $g$  widens the posterior without moving it (Web Table S6); the fractional rung at 0.9 of the ceiling gives the same point estimates with wider intervals and coverage 0.82 to 0.99, reflecting the role of calibration in controlling prior uncertainty.

The  $w = 0.9$  sensitivity has a posterior SD of 1.4 on the Gaussian ATE and of 10 to 34 on the median ratio with power 0 and coverage 1, the point estimate unchanged, because with no RCT controls the slab draws add prior noise of variance  $(1 - w)H_g\tau_1^2$  to the counterfactual control and nothing else, which is why we report  $w = 1$  in single-arm mode. HierLM and HierAFT with the stated hyperprior and no RCT controls draw the control surface from the hierarchy, and their posterior SDs are 8 on the Gaussian ATE and extremely large on the median ratio; these results make the stated model unusable in this setting (Web Appendix G.2).

Table S6: Single-arm design, all methods,  $R = 500$ . Columns as in Table 4; LRC-BART (100) and LRC-BART ( $w = 0.9$ ) use the absolute target 100, capped at the ceiling where infeasible, and LRC-BART (0.9,  $w = 0.9$ ) the fractional target with the slab.

| Design                                      | Method                     | Treatment effect |          |          |      |      | Control  |          |       |     |
|---------------------------------------------|----------------------------|------------------|----------|----------|------|------|----------|----------|-------|-----|
|                                             |                            | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior | ESS |
| Gaussian, $n_T = 200$<br>ATE<br>Sc1         | LM-CP                      | 0.007            | 0.259    | 0.254    | 0.95 | 0.97 | -0.003   | 0.233    | —     |     |
|                                             | HierLM                     | -0.014           | 8.239    | 0.497    | 1.00 | 0.00 | 0.018    | 0.487    | —     |     |
|                                             | BART-CP                    | 0.010            | 0.163    | 0.162    | 0.94 | 1.00 | -0.003   | 0.127    | —     |     |
|                                             | LRC-BART (pooled)          | 0.007            | 0.165    | 0.162    | 0.94 | 1.00 | 0.001    | 0.127    | 172.3 |     |
|                                             | LRC-BART (100)             | 0.006            | 0.252    | 0.161    | 1.00 | 1.00 | 0.001    | 0.127    | 81.9  |     |
|                                             | LRC-BART (0.9)             | 0.007            | 0.202    | 0.163    | 0.98 | 1.00 | 0.000    | 0.128    | 115.8 |     |
|                                             | LRC-BART ( $w = 0.9$ )     | 0.010            | 1.383    | 0.169    | 1.00 | 0.00 | -0.003   | 0.136    | 81.9  |     |
|                                             | LRC-BART (0.9, $w = 0.9$ ) | 0.006            | 1.372    | 0.169    | 1.00 | 0.00 | 0.001    | 0.133    | 116.4 |     |
| Gaussian, $n_T = 200$<br>ATE<br>Sc2         | LM-CP                      | 1.682            | 0.417    | 1.734    | 0.01 | 1.00 | -1.676   | 1.726    | —     |     |
|                                             | HierLM                     | 1.642            | 8.317    | 1.754    | 1.00 | 0.00 | -1.636   | 1.747    | —     |     |
|                                             | BART-CP                    | 0.408            | 0.310    | 0.506    | 0.75 | 1.00 | -0.399   | 0.492    | —     |     |
|                                             | LRC-BART (pooled)          | 0.420            | 0.316    | 0.515    | 0.75 | 1.00 | -0.412   | 0.501    | 34.8  |     |
|                                             | LRC-BART (100)             | 0.421            | 0.316    | 0.515    | 0.76 | 1.00 | -0.413   | 0.501    | 34.7  |     |
|                                             | LRC-BART (0.9)             | 0.417            | 0.416    | 0.511    | 0.89 | 0.98 | -0.409   | 0.498    | 23.1  |     |
|                                             | LRC-BART ( $w = 0.9$ )     | 0.416            | 1.329    | 0.511    | 1.00 | 0.00 | -0.408   | 0.498    | 34.8  |     |
|                                             | LRC-BART (0.9, $w = 0.9$ ) | 0.420            | 1.345    | 0.514    | 1.00 | 0.00 | -0.411   | 0.500    | 23.0  |     |
| Survival, $n_T = 30$<br>median ratio<br>Sc1 | AFT-CP                     | 1.540            | 4.678    | 4.444    | 0.92 | 0.15 | -0.006   | 0.204    | —     |     |
|                                             | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —     |     |
|                                             | BART-CP                    | -0.013           | 0.580    | 0.482    | 0.97 | 0.05 | -0.026   | 0.103    | —     |     |
|                                             | LRC-BART (pooled)          | 0.182            | 0.918    | 0.686    | 0.98 | 0.02 | -0.017   | 0.102    | 80.2  |     |
|                                             | LRC-BART (100)             | 0.180            | 0.917    | 0.686    | 0.98 | 0.03 | -0.017   | 0.099    | 79.0  |     |

*continued on the next page*

Web Table S6, continued

| Design                                       | Method                     | Treatment effect |          |          |      |      | Control  |          |           |
|----------------------------------------------|----------------------------|------------------|----------|----------|------|------|----------|----------|-----------|
|                                              |                            | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior ESS |
|                                              | LRC-BART (0.9)             | 0.222            | 1.174    | 0.711    | 0.99 | 0.01 | -0.005   | 0.133    | 51.8      |
|                                              | LRC-BART ( $w = 0.9$ )     | 1.423            | 11.199   | 1.959    | 1.00 | 0.00 | 0.318    | 0.433    | 79.3      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 1.555            | 13.493   | 2.257    | 1.00 | 0.00 | 0.335    | 0.469    | 52.2      |
| Survival, $n_T = 30$<br>median ratio<br>Sc2  | AFT-CP                     | 20.944           | 46.014   | 51.645   | 0.19 | 0.89 | -0.441   | 0.526    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 0.814            | 1.109    | 1.231    | 0.92 | 0.25 | -0.199   | 0.277    | —         |
|                                              | LRC-BART (pooled)          | 1.324            | 1.859    | 1.864    | 0.95 | 0.21 | -0.207   | 0.284    | 29.2      |
|                                              | LRC-BART (100)             | 1.329            | 1.865    | 1.875    | 0.95 | 0.20 | -0.208   | 0.284    | 29.1      |
|                                              | LRC-BART (0.9)             | 1.511            | 3.461    | 2.055    | 0.97 | 0.12 | -0.180   | 0.279    | 19.3      |
|                                              | LRC-BART ( $w = 0.9$ )     | 4.243            | 28.669   | 5.305    | 1.00 | 0.00 | 0.179    | 0.537    | 28.9      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 4.544            | 33.228   | 6.324    | 1.00 | 0.00 | 0.193    | 0.446    | 19.0      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc1    | AFT-CP                     | 0.036            | 0.178    | 0.202    | 0.90 | 0.16 | 0.079    | 0.187    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.311    | 0.355    | —         |
|                                              | BART-CP                    | -0.111           | 0.176    | 0.186    | 0.92 | 0.02 | -0.021   | 0.089    | —         |
|                                              | LRC-BART (pooled)          | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.085    | 80.2      |
|                                              | LRC-BART (100)             | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.084    | 79.0      |
|                                              | LRC-BART (0.9)             | -0.040           | 0.218    | 0.156    | 0.98 | 0.01 | 0.002    | 0.085    | 51.8      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.080            | 0.998    | 0.200    | 1.00 | 0.00 | 0.020    | 0.089    | 79.3      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.086            | 1.077    | 0.204    | 1.00 | 0.00 | 0.020    | 0.090    | 52.2      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc2    | AFT-CP                     | 1.216            | 0.522    | 1.336    | 0.13 | 0.94 | -0.518   | 0.559    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.017    | 0.201    | —         |
|                                              | BART-CP                    | 0.149            | 0.271    | 0.286    | 0.95 | 0.14 | -0.219   | 0.260    | —         |
|                                              | LRC-BART (pooled)          | 0.234            | 0.297    | 0.339    | 0.94 | 0.19 | -0.210   | 0.251    | 29.2      |
|                                              | LRC-BART (100)             | 0.235            | 0.297    | 0.340    | 0.94 | 0.19 | -0.211   | 0.251    | 29.1      |
|                                              | LRC-BART (0.9)             | 0.256            | 0.452    | 0.358    | 0.97 | 0.13 | -0.207   | 0.248    | 19.3      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.527            | 2.689    | 0.643    | 1.00 | 0.00 | -0.185   | 0.228    | 28.9      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.560            | 3.085    | 0.731    | 1.00 | 0.00 | -0.184   | 0.228    | 19.0      |
| Survival, $n_T = 200$<br>median ratio<br>Sc1 | AFT-CP                     | 0.168            | 0.470    | 0.480    | 0.95 | 0.23 | 0.002    | 0.098    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 0.108            | 0.273    | 0.286    | 0.95 | 0.57 | -0.026   | 0.058    | —         |
|                                              | LRC-BART (pooled)          | 0.076            | 0.272    | 0.262    | 0.97 | 0.49 | -0.018   | 0.055    | 169.5     |
|                                              | LRC-BART (100)             | 0.108            | 0.565    | 0.284    | 1.00 | 0.17 | -0.008   | 0.055    | 81.7      |
|                                              | LRC-BART (0.9)             | 0.087            | 0.350    | 0.268    | 0.99 | 0.34 | -0.014   | 0.055    | 114.0     |
|                                              | LRC-BART ( $w = 0.9$ )     | 1.492            | 15.408   | 4.227    | 1.00 | 0.00 | 0.318    | 0.402    | 81.8      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 1.302            | 10.341   | 1.607    | 1.00 | 0.00 | 0.312    | 0.428    | 113.7     |
| Survival, $n_T = 200$<br>median ratio<br>Sc2 | AFT-CP                     | 9.478            | 5.162    | 10.850   | 0.01 | 1.00 | -0.407   | 0.419    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 1.136            | 0.778    | 1.359    | 0.60 | 0.82 | -0.189   | 0.214    | —         |
|                                              | LRC-BART (pooled)          | 1.195            | 0.851    | 1.409    | 0.64 | 0.82 | -0.196   | 0.219    | 35.9      |
|                                              | LRC-BART (100)             | 1.193            | 0.856    | 1.410    | 0.65 | 0.81 | -0.196   | 0.219    | 35.9      |
|                                              | LRC-BART (0.9)             | 1.336            | 2.306    | 1.562    | 0.82 | 0.57 | -0.181   | 0.207    | 23.9      |
|                                              | LRC-BART ( $w = 0.9$ )     | 3.935            | 26.373   | 4.611    | 1.00 | 0.00 | 0.142    | 0.396    | 36.2      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 4.380            | 34.059   | 7.322    | 1.00 | 0.00 | 0.158    | 0.379    | 23.8      |
|                                              | AFT-CP                     | -0.023           | 0.101    | 0.098    | 0.94 | 0.15 | 0.074    | 0.114    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.305    | 0.317    | —         |

Survival,  $n_T = 200$   
RMST ratio  
Sc1

continued on the next page

Web Table S6, continued

| Design                                     | Method                     | Treatment effect |          |          |      |      | Control |       |           |
|--------------------------------------------|----------------------------|------------------|----------|----------|------|------|---------|-------|-----------|
|                                            |                            | Bias             | SD       | RMSE     | Cov  | Pow  | Bias    | RMSE  | Prior ESS |
|                                            | BART-CP                    | −0.001           | 0.084    | 0.082    | 0.97 | 0.32 | −0.021  | 0.061 | —         |
|                                            | LRC-BART (pooled)          | −0.024           | 0.078    | 0.076    | 0.95 | 0.25 | 0.001   | 0.054 | 169.5     |
|                                            | LRC-BART (100)             | −0.020           | 0.121    | 0.076    | 0.99 | 0.09 | 0.003   | 0.054 | 81.7      |
|                                            | LRC-BART (0.9)             | −0.022           | 0.094    | 0.076    | 0.99 | 0.17 | 0.002   | 0.054 | 114.0     |
|                                            | LRC-BART ( $w = 0.9$ )     | 0.109            | 1.179    | 0.275    | 1.00 | 0.00 | 0.021   | 0.057 | 81.8      |
|                                            | LRC-BART (0.9, $w = 0.9$ ) | 0.097            | 0.860    | 0.138    | 1.00 | 0.00 | 0.019   | 0.057 | 113.7     |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc2 | AFT-CP                     | 1.164            | 0.428    | 1.244    | 0.01 | 1.00 | −0.535  | 0.550 | —         |
|                                            | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.001   | 0.101 | —         |
|                                            | BART-CP                    | 0.305            | 0.206    | 0.361    | 0.61 | 0.77 | −0.233  | 0.261 | —         |
|                                            | LRC-BART (pooled)          | 0.284            | 0.206    | 0.336    | 0.70 | 0.74 | −0.224  | 0.250 | 35.9      |
|                                            | LRC-BART (100)             | 0.284            | 0.207    | 0.336    | 0.69 | 0.73 | −0.224  | 0.250 | 35.9      |
|                                            | LRC-BART (0.9)             | 0.301            | 0.350    | 0.353    | 0.85 | 0.50 | −0.222  | 0.248 | 23.9      |
|                                            | LRC-BART ( $w = 0.9$ )     | 0.562            | 2.419    | 0.633    | 1.00 | 0.00 | −0.200  | 0.228 | 36.2      |
|                                            | LRC-BART (0.9, $w = 0.9$ ) | 0.585            | 2.496    | 0.662    | 1.00 | 0.00 | −0.198  | 0.226 | 23.8      |